

---

DELPHIX MASKING · PRE-CONFIGURED PROFILE SET

# Panama — Ley 81 de 2019

Discovery and masking of Panamanian personal data: cédula and RUC with a valid check digit, quasi-identifiers, the sensitive data of article 4, economic and credit data, and criminal records.

---

76

DOMAINS

135

CLASSIFIERS

142

ALGORITHMS

# Contents

The first sections explain what the set is for and how it works. The domain chapters then describe every domain: what it is, how its data is found and how it is masked, with examples. The appendices list every algorithm, file and classifier the set contains.

|          |                                         |
|----------|-----------------------------------------|
| <b>1</b> | <b>About this profile set</b>           |
| <b>2</b> | <b>Law 81 of 2019 at a glance</b>       |
| <b>3</b> | <b>Using the set</b>                    |
| <b>4</b> | <b>How sensitive data is discovered</b> |
| <b>5</b> | <b>How data is masked</b>               |
| <b>6</b> | <b>Domains at a glance</b>              |

## PART 7 · DIRECT IDENTIFIERS

|             |                                         |                             |
|-------------|-----------------------------------------|-----------------------------|
| <b>7.1</b>  | <b>Cédula (national ID)</b>             | PA_L1_CEDULA                |
| <b>7.2</b>  | <b>RUC (taxpayer number)</b>            | PA_L1_RUC                   |
| <b>7.3</b>  | <b>DV in its own column</b>             | PA_L1_RUC_DV                |
| <b>7.4</b>  | <b>Passport</b>                         | PA_L1_PASAPORTE             |
| <b>7.5</b>  | <b>Immigration document</b>             | PA_L1_DOCUMENTO_MIGRATORIO  |
| <b>7.6</b>  | <b>Social security number</b>           | PA_L1_SEGURO_SOCIAL         |
| <b>7.7</b>  | <b>Professional license (idoneidad)</b> | PA_L1_IDONEIDAD             |
| <b>7.8</b>  | <b>Given names</b>                      | PA_L1_NOMBRE                |
| <b>7.9</b>  | <b>Surnames</b>                         | PA_L1_APELLIDO              |
| <b>7.10</b> | <b>Full name</b>                        | PA_L1_NOMBRE_COMPLETO       |
| <b>7.11</b> | <b>E-mail</b>                           | PA_L1_EMAIL                 |
| <b>7.12</b> | <b>Phone</b>                            | PA_L1_TELEFONO              |
| <b>7.13</b> | <b>Address</b>                          | PA_L1_DIRECCION             |
| <b>7.14</b> | <b>Address detail</b>                   | PA_L1_DIRECCION_COMPLEMENTO |
| <b>7.15</b> | <b>Bank account</b>                     | PA_L1_CUENTA_BANCARIA       |
| <b>7.16</b> | <b>Payment card</b>                     | PA_L1_TARJETA               |
| <b>7.17</b> | <b>IP address</b>                       | PA_L1_IP                    |
| <b>7.18</b> | <b>Device identifier</b>                | PA_L1_DISPOSITIVO           |
| <b>7.19</b> | <b>License plate</b>                    | PA_L1_PLACA                 |
| <b>7.20</b> | <b>Vehicle identification number</b>    | PA_L1_VIN                   |
| <b>7.21</b> | <b>Property number (finca)</b>          | PA_L1_FINCA                 |
| <b>7.22</b> | <b>Contract or customer number</b>      | PA_L1_NUMERO_CONTRATO       |
| <b>7.23</b> | <b>User name or social network</b>      | PA_L1_USUARIO_RRSS          |
| <b>7.24</b> | <b>Credential</b>                       | PA_L1_CREDENCIAL            |
| <b>7.25</b> | <b>Geolocation</b>                      | PA_L1_GEOLOCALIZACION       |

|      |           |                   |
|------|-----------|-------------------|
| 7.26 | Free text | PA_L1_TEXTO_LIBRE |
|------|-----------|-------------------|

## PART 8 · QUASI-IDENTIFIERS

|      |                                |                        |
|------|--------------------------------|------------------------|
| 8.1  | Birth date                     | PA_L2_FECHA_NACIMIENTO |
| 8.2  | Birth year                     | PA_L2_ANIO_NACIMIENTO  |
| 8.3  | Age                            | PA_L2_EDAD             |
| 8.4  | Life event dates               | PA_L2_FECHA_EVENTO     |
| 8.5  | Sex                            | PA_L2_SEXO             |
| 8.6  | District                       | PA_L2_DISTRITO         |
| 8.7  | Corregimiento                  | PA_L2_CORREGIMIENTO    |
| 8.8  | Corregimiento or district code | PA_L2_CODIGO_LUGAR     |
| 8.9  | Populated place                | PA_L2_LUGAR_POBLADO    |
| 8.10 | Barriada or neighborhood       | PA_L2_BARRIADA         |
| 8.11 | Postal code                    | PA_L2_CODIGO_POSTAL    |
| 8.12 | Nationality                    | PA_L2_NACIONALIDAD     |
| 8.13 | Marital status                 | PA_L2_ESTADO_CIVIL     |
| 8.14 | Occupation                     | PA_L2_OCUPACION        |
| 8.15 | Employer                       | PA_L2_EMPLEADOR        |
| 8.16 | Schooling                      | PA_L2_ESCOLARIDAD      |
| 8.17 | School                         | PA_L2_ESCUELA          |
| 8.18 | Children and dependents        | PA_L2_DEPENDIENTES     |

## PART 9 · SENSITIVE DATA

|      |                                |                                 |
|------|--------------------------------|---------------------------------|
| 9.1  | Ethnic origin                  | PA_L3_ETNIA                     |
| 9.2  | Indigenous language            | PA_L3 LENGUA_INDIGENA           |
| 9.3  | Diagnosis                      | PA_L3_SALUD_DIAGNOSTICO         |
| 9.4  | Medication                     | PA_L3_SALUD_MEDICAMENTO         |
| 9.5  | Clinical identifier            | PA_L3_SALUD_IDENTIFICADOR       |
| 9.6  | Clinical text                  | PA_L3_SALUD_TEXTO_CLINICO       |
| 9.7  | Test result                    | PA_L3_SALUD_RESULTADO_EXAMEN    |
| 9.8  | Blood group                    | PA_L3_SALUD_GRUPO_SANGUINEO     |
| 9.9  | Disability                     | PA_L3_SALUD_DISCAPACIDAD        |
| 9.10 | Sexual and reproductive health | PA_L3_SALUD_SEXUAL_REPRODUCTIVA |
| 9.11 | Mental health and addiction    | PA_L3_SALUD_MENTAL_ADICCIONES   |
| 9.12 | Biometric data                 | PA_L3_BIOMETRICO                |
| 9.13 | Genetic data                   | PA_L3_INFORMACION_GENETICA      |
| 9.14 | Sexual orientation             | PA_L3_ORIENTACION_SEXUAL        |
| 9.15 | Gender identity                | PA_L3_IDENTIDAD_GENERO          |

|      |                                 |                                 |
|------|---------------------------------|---------------------------------|
| 9.16 | Sexual life                     | PA_L3_VIDA_SEXUAL               |
| 9.17 | Religion                        | PA_L3_RELIGION                  |
| 9.18 | Philosophical and moral beliefs | PA_L3_CREENCIA_FILOSOFICA_MORAL |
| 9.19 | Political opinion               | PA_L3_OPINION_POLITICA          |
| 9.20 | Political affiliation           | PA_L3_AFILIACION_POLITICA       |
| 9.21 | Union membership                | PA_L3_AFILIACION_SINDICAL       |

#### PART 10 · ECONOMIC AND CREDIT DATA

|      |                                |                             |
|------|--------------------------------|-----------------------------|
| 10.1 | Credit reference               | PA_FIN_REFERENCIA_CREDITO   |
| 10.2 | Debt amount                    | PA_FIN_MONTO_DEUDA          |
| 10.3 | Salary and income              | PA_FIN_INGRESO_MENSUAL      |
| 10.4 | Assets                         | PA_FIN_PATRIMONIO           |
| 10.5 | Pension and individual account | PA_FIN_PENSION              |
| 10.6 | Social security coverage       | PA_FIN_SEGURO_SOCIAL        |
| 10.7 | Social program                 | PA_FIN_PROGRAMA_SOCIAL      |
| 10.8 | Socioeconomic level            | PA_FIN_NIVEL_SOCIOECONOMICO |
| 10.9 | Housing tenure                 | PA_FIN_VIVIENDA             |

#### PART 11 · CRIMINAL RECORDS

|      |                                                 |                       |
|------|-------------------------------------------------|-----------------------|
| 11.1 | Criminal record                                 | PA_PENAL_ANTECEDENTES |
| 11.2 | Case number                                     | PA_PENAL_CAUSA        |
| 12   | The RUC check digit, computed by the set        |                       |
| 13   | Where people live: districts and corregimientos |                       |
| 14   | Limits                                          |                       |
| 15   | Verification                                    |                       |
| 16   | Sources                                         |                       |
| A    | Algorithms                                      |                       |
| B    | Files                                           |                       |
| C    | Exclusion lists                                 |                       |
| D    | Profile set and classifiers                     |                       |

## About this profile set

This profile set finds the personal data in a Panamanian database and masks it, so that a non-production copy — for testing, development, training or analytics — no longer identifies anyone.

Masking only names and cédulas does not get there: a birth date, a sex and a small corregimiento are enough to point at one person. The set works on the same layers the risk is in:

- **Direct identifiers** — cédula, RUC and its DV, names, contact, accounts and cards, passport and immigration documents, professional licenses, property, devices, vehicles, free text.
- **Quasi-identifiers** — birth date, age, sex, district, corregimiento, barriada, postal code, occupation, employer, schooling.
- **Sensitive data** — ethnic origin and indigenous language, health, biometric and genetic data, religious, philosophical and moral beliefs, political opinions and affiliation, union membership, sexual orientation.
- **Economic and credit data** — APC credit references, debt, salary, assets, pensions, social security coverage, social programs, poverty, housing tenure.
- **Criminal records** — police records and case numbers.

Everything the set needs travels with it: loading it creates the domains, classifiers, algorithms and lookup files, and sending it to a Masking Engine sends all of them.

## Law 81 of 2019 at a glance

Law 81 of 26 March 2019, on Personal Data Protection, was published in Gaceta Oficial 28743-A of 29 March 2019 and has been in force since 29 March 2021, two years after its enactment (art. 47). Executive Decree 285 of 28 May 2021 regulates it. The supervisory authority is the **National Authority for Transparency and Access to Information (ANTAI)**, through its Personal Data Protection Directorate.

What matters for a copy of production data:

- **Personal data** is any information concerning natural persons that identifies them or makes them identifiable (art. 4, 9). The law protects natural persons: a legal entity's data is outside its scope.
- **Sensitive data** is data that refers to the data subject's intimate sphere, or whose misuse may give rise to discrimination or carry a serious risk. As examples, not as a closed list, the law names data that may reveal racial or ethnic origin; religious, philosophical and moral beliefs or convictions; union membership; political opinions; data concerning health, life, sexual preference or orientation, genetic data or biometric data (art. 4, 11). Decree 285 defines biometric, genetic and health data (art. 4, 2 to 4).
- **Anonymized data is out of scope.** Processing of information obtained through a prior dissociation or anonymization procedure, such that the result cannot be associated with the data subject, is excluded from the law (art. 3, 5). *Dissociated data* cannot be associated with the data subject nor, by its structure, content or level of detail, allow their identification (art. 4, 10).
- **Purpose.** Data is used for the purposes it was authorized for; any other use needs consent, a special law or another legal basis (arts. 2, 2 and 11). Further processing for research or studies needs no consent when the data is anonymized (art. 12).
- **Health and sensitive data.** Consent for health data must be prior, irrefutable and express (art. 8; Decree 285, art. 18). Sensitive data may not be transferred save for exceptions, and for historical, statistical or scientific purposes it must be dissociated (art. 13).

- **Economic, financial, banking or commercial data** is processed with prior consent (art. 8, 3). Credit references also have a law of their own, Law 24 of 2002.
- **Criminal convictions** have their own disclosure rules (art. 30; Decree 285, art. 7).
- **Security.** Proportionate technical and organizational measures, especially for sensitive data (art. 2, 5; Decree 285, art. 36). A security breach is notified to ANTAI and to the affected data subjects within seventy-two hours (Decree 285, art. 37).
- **Sanctions.** Fines from one thousand to ten thousand balboas set by ANTAI (art. 36). Disregarding the rules on sensitive data is a very serious infringement, which can lead to closing the database's records or suspending the processing activity (arts. 41 and 43).

A test environment with production data is processing for a purpose the data subject did not authorize, and it holds sensitive and financial data outside the controls built for them. A copy that is effectively anonymized falls, under article 3, outside the law: that is what the set is for.

The list of sensitive data is not exhaustive. The set also treats gender identity and sexual life as sensitive, since they expose people to discrimination in the same way. Reform proposals are under discussion, among them higher fines; this document follows the text in force in September 2026.

This document describes technical controls. Whether a copy counts as anonymized is a legal assessment of the whole environment — access, retention, what else it is combined with — not of the profile set alone.

---

### 3

## Using the set

- **Load** — *Settings* → *Profile Sets* → *Load*. Loading again resets every object of the set to how it ships.
- **Send to Delphix** — from the profile set, or with *Send everything to Delphix*. Domains, algorithms and files go first.
- **Profile** — run a profiling job with the set and review the assigned domains before masking.
- **Mask** — run the masking job. Use the same engine key in every environment so masked values stay consistent across systems.

The assignment threshold is 60%. An unambiguous column name reaches it alone; an ambiguous one — `NOMBRE` , `RESULTADO` , `CIUDAD` — needs the values to agree. A column with a meaningless name ( `CAMP01` , `OBS` ) is still found when its values have a recognizable shape or vocabulary: a cédula, a RUC with its DV, a plate, a corregimiento.

Column names are recognized in Spanish, in the abbreviations of Panamanian systems ( `NUM_CEDULA` , `CIP` , `FEC_NAC` , `COD_CORREG` , `NIS` ), in the fields of the electronic invoice ( `dRuc` , `dDV` ) and in English.

Before masking, inventory the binary columns (BLOBs, images, PDFs). Onboarding processes often store a photo of the cédula: the number is still readable in the image. Report those columns as residual risk.

---

### 4

## How sensitive data is discovered

A profiling job looks at every column and asks each classifier of the profile set how confident it is that the column belongs to its domain. Classifiers come in four kinds:

- **Column name** — regular expressions on the column name. Each rule has a strength: the confidence it gives when it matches.
- **Data type** — accepts the column type and length, or rules the domain out. It never assigns a domain on its own.
- **Value format** — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per domain, and the column is assigned to the domain with the highest confidence when it reaches the threshold of the profile set — **60%** here. Values that do not match a format or a list count against their domain by the classifier's reject strength.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters:

`rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

Profiling proposes; it does not decide. Review the assigned domains before running a masking job, and look for columns the set left unassigned.

---

## 5

# How data is masked

Each domain names one algorithm, applied by the masking job to every column assigned to the domain. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each domain shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for the engine key: the same value masks the same way in every table, database and job, so joins between masked systems keep working.

The examples in this document were produced by running the algorithms with this tool's key. On an engine with another key the masked values differ; their shape does not.

# Domains at a glance

Every domain of the set, how it is found and the algorithm that masks it.

| DOMAIN                                                     | FOUND BY    |              |              | ALGORITHM          |
|------------------------------------------------------------|-------------|--------------|--------------|--------------------|
| DIRECT IDENTIFIERS                                         |             |              |              |                    |
| <b>Cédula (national ID)</b><br>PA_L1_CEDULA                | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PA_CEDULA          |
| <b>RUC (taxpayer number)</b><br>PA_L1_RUC                  | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PA_RUC             |
| <b>DV in its own column</b><br>PA_L1_RUC_DV                | COLUMN NAME | DATA TYPE    |              | PA_DV              |
| <b>Passport</b><br>PA_L1_PASAPORTE                         | COLUMN NAME | VALUE FORMAT |              | PA_CM_ALFANUM      |
| <b>Immigration document</b><br>PA_L1_DOCUMENTO_MIGRATORIO  | COLUMN NAME |              |              | PA_CM_ALFANUM      |
| <b>Social security number</b><br>PA_L1_SEGURO_SOCIAL       | COLUMN NAME |              |              | PA_CM_ALFANUM      |
| <b>Professional license (idoneidad)</b><br>PA_L1_IDONEIDAD | COLUMN NAME |              |              | PA_CM_ALFANUM      |
| <b>Given names</b><br>PA_L1_NOMBRE                         | COLUMN NAME | KNOWN VALUES |              | PA_NOMBRES         |
| <b>Surnames</b><br>PA_L1_APELLIDO                          | COLUMN NAME | KNOWN VALUES |              | PA_APELLIDOS       |
| <b>Full name</b><br>PA_L1_NOMBRE_COMPLETO                  | COLUMN NAME | KNOWN VALUES |              | PA_NOMBRE_COMPLETO |
| <b>E-mail</b><br>PA_L1_EMAIL                               | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PA_EMAIL           |
| <b>Phone</b><br>PA_L1_TELEFONO                             | COLUMN NAME | VALUE FORMAT |              | PA_TELEFONO        |
| <b>Address</b><br>PA_L1_DIRECCION                          | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PA_DIRECCION       |
| <b>Address detail</b><br>PA_L1_DIRECCION_COMPLEMENTO       | COLUMN NAME |              |              | PA_CM_ALFANUM      |
| <b>Bank account</b><br>PA_L1_CUENTA_BANCARIA               | COLUMN NAME |              |              | PA_CM_ALFANUM      |
| <b>Payment card</b><br>PA_L1_TARJETA                       | COLUMN NAME | VALUE FORMAT |              | PA_TARJETA         |
| <b>IP address</b><br>PA_L1_IP                              | COLUMN NAME | VALUE FORMAT |              | PA_IP              |
| <b>Device identifier</b><br>PA_L1_DISPOSITIVO              | COLUMN NAME | VALUE FORMAT |              | PA_CM_HEX          |
| <b>License plate</b><br>PA_L1_PLACA                        | COLUMN NAME | VALUE FORMAT |              | PA_PLACA           |

| DOMAIN                                                      | FOUND BY    |              |              | ALGORITHM           |
|-------------------------------------------------------------|-------------|--------------|--------------|---------------------|
| <b>Vehicle identification number</b><br>PA_L1_VIN           | COLUMN NAME | VALUE FORMAT |              | PA_CM_ALFANUM       |
| <b>Property number (finca)</b><br>PA_L1_FINCA               | COLUMN NAME |              |              | PA_CM_ALFANUM       |
| <b>Contract or customer number</b><br>PA_L1_NUMERO_CONTRATO | COLUMN NAME |              |              | PA_CM_ALFANUM       |
| <b>User name or social network</b><br>PA_L1_USUARIO_RRSS    | COLUMN NAME |              |              | PA_CM_ALFANUM       |
| <b>Credential</b><br>PA_L1_CREDENCIAL                       | COLUMN NAME |              |              | PA_REDACTAR         |
| <b>Geolocation</b><br>PA_L1_GEOLOCALIZACION                 | COLUMN NAME | VALUE FORMAT |              | PA_COORDENADA       |
| <b>Free text</b><br>PA_L1_TEXTO_LIBRE                       | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PA_TEXTO_LIBRE      |
| QUASI-IDENTIFIERS                                           |             |              |              |                     |
| <b>Birth date</b><br>PA_L2_FECHA_NACIMIENTO                 | COLUMN NAME | DATA TYPE    |              | PA_FECHA_NACIMIENTO |
| <b>Birth year</b><br>PA_L2_ANIO_NACIMIENTO                  | COLUMN NAME | DATA TYPE    |              | PA_ANIO             |
| <b>Age</b><br>PA_L2_EDAD                                    | COLUMN NAME | DATA TYPE    |              | PA_EDAD             |
| <b>Life event dates</b><br>PA_L2_FECHA_EVENTO               | COLUMN NAME | DATA TYPE    |              | PA_FECHA_EVENTO     |
| <b>Sex</b><br>PA_L2_SEX0                                    | COLUMN NAME | KNOWN VALUES |              | PA_SEX0             |
| <b>District</b><br>PA_L2_DISTRITO                           | COLUMN NAME | KNOWN VALUES |              | PA_DISTRITO         |
| <b>Corregimiento</b><br>PA_L2_CORREGIMIENTO                 | COLUMN NAME | KNOWN VALUES |              | PA_CORREGIMIENTO    |
| <b>Corregimiento or district code</b><br>PA_L2_CODIGO_LUGAR | COLUMN NAME | DATA TYPE    | KNOWN VALUES | PA_CODIGO_LUGAR     |
| <b>Populated place</b><br>PA_L2_LUGAR_POBLADO               | COLUMN NAME |              |              | PA_LUGAR_POBLADO    |
| <b>Barriada or neighborhood</b><br>PA_L2_BARRIADA           | COLUMN NAME |              |              | PA_BARRIADA         |
| <b>Postal code</b><br>PA_L2_CODIGO_POSTAL                   | COLUMN NAME | VALUE FORMAT |              | PA_CODIGO_POSTAL    |
| <b>Nationality</b><br>PA_L2_NACIONALIDAD                    | COLUMN NAME | KNOWN VALUES |              | PA_NACIONALIDAD     |
| <b>Marital status</b><br>PA_L2_ESTADO_CIVIL                 | COLUMN NAME | KNOWN VALUES |              | PA_ESTADO_CIVIL     |
| <b>Occupation</b><br>PA_L2_OCUPACION                        | COLUMN NAME |              |              | PA_OCUPACION        |
| <b>Employer</b><br>PA_L2_EMPLEADOR                          | COLUMN NAME |              |              | PA_EMPLEADOR        |

| DOMAIN                                                                    | FOUND BY    |              | ALGORITHM              |
|---------------------------------------------------------------------------|-------------|--------------|------------------------|
| <b>Schooling</b><br>PA_L2_ESCOLARIDAD                                     | COLUMN NAME | KNOWN VALUES | PA_ESCOLARIDAD         |
| <b>School</b><br>PA_L2_ESCUELA                                            | COLUMN NAME |              | PA_ESCUELA             |
| <b>Children and dependents</b><br>PA_L2_DEPENDIENTES                      | COLUMN NAME | DATA TYPE    | PA_CANTIDAD_ACOTADA    |
| SENSITIVE DATA                                                            |             |              |                        |
| <b>Ethnic origin</b><br>PA_L3_ETNIA                                       | COLUMN NAME | KNOWN VALUES | PA_PUEBLO              |
| <b>Indigenous language</b><br>PA_L3 LENGUA_INDIGENA                       | COLUMN NAME | KNOWN VALUES | PA LENGUA_INDIGENA     |
| <b>Diagnosis</b><br>PA_L3_SALUD_DIAGNOSTICO                               | COLUMN NAME | VALUE FORMAT | PA_DIAGNOSTICO         |
| <b>Medication</b><br>PA_L3_SALUD_MEDICAMENTO                              | COLUMN NAME | KNOWN VALUES | PA_MEDICAMENTO         |
| <b>Clinical identifier</b><br>PA_L3_SALUD_IDENTIFICADOR                   | COLUMN NAME |              | PA_CM_ALFANUM          |
| <b>Clinical text</b><br>PA_L3_SALUD_TEXTO_CLINICO                         | COLUMN NAME | DATA TYPE    | PA_TEXTO_LIBRE         |
| <b>Test result</b><br>PA_L3_SALUD_RESULTADO_EXAMEN                        | COLUMN NAME |              | PA_CM_ALFANUM          |
| <b>Blood group</b><br>PA_L3_SALUD_GRUPO_SANGUINEO                         | COLUMN NAME | KNOWN VALUES | PA_GRUPO_SANGUINEO     |
| <b>Disability</b><br>PA_L3_SALUD_DISCAPACIDAD                             | COLUMN NAME | KNOWN VALUES | PA_DISCAPACIDAD        |
| <b>Sexual and reproductive health</b><br>PA_L3_SALUD_SEXUAL_REPRODUCTIVA  | COLUMN NAME |              | PA_CATEGORIA_SUPRIMIDA |
| <b>Mental health and addiction</b><br>PA_L3_SALUD_MENTAL_ADICCIONES       | COLUMN NAME |              | PA_CATEGORIA_SUPRIMIDA |
| <b>Biometric data</b><br>PA_L3_BIOMETRICO                                 | COLUMN NAME | DATA TYPE    | PA_REDACTAR            |
| <b>Genetic data</b><br>PA_L3_INFORMACION_GENETICA                         | COLUMN NAME |              | PA_REDACTAR            |
| <b>Sexual orientation</b><br>PA_L3_ORIENTACION_SEXUAL                     | COLUMN NAME | KNOWN VALUES | PA_ORIENTACION_SEXUAL  |
| <b>Gender identity</b><br>PA_L3_IDENTIDAD_GENERO                          | COLUMN NAME | KNOWN VALUES | PA_IDENTIDAD_GENERO    |
| <b>Sexual life</b><br>PA_L3_VIDA_SEXUAL                                   | COLUMN NAME |              | PA_CATEGORIA_SUPRIMIDA |
| <b>Religion</b><br>PA_L3_RELIGION                                         | COLUMN NAME | KNOWN VALUES | PA_RELIGION            |
| <b>Philosophical and moral beliefs</b><br>PA_L3_CREENCIA_FILOSOFICA_MORAL | COLUMN NAME |              | PA_CATEGORIA_SUPRIMIDA |
| <b>Political opinion</b><br>PA_L3_OPINION_POLITICA                        | COLUMN NAME | KNOWN VALUES | PA_OPINION_POLITICA    |

| DOMAIN                                                    | FOUND BY    |              | ALGORITHM               |
|-----------------------------------------------------------|-------------|--------------|-------------------------|
| <b>Political affiliation</b><br>PA_L3_AFILIACION_POLITICA | COLUMN NAME | KNOWN VALUES | PA_PARTIDO_POLITICO     |
| <b>Union membership</b><br>PA_L3_AFILIACION_SINDICAL      | COLUMN NAME | KNOWN VALUES | PA_CATEGORIA_SUPRIMIDA  |
| ECONOMIC AND CREDIT DATA                                  |             |              |                         |
| <b>Credit reference</b><br>PA_FIN_REFERENCIA_CREDITO      | COLUMN NAME |              | PA_APC                  |
| <b>Debt amount</b><br>PA_FIN_MONTO_DEUDA                  | COLUMN NAME | DATA TYPE    | PA_MONTO_DEUDA          |
| <b>Salary and income</b><br>PA_FIN_INGRESO_MENSUAL        | COLUMN NAME | DATA TYPE    | PA_INGRESO_MENSUAL      |
| <b>Assets</b><br>PA_FIN_PATRIMONIO                        | COLUMN NAME | DATA TYPE    | PA_PATRIMONIO           |
| <b>Pension and individual account</b><br>PA_FIN_PENSION   | COLUMN NAME | DATA TYPE    | PA_MONTO_PENSION        |
| <b>Social security coverage</b><br>PA_FIN_SEGURO_SOCIAL   | COLUMN NAME | KNOWN VALUES | PA_SEGURO_SOCIAL        |
| <b>Social program</b><br>PA_FIN_PROGRAMA_SOCIAL           | COLUMN NAME | KNOWN VALUES | PA_PROGRAMA_SOCIAL      |
| <b>Socioeconomic level</b><br>PA_FIN_NIVEL_SOCIOECONOMICO | COLUMN NAME | KNOWN VALUES | PA_NIVEL_SOCIOECONOMICO |
| <b>Housing tenure</b><br>PA_FIN_VIVIENDA                  | COLUMN NAME | KNOWN VALUES | PA_VIVIENDA             |
| CRIMINAL RECORDS                                          |             |              |                         |
| <b>Criminal record</b><br>PA_PENAL_ANTECEDENTES           | COLUMN NAME |              | PA_ANTECEDENTES         |
| <b>Case number</b><br>PA_PENAL_CAUSA                      | COLUMN NAME | VALUE FORMAT | PA_CM_ALFANUM           |

# Direct identifiers

Data that identifies a person on its own. The cédula is the key of almost every Panamanian record: it is a natural person's RUC, the number of their driver's license and their identification at the Social Security Fund. It is masked the same way everywhere, and the RUC keeps a valid check digit, because applications validate it.

## 7.1 Cédula (national ID)

PA\_L1\_CEDULA

COLUMN NAME DATA TYPE VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The number of the identity card issued by the Electoral Tribunal: province of birth (1 to 13; 10, 11 and 12 are the Guna Yala, Emberá-Wounaan and Ngäbe-Buglé comarcas), tomo and asiento of the civil registry entry. 8–442–445 . Letters mark other cases: E resident foreigner, N naturalized, PE Panamanian born abroad, AV registered before the civil registry law, PI indigenous Panamanian; 8–NT–12–3456 is the tax number of someone without a cédula. In Panama the driver's license carries the cédula number, so license columns fall here too.

HOW IT IS FOUND

| COLUMN NAME | PA_L1_CEDULA – Path                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90%         | <code>cedula[a-z0-9]* c[eé]dula ced cip n(o um ro umero)?(de_)?cedula cedula_?</code><br><code>(identidad personal cliente empleado paciente asegurado beneficiario titular conyuge acudiente</code><br><code>) (cliente empleado paciente asegurado beneficiario titular)?cedula documento?(de_)?</code><br><code>identidad doc?identidad id?personal identificacion?personal n(o um ro)?</code><br><code>identificacion numero?documento num?doc nro?doc national?id dni</code> |
| 80%         | <code>licencia?(de_)?(conducir manejo chofer conductor) n(o um ro)?licencia(_?conducir)?</code><br><code>licencia driver?license drivers?license</code>                                                                                                                                                                                                                                                                                                                           |

DATA TYPE PA\_L1\_CEDULA – Type  
Accepts text (length 5 or more); any other type rules the domain out.

|                                                                                        |                      |          |       |
|----------------------------------------------------------------------------------------|----------------------|----------|-------|
| VALUE FORMAT                                                                           | PA_L1_CEDULA - Regex |          |       |
| PATTERN                                                                                |                      | STRENGTH | NOTES |
| (?!\\d{1,2}-\\d{1,2}-(19 20)\\d{2}\$)(([1-9] 1[0-3])(AV PI)? E N PE)-\\d{1,4}-\\d{1,6} |                      | 100%     |       |
| ([1-9] 1[0-3])-NT-\\d{1,4}-\\d{1,6}                                                    |                      | 90%      |       |

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

The province becomes another province between 1 and 13, and tomo and asiento are mapped digit by digit keeping their length; the letters stay, and so does the document type. The mapping is one-to-one: a cédula used as a key does not collide. The same algorithm masks the cédula inside the RUC, so a person's cédula and RUC still agree. Text that is not shaped like a cédula is masked character by character.

PA\_CEDULA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

|                                              |                   |
|----------------------------------------------|-------------------|
| PA_PROVINCIA_CEDULA                          | NUMERIC MAPPING   |
| Maps the number to another between 1 and 13. |                   |
| PA_CM_ALFANUM                                | CHARACTER MAPPING |

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE       |   | MASKED      |
|-------------|---|-------------|
| 8-442-445   | → | 1-433-300   |
| E-8-123456  | → | E-1-904292  |
| 4PI-234-123 | → | 8PI-158-957 |

## 7.2 RUC (taxpayer number)

PA\_L1\_RUC

COLUMN NAME DATA TYPE VALUE FORMAT REGEX DECOMPOSE

#### WHAT IT IS

The DGI's Single Taxpayer Registry number. A natural person's RUC is their cédula, with a two-digit check digit next to it: 8-442-445 DV 08 . A legal entity's RUC is its Public Registry number: 155596713-2-2015 DV 59 . One column often mixes both.

#### HOW IT IS FOUND

COLUMN NAME PA\_L1\_RUC - Path

90% `ruc[a-z0-9]* d_?ruc n(o|um|ro|umero)_?(de_)?ruc registro_?unico(_?de)?(_? contribuyentes?)? ruc_? (cliente|proveedor|emisor|receptor|contribuyente|persona|natural|empleado|beneficiario|titular ) (cliente|proveedor|emisor|receptor)_?ruc tax_?id id_?fiscal nit`

DATA TYPE PA\_L1\_RUC - Type

Accepts text (length 5 or more); any other type rules the domain out.

VALUE FORMAT PA\_L1\_RUC - Regex

| PATTERN                                                                                                       | STRENGTH | NOTES |
|---------------------------------------------------------------------------------------------------------------|----------|-------|
| <code>(?:([1-9] 1[0-3])(?:AV PI)?-\d{1,4}-\d{1,6} (?:E N PE)-\d{1,4}-\d{1,6})(?: +DV +  +DV -  +)\d{2}</code> | 100%     |       |
| <code>(?!\\d{1,2}-\\d{1,2}-(19 20)\\d{2}\$)(([1-9] 1[0-3])(AV PI)? E N PE)-\\d{1,4}-\\d{1,6}</code>           | 60%      |       |
| <code>\\d{3,10}-\\d{1,4}-\\d{1,7}(\\s*DV\\s*\\d{2})?</code>                                                   | 60%      |       |

A value that matches none counts 20% against the domain.

#### HOW IT IS MASKED

A natural person's RUC is masked as their cédula and the DV is recomputed, so every masked RUC passes DGI validation. The separators DV , DV , - and a space are recognized. A legal entity's RUC stays as it is: it is not personal data under article 4, and keeping it keeps joins with supplier catalogs working. See *The RUC check digit, computed by the set*.

PA\_RUC REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

PA\_RUC\_DV STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_1 → PA\_RUC\_DV\_2 → PA\_RUC\_DV\_3 → PA\_RUC\_DV\_4 → PA\_RUC\_DV\_5 , in this order. It is made of 46 algorithms that repeat the same step position by position; each one is in Appendix A.

PA\_CEDULA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

EXAMPLES

| VALUE                  |   | MASKED                 |
|------------------------|---|------------------------|
| 8-442-445 DV 08        | → | 1-433-300 DV 12        |
| 1AV-432-658 DV 96      | → | 9AV-920-061 DV 24      |
| 155596713-2-2015 DV 59 | → | 155596713-2-2015 DV 59 |

7.3 DV in its own column

PA\_L1\_RUC\_DV

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

The RUC check digit kept apart, as in the electronic invoice ( dDV ) or a DV column.

HOW IT IS FOUND

COLUMN NAME PA\_L1\_RUC\_DV - Path

90% dv d\_?dv digito\_?verificador dig\_?verif dv\_?ruc ruc\_?dv check\_?digit

DATA TYPE PA\_L1\_RUC\_DV - Type

Accepts text (length up to 3) or number (length up to 2); any other type rules the domain out.

HOW IT IS MASKED

The two digits are mapped. They cannot be recomputed: an algorithm sees one column, and the RUC is in another. Masking it keeps it from helping link the row to the real RUC.

PA\_DV REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 08    | → | 57     |

7.4 Passport

PA\_L1\_PASAPORTE

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

Panamanian and foreign passport numbers.

HOW IT IS FOUND

COLUMN NAME PA\_L1\_PASAPORTE - Path

85% pasaporte[a-z\_]\* passport[a-z\_]\* n(o|um|ro|umero)?pasaporte pasap

|                   |                         |       |
|-------------------|-------------------------|-------|
| VALUE FORMAT      | PA_L1_PASAPORTE – Regex |       |
| PATTERN           | STRENGTH                | NOTES |
| [A-Z]{1,2}\d{6,8} |                         | 30%   |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE     |   | MASKED    |
|-----------|---|-----------|
| PA0123456 | → | PU0730277 |

## 7.5 Immigration document

PA\_L1\_DOCUMENTO\_MIGRATORIO

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

Resident cards, residence and work permits, visas and foreign identification numbers. A permanent resident's E cédula is in the cédula domain.

#### HOW IT IS FOUND

##### COLUMN NAME PA\_L1\_DOCUMENTO\_MIGRATORIO - Path

80% carne\_?(migratorio|de\_?migracion|residente|de\_?residente|residencia|refugiado|trabajo)  
carnet\_?(migratorio|residente|residencia) permiso\_?(de\_?)?(residencia|trabajo|migratorio)  
n(o|u|m|ro)\_?carne(t)? documento\_?migratorio doc\_?migratorio filiacion\_?migratoria visa  
n(o|u|m|ro)\_?visa id\_?extranjero documento\_?extranjero identificacion\_?extranjera

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE     |   | MASKED    |
|-----------|---|-----------|
| 123456789 | → | 110858568 |

## 7.6 Social security number

PA\_L1\_SEGURO\_SOCIAL

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The insured person's or card number of the Social Security Fund (CSS) kept by older systems. Today the CSS identifies the insured by their cédula.

#### HOW IT IS FOUND

##### COLUMN NAME PA\_L1\_SEGURO\_SOCIAL - Path

85% n(o|u|m|ro|umero)\_?(de\_?)?seguro\_?social seguro\_?social\_?(num|no|nro|numero) nss  
n(o|u|m|ro)\_?css css\_?(num|no|numero|asegurado) n(o|u|m|ro)\_?asegurado num\_?asegurado  
n(o|u|m|ro)\_?(de\_?)?afiliado(\_?css)? carne\_?(del\_?)?seguro(\_?social)? ss\_?number social\_?  
security

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE    |   | MASKED   |
|----------|---|----------|
| 123-4567 | → | 564-9285 |

## 7.7 Professional license (idoneidad)

PA\_L1\_IDONEIDAD

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The idoneidad number that allows someone to practice a regulated profession — medicine, nursing, law, engineering. The registries are public: the number leads straight to the person's name.

#### HOW IT IS FOUND

##### COLUMN NAME PA\_L1\_IDONEIDAD - Path

85% idoneidad n(o|um|ro|umero)?(de\_)?idoneidad registro?(medico|profesional|de\_?idoneidad) licencia\_?profesional codigo?(medico|profesional) colegiatura n(o|um|ro)?registro\_? profesional

#### HOW IT IS MASKED

Character by character, keeping the length.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 12345 | → | 31564  |

## 7.8 Given names

PA\_L1\_NOMBRE

#### COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

#### WHAT IT IS

First and middle names ( NOMBRES , PRIMER\_NOMBRE ), including compound names such as Luis Carlos .

#### HOW IT IS FOUND

##### COLUMN NAME PA\_L1\_NOMBRE - Path

85% primer\_?nombre segundo\_?nombre nombres?\_(de\_)?pila nombre\_?usual first\_?name given\_? names? fname name\_?first middle\_?name nombre[12] nombres (except when followed by list E1) nom\_? (pila|trab|emp)

50% nombre (except when followed by list E2) nomb? (except when followed by list E3) name (except when followed by list E4)

**KNOWN VALUES** PA\_L1\_NOMBRE – List

| LIST                    | VALUES | STRENGTH | FIRST VALUES                           |
|-------------------------|--------|----------|----------------------------------------|
| pa-detectar-nombres.txt | 291    | 70%      | María, Maria, José, Jose, Luis, Carlos |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Each word is replaced by a name from a list of 253 names used in Panama, keeping the number of words and the case. More than three words are masked character by character.

**PA\_NOMBRES** **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

**PA\_NOMBRE\_PILA** **NAME**

Replaces the name with one of the 253 names in pa-nombres.txt .

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE       |   | MASKED           |
|-------------|---|------------------|
| Luis Carlos | → | Teodoro Norberto |
| YAMILETH    | → | ZULEIKA          |

## 7.9 Surnames

PA\_L1\_APELLIDO

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

#### WHAT IT IS

Paternal and maternal surnames, in one column or two, and a married woman's surname: González de Pérez .

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_APELLIDO – Path

85%  
apellidos? apellido?(paterno|materno|pat|mat|uno|dos|casada|de?casada|[12]) ape?  
(pat|mat|paterno|materno|casada|[12]) apepat apemat ap?(pat|mat|paterno|materno) a?  
paterno a?materno paterno materno primer?apellido segundo?apellido apellido?  
usual last?name surname lname family?name second?last?name married?name

**KNOWN VALUES** PA\_L1\_APELLIDO – List

| LIST                      | VALUES | STRENGTH | FIRST VALUES                                               |
|---------------------------|--------|----------|------------------------------------------------------------|
| pa-detectar-apellidos.txt | 258    | 70%      | Rodríguez, Rodriguez, González, Gonzalez, Sánchez, Sanchez |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Each surname is replaced by one from a list of 200 surnames common in Panama — led by Rodríguez, González, Sánchez, Martínez, Castillo and Ábrego —, keeping the number of words and the case. The de of a married surname stays.

**PA\_APELLIDOS** **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

#### PA\_APELLIDO NAME

Replaces the name with one of the 200 names in `pa-apellidos.txt`.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

##### VALUE

González de Pérez

##### MASKED

→ **Tapia de Aguilar**

## 7.10 Full name

PA\_L1\_NOMBRE\_COMPLETO

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

#### WHAT IT IS

Given names and surnames in one column, in Panamanian order, with `de` and the husband's surname where present, or as `APELLIDOS`, `NOMBRES`. Also the name of an account holder, beneficiary, insured person or guardian (*acudiente*).

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_L1\_NOMBRE\_COMPLETO - Path

90%

`nombre?(completo|compl|full)` `nombres?_?y?apellidos?` `nombres?_?apellidos?`  
`nombreapellido` `nom?completo` `nomcompleto` `nombrecompleto` `full?name` `fullname` `person_?`  
`name` `customer?name` `employee?name` `patient?name` `nombre_?`  
`(cliente|cte|cli|paciente|pac|trabajador|trab|empleado|colaborador|funcionario|servidor_?`  
`publico|alumno|estudiante|afiliado|asegurado|cotizante|jubilado|pensionado|beneficiario|benef|`  
`titular|madre|padre|acudiente|tutor|conyuge|pareja|medico|doctor|profesional|responsable|conta`  
`cto|emergencia|fiador|codeudor|deudor|remitente|destinatario|receptor|firmante|testigo|represe`  
`ntante|apoderado|propietario|arrendatario|inquilino|contratante|victima|denunciante|imputado|s`  
`ocio|donante|heredero|persona|pers|conductor|chofer|vendedor|ejecutivo|agente|solicitante|ciud`  
`adano|elector|votante|contribuyente|jefe?(de_?)?hogar|hijo|hija|familiar|dependiente)`

60%

`titular` `beneficiario` `acudiente` `contacto?emergencia` `conyuge` `tutor?legal`  
`representante?legal` `apoderado` `paciente` `victima` `asegurado` `contratante` `fiador`  
`codeudor` `jubilado` `pensionado`

50%

`nombre` (except when followed by *list E2*) `name` (except when followed by *list E4*)

##### KNOWN VALUES

PA\_L1\_NOMBRE\_COMPLETO - List

##### LIST

##### VALUES

##### STRENGTH

##### FIRST VALUES

`pa-detectar-nombres.txt`

291

60%

María, Maria, José, Jose, Luis, Carlos

`pa-detectar-apellidos.txt`

258

60%

Rodríguez, Rodriguez, González, Gonzalez, Sánchez, Sanchez

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Given names become given names and surnames become surnames, by position: one word is a given name, two are a given name and a surname, three are a given name and two surnames, four are two given names

and two surnames. `María Elena González de Pérez` keeps its `de` and its structure. A comma separates surnames from given names.

`PA_NOMBRE_COMPLETO` **REGEX DECOMPOSE**

Tries 8 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_APELLIDOS` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_APELLIDO` **NAME**

Replaces the name with one of the 200 names in `pa-apellidos.txt` .

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_NOMBRES` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_NOMBRE_PILA` **NAME**

Replaces the name with one of the 253 names in `pa-nombres.txt` .

**EXAMPLES**

| VALUE                         |   | MASKED                           |
|-------------------------------|---|----------------------------------|
| María Elena González de Pérez | → | Édgar Evelyn Tapia de Aguilar    |
| SÁNCHEZ ÁBREGO, LUIS CARLOS   | → | VALDÉS BATISTA, TEODORO NORBERTO |

7.11 E-mail

`PA_L1_EMAIL`

`COLUMN NAME` `DATA TYPE` `VALUE FORMAT` `REGEX DECOMPOSE`

**WHAT IT IS**

Personal and work e-mail addresses.

**HOW IT IS FOUND**

`COLUMN NAME` `PA_L1_EMAIL` - Path

85% `e_?mail[a-z0-9_]*` `mail[a-z0-9_]*` `correo(_?electronico)?[a-z0-9_]*` `email_?address`  
`correo_?(personal|laboral|contacto|receptor|cliente)`

`DATA TYPE` `PA_L1_EMAIL` - Type

Accepts text (length 5 or more); any other type rules the domain out.

`VALUE FORMAT` `PA_L1_EMAIL` - Regex

| PATTERN                                                              | STRENGTH | NOTES |
|----------------------------------------------------------------------|----------|-------|
| <code>[A-Za-z0-9._%+'\-\-]+@[A-Za-z0-9.\-\-]+\.[A-Za-z]{2,24}</code> | 100%     |       |

A value that matches none counts 20% against the domain.

**HOW IT IS MASKED**

The name and domain are mapped character by character and the top-level domain becomes `.test` , reserved for testing; a masked address never receives mail.

`PA_EMAIL` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

## EXAMPLES

## VALUE

yamileth.gonzalez@gmail.com

## MASKED

→ tificfk.yedsiquv@pgoeb.test

## 7.12 Phone

PA\_L1\_TELEFONO

## COLUMN NAME

## VALUE FORMAT

## REGEX DECOMPOSE

## WHAT IT IS

Seven-digit landlines ( 223–4567 ) and eight-digit mobiles starting with 6 ( 6123–4567 ), with or without +507 . Panama has no area codes.

## HOW IT IS FOUND

## COLUMN NAME

PA\_L1\_TELEFONO – Path

85%

tel telefono[a-z0-9]\* tel[eé]fono tel\_?  
(cel|casa|residencia|oficina|trabajo|movil|contacto|fijo|emergencia) celular[a-z0-9]\* cel  
movil m[oó]vil whatsapp wa fax phone[a-z0-9]\* mobile cellphone n(o|um|ro)?  
tel(efono)? telef tlf extension

## VALUE FORMAT

PA\_L1\_TELEFONO – Regex

## PATTERN

## STRENGTH

## NOTES

`(\+?(?507\)?[\s-]?)?6\d{3}[\s-]?\d{4}`

90%

`(\+?(?507\)?[\s-]?)?[2-9]\d{2}[\s-]?\d{4}`

70%

`\+?\d[\d\s()\-]{6,15}`

30%

A value that matches none counts 30% against the domain.

## HOW IT IS MASKED

+507 , the separators and the first digit — the 6 of a mobile, the region of a landline — stay; the rest is masked digit by digit. A number not shaped like a Panamanian number is masked whole.

## PA\_TELEFONO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

## PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

## EXAMPLES

## VALUE

## MASKED

+507 6123–4567

→ +507 6957–9910

223–4567

→ 246–9910

## 7.13 Address

PA\_L1\_DIRECCION

## COLUMN NAME

## DATA TYPE

## VALUE FORMAT

## SECURE LOOKUP

## WHAT IT IS

The address line as written in Panama, which rarely has a street number: street or road, building, floor and apartment, or barriada, street and house.

## HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_DIRECCION - Path

**80%** direccion[a-z]\* (except when followed by list E5) direcci[oó]n dir domicilio[a-z]\* calle(\_?y\_? casa)? avenida address street addr dir\_? (residencial|residencia|particular|comercial|laboral|envio|entrega|facturacion|cliente|paciente|trabajo) residencia lugar\_?(de\_)?residencia home\_?address ubicacion\_? (domicilio|vivienda) senas punto\_?de\_?referencia

**DATA TYPE** PA\_L1\_DIRECCION - Type

Accepts text (length 6 or more); any other type rules the domain out.

**VALUE FORMAT** PA\_L1\_DIRECCION - Regex

| PATTERN                                                                                               | STRENGTH | NOTES |
|-------------------------------------------------------------------------------------------------------|----------|-------|
| (?i)(v[íí]a ave?\.? avenida calle c\. carretera paseo boulevard blvd\.? corredor camino)\s+.          | 70%      |       |
| (?i).*(edificio edif\.? torre ph)\s+.(piso apto\.? apartamento oficina local)\s*\w+.                  | 80%      |       |
| (?i).*(barriada urbanizaci[oó]n urb\. residencial sector manzana)\s+.(casa lote)\s*(n[oó]\.?\s*)?\w+. | 80%      |       |

A value that matches none counts 20% against the domain.

## HOW IT IS MASKED

It is replaced by one of 3,000 fictitious Panamanian addresses, with common streets, roads, barriadas and buildings in realistic proportions.

**PA\_DIRECCION** **SECURE LOOKUP**

Replaces the value with one of the 3,000 values in `pa-direcciones.txt`, chosen by a keyed hash.

## EXAMPLES

| VALUE                                                 | MASKED                      |
|-------------------------------------------------------|-----------------------------|
| Calle 50, Edificio Global Bank, Piso 21, Oficina 2104 | → Las Cumbres, Casa No. 242 |

## 7.14 Address detail

PA\_L1\_DIRECCION\_COMPLEMENTO

**COLUMN NAME** **CHARACTER MAPPING**

## WHAT IT IS

House, apartment, floor, building, tower or lot numbers in columns of their own.

## HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_DIRECCION\_COMPLEMENTO - Path

**70%** n(o|um|ro|umero)\_?casa casa\_?(num|no|numero) apto apartamento n(o|um|ro)\_?apto piso edificio torre ph lote manzana local oficina\_?num house\_?number apartment apt unit\_?number floor

## HOW IT IS MASKED

Character by character, keeping length and format.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

##### VALUE

Casa 23

→

##### MASKED

Miya 82

## 7.15 Bank account

PA\_L1\_CUENTA\_BANCARIA

#### COLUMN NAME

#### CHARACTER MAPPING

#### WHAT IT IS

Checking, savings and payroll account numbers, in each bank's formats.

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_L1\_CUENTA\_BANCARIA - Path

85%

cuenta\_?

(bancaria|banco|corriente|ahorro|ahorros|deposito|abono|pago|destino|origen|planilla|debito|ac  
h|plazo\_?fijo) cta\_?(bancaria|banco|corriente|ahorro|planilla|ach) n(o|um|ro|umero)?\_  
(de\_)?cuenta(\_?(banco|bancaria|corriente|ahorro|ach))? account\_?(number|no|num|nbr) bank\_  
account num\_?cta no\_?cta iban

#### HOW IT IS MASKED

Character by character, keeping length and format.

##### PA\_CM\_ALFANUM

#### CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

##### VALUE

04-72-01-123456-7

→

##### MASKED

25-71-81-450676-9

## 7.16 Payment card

PA\_L1\_TARJETA

#### COLUMN NAME

#### VALUE FORMAT

#### REGEX DECOMPOSE

#### WHAT IT IS

Credit, debit and prepaid card numbers, including the Tarjeta Clave Social through which MIDES pays its programs.

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_L1\_TARJETA - Path

85%

tarjeta(\_?(credito|debito|cred|deb|num|no|numero|bancaria|clave|clave\_?social|prepago))?  
clave\_?social n(o|um|ro|umero)?\_?tarjeta pan card\_?(number|no|num) credit\_?card cc\_  
(number|num) tdc tdd num\_?tdc plastico

##### VALUE FORMAT

PA\_L1\_TARJETA - Regex

##### PATTERN

##### STRENGTH

##### NOTES

\d{13,19}

90%

LUHN check digit; after removing [\s\~]

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by another card number that still passes the Luhn check. Values not shaped like a card pass unchanged instead of failing the row.

PA\_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

PA\_TARJETA\_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

EXAMPLES

| VALUE               | MASKED                |
|---------------------|-----------------------|
| 4916 3131 2345 6780 | → 1374 9602 1069 5000 |

7.17 IP address

PA\_L1\_IP

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses of users and devices.

HOW IT IS FOUND

COLUMN NAME PA\_L1\_IP - Path

85%

ip ip\_?

(address|addr|origen|destino|cliente|usuario|remota|publica|privada|acceso|conexion|login)

direccion\_?ip dir\_?ip ipv[46] remote\_?addr client\_?ip host\_?ip

VALUE FORMAT PA\_L1\_IP - Regex

PATTERN

STRENGTH

NOTES

((25[0-5]|2[0-4]\d|1?\d?\d)\.){3}(25[0-5]|2[0-4]\d|1?\d?\d)

100%

[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7}

90%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Each IPv4 octet is mapped to a number from 1 to 254; an IPv6 address is mapped as hexadecimal.

PA\_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_HEX .

PA\_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

PA\_CM\_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

| VALUE         | MASKED          |
|---------------|-----------------|
| 190.140.22.10 | → 164.230.40.16 |

## 7.18 Device identifier

PA\_L1\_DISPOSITIVO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

### WHAT IT IS

IMEI, MAC address, advertising and session identifiers, equipment serial numbers.

### HOW IT IS FOUND

COLUMN NAME PA\_L1\_DISPOSITIVO - Path

80% imei imsi iccid mac mac\_address direccion\_mac device\_id id\_dispositivo  
dispositivo\_id uuid\_dispositivo advertising\_id idfa gaid cookie(\_id)? session\_  
id id\_sesion serie\_equipo n(om|ro)\_serie\_(equipo|celular|telefono)

VALUE FORMAT PA\_L1\_DISPOSITIVO - Regex

| PATTERN                                             | STRENGTH | NOTES            |
|-----------------------------------------------------|----------|------------------|
| <code>([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2}</code> | 100%     |                  |
| <code>\d{15}</code>                                 | 80%      | LUHN check digit |

A value that matches none counts 30% against the domain.

### HOW IT IS MASKED

Hex stays hex and digits stay digits, so the value keeps its format.

PA\_CM\_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

### EXAMPLES

| VALUE             | MASKED              |
|-------------------|---------------------|
| 3c:22:fb:9a:10:e4 | → 5e:83:bb:3c:31:f9 |

## 7.19 License plate

PA\_L1\_PLACA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

### WHAT IT IS

Plates in the format used since 2013, two letters and four digits ( AB1234 ), the earlier six-digit ones, and category plates: M motorcycle, T taxi, B bus, G government, CD diplomatic corps, PR press and others.

### HOW IT IS FOUND

COLUMN NAME PA\_L1\_PLACA - Path

85% placas?(?(\vehiculo|auto|circulacion|vehicular|num|no|numero))? n(om|ro)\_placas?  
matricula(?(\vehiculo|auto|vehicular))? license\_plate plate(\_number)? registro\_  
vehicular revisado

VALUE FORMAT PA\_L1\_PLACA - Regex

| PATTERN                                                           | STRENGTH | NOTES |
|-------------------------------------------------------------------|----------|-------|
| <code>[A-Z]{2}[\s\~]?[0-9]{4}</code>                              | 80%      |       |
| <code>(ADM MB MI PR BC CD CC M T B G D E)[\s\~]?[0-9]{3,6}</code> | 70%      |       |
| <code>\d{6}</code>                                                | 30%      |       |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Serial letters and digits are mapped; category letters stay, so a taxi plate is still a taxi plate.

#### PA\_PLACA REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE  |   | MASKED |
|--------|---|--------|
| AB1234 | → | EX9707 |
| T12345 | → | T31564 |

## 7.20 Vehicle identification number

PA\_L1\_VIN

### COLUMN NAME VALUE FORMAT CHARACTER MAPPING

#### WHAT IT IS

The VIN, and engine and chassis numbers.

#### HOW IT IS FOUND

#### COLUMN NAME PA\_L1\_VIN - Path

80%  
vin n(o|um|ro|umero)?(de\_)?(serie|identificacion)?(vehicular|vehiculo) serie?  
vehicul(o|ar) n(o|um|ro)?serie numero?serie n(o|um|ro)?motor chasis n(o|um|ro)?  
chasis vin?number

#### VALUE FORMAT PA\_L1\_VIN - Regex

| PATTERN             | STRENGTH | NOTES |
|---------------------|----------|-------|
| [A-HJ-NPR-Z0-9]{17} |          | 70%   |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE             |   | MASKED            |
|-------------------|---|-------------------|
| 3VWFE21C04M000001 | → | 7VVCU97M86J091175 |

## 7.21 Property number (finca)

PA\_L1\_FINCA

### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

A property's finca number, folio real or cadastral location code. The Public Registry can be searched by anyone: the finca leads to its owner.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_FINCA - Path

**85%** finca n(o|um|ro|umero)?(de\_)?finca folio\_?real codigo\_?(de\_)?ubicacion numero\_?catastral n(o|um|ro)?catastro catastro inmueble\_?(num|no|numero|registro) registro\_?(de\_?la\_)?propiedad

#### HOW IT IS MASKED

Character by character, keeping length and format.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE    |   | MASKED          |
|----------|---|-----------------|
| 30162456 | → | <b>39020887</b> |

## 7.22 Contract or customer number

PA\_L1\_NUMERO\_CONTRATO

**COLUMN NAME** **CHARACTER MAPPING**

#### WHAT IT IS

Contract, policy, loan, customer, employee, payroll or claim numbers, and the electricity service's NIS.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_NUMERO\_CONTRATO - Path

**70%** contrato n(o|um|ro|umero)?contrato p[ó]liza n(o|um|ro|umero)?poliza n(o|um|ro)?(de\_)?(prestamo|credito|hipoteca) n(o|um|ro)?operacion n(o|um|ro|umero)?(de\_)?socio n(o|um|ro|umero)?(de\_)?cliente id\_cliente cod(igo)?\_cliente nis n(o|um|ro)?(de\_)?medidor n(o|um|ro|umero)?(de\_)?empleado num\_emp no\_emp n(o|um|ro)?planilla posicion\_planilla n(o|um|ro)?sinistro n(o|um|ro)?(de\_)?matricula n(o|um|ro)?expediente\_(cliente|laboral|personal) folio\_(contrato|poliza|credito|cliente)

#### HOW IT IS MASKED

Character by character, keeping length and format.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE      |   | MASKED            |
|------------|---|-------------------|
| POL-123456 | → | <b>MIF-194239</b> |

## 7.23 User name or social network

PA\_L1\_USUARIO\_RRSS

**COLUMN NAME** **CHARACTER MAPPING**

#### WHAT IT IS

User names, aliases and social network profiles.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_USUARIO\_RRSS - Path

**70%** usuario username user\_?name login nick(name)? alias cuenta\_?usuario usr facebook  
instagram twitter tiktok linkedin red\_?social perfil\_  
(facebook|instagram|twitter|linkedin|tiktok) telegram handle

#### HOW IT IS MASKED

Character by character, keeping length and format.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

**VALUE**

**MASKED**

@yami\_gonzalez

→

@denu\_colwawem

## 7.24 Credential

PA\_L1\_CREDENCIAL

**COLUMN NAME** **REGEX DECOMPOSE**

#### WHAT IT IS

Passwords, PINs, password hashes, tokens and API keys.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L1\_CREDENCIAL - Path

**90%** password passwd pwd pass contrasen[aa] contraseña clave[a-z]\* (except when followed by list E6)  
pin nip hash\_?password password\_?hash secret token\_?acceso access\_?token refresh\_  
token api\_?key otp respuesta\_?secreta pregunta\_?secreta

#### HOW IT IS MASKED

Every character becomes X . A credential has no reason to look real in a test environment.

**PA\_REDACTAR** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

#### EXAMPLES

**VALUE**

**MASKED**

Secreta#2024

→

XXXXXXXXXXXX

## 7.25 Geolocation

PA\_L1\_GEOLOCALIZACION

**COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

#### WHAT IT IS

Latitude and longitude, together or in separate columns. Panama lies between 7° and 10° north and 77° and 83° west.

HOW IT IS FOUND

COLUMN NAME

PA\_L1\_GEOLOCALIZACION - Path

85%

lat

latitud

latitude

lon

lng

longitude

coordenadas?

coord\_?[xy]

geo\_?

(lat|lon|lng|loc|localizacion|posicion)

geolocalizacion

ubicacion?gps

gps

posicion\_?

gps

50%

longitud

long

VALUE FORMAT

PA\_L1\_GEOLOCALIZACION - Regex

| PATTERN                                      | STRENGTH | NOTES |
|----------------------------------------------|----------|-------|
| [7-9]\.\d{3,}\s*,\s*-(7[7-9] 8[0-3])\.\d{3,} | 90%      |       |
| [7-9]\.\d{3,}                                | 60%      |       |
| -(7[7-9] 8[0-3])\.\d{3,}                     | 70%      |       |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the rest is masked: the place is generalized, not moved.

PA\_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

| PA_CM_ALFANUM                                                                      | CHARACTER MAPPING |
|------------------------------------------------------------------------------------|-------------------|
| Maps each character to another of its group (7 groups), keeping length and format. |                   |

EXAMPLES

| VALUE               | MASKED                |
|---------------------|-----------------------|
| 8.983333,-79.516667 | → 8.903230,-79.596978 |

7.26 Free text

PA\_L1\_TEXT0\_LIBRE

| COLUMN NAME | DATA TYPE | VALUE FORMAT | FREE TEXT REDACTION |
|-------------|-----------|--------------|---------------------|
|-------------|-----------|--------------|---------------------|

WHAT IT IS

Remarks, comments, complaints and notes where personal data is written without structure.

HOW IT IS FOUND

| COLUMN NAME                                                            | PA_L1_TEXT0_LIBRE – Path                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60%                                                                    | observaciones? obs comentarios? notas? detalle descripcion?<br>(caso queja reclamo incidente solicitud problema atencion hechos) texto?libre mensaje<br>queja reclamo referencias? motivo?(texto detalle) narrativa relato resumen?caso<br>remarks comments? notes? free?text memo bitacora |
| DATA TYPE                                                              | PA_L1_TEXT0_LIBRE – Type                                                                                                                                                                                                                                                                    |
| Accepts text (length 30 or more); any other type rules the domain out. |                                                                                                                                                                                                                                                                                             |

**VALUE FORMAT** PA\_L1\_TEXT0\_LIBRE – Regex

| PATTERN                                                                | STRENGTH | NOTES                 |
|------------------------------------------------------------------------|----------|-----------------------|
| <code>(( [1-9]   1[0-3] ) (AV PI)?  E N PE) - \d{1,4} - \d{1,6}</code> | 60%      | anywhere in the value |
| <code>[\w.+\-]+\@[ \w\-\-]+\.[A-Za-z]{2,}</code>                       | 70%      | anywhere in the value |
| <code>(\+?507\s?)?6\d{3}-?\d{4}</code>                                 | 50%      | anywhere in the value |

A value that matches none does not count against the domain.

**HOW IT IS MASKED**

Free Text Redaction: cédulas, e-mails, mobiles, hyphenated landlines, card numbers, plates and IPs become [DAT0] , and the given names and surnames of the lists become [NOMBRE] .

**PA\_TEXT0\_LIBRE FREE TEXT REDACTION**

Replaces the text matching 7 patterns with [DAT0] , and the 1,094 terms of `pa-texto-nombres.txt` with [NOMBRE] .

**EXAMPLES**

| VALUE                                                                           |   | MASKED                                                      |
|---------------------------------------------------------------------------------|---|-------------------------------------------------------------|
| Cliente González, cédula 8-442-445, correo y.gonzalez@gmail.com, tel 6123-4567. | → | Cliente [NOMBRE], cédula [DAT0], correo [DAT0], tel [DAT0]. |

# Quasi-identifiers

Data that identifies no one on its own, but does in combination. Panama has 699 corregimientos, and 616 of them had fewer than 10,000 inhabitants in the 2023 census: a birth date, a sex and one of those corregimientos point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

## 8.1 Birth date

PA\_L2\_FECHA\_NACIMIENTO

COLUMN NAME DATA TYPE DATE SHIFT

WHAT IT IS

The birth date, in date, timestamp or text columns.

HOW IT IS FOUND

COLUMN NAME PA\_L2\_FECHA\_NACIMIENTO – Path

90% fecha\_?(de\_)?nac(imiento)? fec\_?nac(imiento)? fch\_?nac f\_?nac(imiento|im)? fnac fechanac(imiento)? fecnac nacimiento dob birth\_?date date\_?of\_?birth birthdate birthday cumplea(n|ñ)os

DATA TYPE PA\_L2\_FECHA\_NACIMIENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 60 days back or forward: it breaks the link to the person and keeps ages and cohorts almost intact.

PA\_FECHA\_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

EXAMPLES

| VALUE      |   | MASKED           |
|------------|---|------------------|
| 1985-07-20 | → | 1985-08-05T00:00 |

## 8.2 Birth year

PA\_L2\_ANIO\_NACIMIENTO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

The birth year alone.

HOW IT IS FOUND

COLUMN NAME PA\_L2\_ANIO\_NACIMIENTO – Path

90% an(i)?o\_?(de\_)?nac(imiento)? año\_?(de\_)?nac(imiento)? birth\_?year year\_?of\_?birth ano\_?nacim

DATA TYPE PA\_L2\_ANIO\_NACIMIENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

#### HOW IT IS MASKED

The decade stays and the last digit is mapped: 1985 becomes another year of the eighties.

##### PA\_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

##### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 1985  | → | 1980   |

### 8.3

## Age

PA\_L2\_EDAD

COLUMN NAME DATA TYPE REGEX DECOMPOSE

#### WHAT IT IS

Ages in years, whole or with decimals.

#### HOW IT IS FOUND

COLUMN NAME PA\_L2\_EDAD – Path

85%

edad(\_?

(actual|paciente|cliente|afiliado|asegurado|beneficiario|ingreso|anos|años|cumplida|jubilacion

))? age anos\_?cumplidos rango\_?(de\_)?edad grupo\_?(de\_)?edad edad\_?en\_?anos

DATA TYPE PA\_L2\_EDAD – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

#### HOW IT IS MASKED

The tens digit stays and the units digit is mapped (37 becomes 3x). From 100 on, 90: extreme ages identify on their own.

##### PA\_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 37    | → | 35     |
| 104   | → | 90     |

### 8.4

## Life event dates

PA\_L2\_FECHA\_EVENTO

COLUMN NAME DATA TYPE DATE SHIFT

#### WHAT IT IS

Dates of death, marriage, hiring and leaving a job, retirement, hospitalization, diagnosis, naturalization and other personal events.

HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_FECHA\_EVENTO – Path

**80%** fecha\_?(de\_)?(defuncion|fallecimiento|muerte|matrimonio|union|divorcio|ingreso|inicio\_?labores|contratacion|contrato|egreso|despido|renuncia|jubilacion|pension|incapacidad|hospitalizacion|alta\_?medica|diagnostico|atencion|consulta|parto|embarazo|detencion|sentencia|vacunacion|cirugia|reingreso|afiliacion|inscripcion\_?css|naturalizacion|residencia) fec\_?(defuncion|ingreso|egreso|contrato|atencion|diagnostico|incapacidad|jubilacion) fch\_?(ingreso|egreso) death\_?date hire\_?date termination\_?date date\_?of\_?death admission\_?date discharge\_?date

**DATA TYPE** PA\_L2\_FECHA\_EVENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 30 days: less than the birth date, since they are often combined with it.

PA\_FECHA\_EVENTO **DATE SHIFT**

Moves the date by -30 to 30 days.

EXAMPLES

| VALUE      |   | MASKED           |
|------------|---|------------------|
| 2021-03-15 | → | 2021-03-05T00:00 |

8.5

Sex

PA\_L2\_SEX0

**COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Registered sex in any notation: F / M , H / M , Femenino , Hombre , Male .

HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_SEX0 – Path

**85%** sexo(\_?(biologico|registral|paciente|cliente|asegurado|trabajador))? sex g[eé]nero (except when followed by identidad , social , autopercebido ) gender cod\_?sexo

**KNOWN VALUES** PA\_L2\_SEX0 – List

| LIST                 | VALUES | STRENGTH | FIRST VALUES                     |
|----------------------|--------|----------|----------------------------------|
| pa-detectar-sexo.txt | 11     | 50%      | h, m, f, hombre, mujer, femenino |

A value that matches none counts 50% against the domain.

HOW IT IS MASKED

Replaced within its own vocabulary: F / M stays F / M .

PA\_SEX0 **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_SEX0\_PALABRA **SECURE LOOKUP**

Replaces the value with one of the 2 values in pa-sexo-palabra.txt , chosen by a keyed hash.

PA\_SEX0\_MUJER\_HOMBRE **SECURE LOOKUP**

Replaces the value with one of the 2 values in pa-sexo-mujer-hombre.txt , chosen by a keyed hash.

#### PA\_SEX0\_INGLES SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-ingles.txt`, chosen by a keyed hash.

#### PA\_SEX0\_F\_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-f-m.txt`, chosen by a keyed hash.

#### PA\_SEX0\_H\_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-h-m.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE     |   | MASKED   |
|-----------|---|----------|
| F         | → | M        |
| Masculino | → | Femenino |

## 8.6 District

PA\_L2\_DISTRITO

### COLUMN NAME

### KNOWN VALUES

### DATA CLEANSING

#### WHAT IT IS

The district name. Panama has 81 inhabited districts — the Guna Yala comarca is not divided into districts —, and 47 had fewer than 20,000 inhabitants in 2023.

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_L2\_DISTRITO - Path

85%

`distrito[a-z]* nom(bre)?_?(dist|distrito) dist municipio mun distrito_?  
(residencia|nacimiento|domicilio)`

##### KNOWN VALUES

PA\_L2\_DISTRITO - List

##### LIST

##### VALUES

##### STRENGTH

##### FIRST VALUES

`pa-detectar-  
distritos.txt`

111

80%

Bocas del Toro, Changuinola, Chiriquí Grande, Chiriqui Grande, Almirante,  
Aguadulce

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

A district under 20,000 inhabitants becomes the nearest district of its province or comarca with at least 20,000; larger ones stay. See *Where people live*.

#### PA\_DISTRITO DATA CLEANSING

Replaces each value found in `pa-distrito-generalizado.txt` (234 pairs) by its counterpart; values not in the file stay as they are.

#### EXAMPLES

| VALUE        |   | MASKED |
|--------------|---|--------|
| Santa Isabel | → | Colón  |
| David        | → | David  |

COLUMN NAME

KNOWN VALUES

DATA CLEANSING

WHAT IT IS

The corregimiento name, the smallest division with a census population: 699 of them, averaging 5,800 inhabitants.

HOW IT IS FOUND

COLUMN NAME

PA\_L2\_CORREGIMIENTO – Path

85%

corregimiento[a-z]\*

nom(bre)?\_?(correg|corregimiento)

correg

corr

corregimiento\_?

(residencia|nacimiento|domicilio)

50%

ciudad[a-z]\*

city

town

poblacion

KNOWN VALUES

PA\_L2\_CORREGIMIENTO – List

| LIST                           | VALUES | STRENGTH | FIRST VALUES                                                                             |
|--------------------------------|--------|----------|------------------------------------------------------------------------------------------|
| pa-detectar-corregimientos.txt | 814    | 80%      | Bocas del Toro, Bastimentos, Punta Laurel, Tierra Oscura, Bocas del Drago, San Cristóbal |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A corregimiento under 10,000 inhabitants becomes the nearest corregimiento of its province or comarca with at least 10,000, or of the country if the province has none. Large ones stay. See *Where people live*.

PA\_CORREGIMIENTO DATA CLEANSING

Replaces each value found in pa-corregimiento-generalizado.txt (1,760 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

| VALUE       | MASKED          |
|-------------|-----------------|
| Cerro Punta | → La Concepción |
| Tocumen     | → Tocumen       |

COLUMN NAME

DATA TYPE

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

INEC geographic codes: six digits for a corregimiento ( 040612 ), four for a district ( 0306 ), also without the leading zero.

HOW IT IS FOUND

COLUMN NAME

PA\_L2\_CODIGO\_LUGAR – Path

85%

cod(igo)?\_?(correg|corregimiento|dist|distrito|lugar(\_?poblado)?|inec|geografico)

cve\_?

(correg|dist|lugar)

id\_?(correg|corregimiento|distrito)

correg\_?(cod|id)

distrito\_?

(cod|id)

codigo\_?censal

DATA TYPE

PA\_L2\_CODIGO\_LUGAR – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

**KNOWN VALUES** PA\_L2\_CODIGO\_LUGAR – List

| LIST                         | VALUES | STRENGTH | FIRST VALUES                                |
|------------------------------|--------|----------|---------------------------------------------|
| pa-detectar-codigo-lugar.txt | 1,405  | 40%      | 010101, 10101, 010102, 10102, 010104, 10104 |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

By the same rule as the names: a small corregimiento or district becomes the code of the large one it maps to.

PA\_CODIGO\_LUGAR **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_CODIGO\_CORREGIMIENTO **DATA CLEANSING**

Replaces each value found in pa-codigo-corregimiento-generalizado.txt (1,331 pairs) by its counterpart; values not in the file stay as they are.

PA\_CODIGO\_DISTRITO **DATA CLEANSING**

Replaces each value found in pa-codigo-distrito-generalizado.txt (148 pairs) by its counterpart; values not in the file stay as they are.

#### EXAMPLES

| VALUE  |   | MASKED |
|--------|---|--------|
| 040612 | → | 040612 |
| 0306   | → | 0301   |

## 8.9 Populated place

PA\_L2\_LUGAR\_POBLADO

**COLUMN NAME** **SECURE LOOKUP**

#### WHAT IT IS

Populated places, hamlets and communities, and the place of birth.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_LUGAR\_POBLADO – Path

**80%** lugar\_?poblado nom(bre)?\_?lugar\_?poblado poblado comunidad (except when followed by *indigena*, *religiosa*) caserio localidad lugar\_?(de\_)?nacimiento pueblo\_?(residencia|origen)

#### HOW IT IS MASKED

Replaced by the name of a corregimiento of more than 20,000 inhabitants: still a place, no longer theirs.

PA\_LUGAR\_POBLADO **SECURE LOOKUP**

Replaces the value with one of the 52 values in pa-lugares-poblados.txt, chosen by a keyed hash.

#### EXAMPLES

| VALUE             |   | MASKED  |
|-------------------|---|---------|
| Los Llanos de Ocú | → | El Coco |

## 8.10 Barriada or neighborhood

PA\_L2\_BARRIADA

**COLUMN NAME** **SECURE LOOKUP**

#### WHAT IT IS

Barriadas, urbanizaciones, residential developments and sectors.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_BARRIADA – Path

85% `barriada[a-z]*` `barrio` `urbanizacion` `urb` `residencial` `sector` `nom(bre)?_?`  
`(barriada|urbanizacion|barrio)` `condominio` `ph?nombre` `nombre?ph`

#### HOW IT IS MASKED

Replaced by one of 75 common *barriadas* and neighborhoods.

PA\_BARRIADA **SECURE LOOKUP**

Replaces the value with one of the 75 values in `pa-barriadas.txt`, chosen by a keyed hash.

#### EXAMPLES

VALUE

MASKED

Villa Guadalupe

→

Las Acacias

## 8.11 Postal code

PA\_L2\_CODIGO\_POSTAL

**COLUMN NAME**

**VALUE FORMAT**

**REGEX DECOMPOSE**

#### WHAT IT IS

Postal codes of Correos y Telégrafos, little used in Panama. The first two digits give the province.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_CODIGO\_POSTAL – Path

85% `cod(igo)?_?postal` `zona?postal` `cp` `zip(_?code)?` `postal?code` `codpostal` `apartado?`  
`postal`

**VALUE FORMAT** PA\_L2\_CODIGO\_POSTAL – Regex

PATTERN

STRENGTH

NOTES

`\d{4,5}`

20%

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

The first two digits stay and the rest becomes zeros.

PA\_CODIGO\_POSTAL **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM.

PA\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

VALUE

MASKED

0801

→

0800

## 8.12 Nationality

PA\_L2\_NACIONALIDAD

**COLUMN NAME**

**KNOWN VALUES**

**SECURE LOOKUP**

#### WHAT IT IS

Nationality or country of birth. In a Panamanian database, an uncommon nationality identifies.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_NACIONALIDAD – Path

**80%** nacionalidad nationality pais\_?(de\_?)(nacimiento|origen) ciudadania citizenship  
country\_of\_birth extranjero

**KNOWN VALUES** PA\_L2\_NACIONALIDAD – List

| LIST                           | VALUES | STRENGTH | FIRST VALUES                                                           |
|--------------------------------|--------|----------|------------------------------------------------------------------------|
| pa-detectar-nacionalidades.txt | 63     | 70%      | Panameña, Panamena, Colombiana, Venezolana, Nicaragüense, Nicaraguense |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by one of the 20 most common nationalities in Panama.

PA\_NACIONALIDAD **SECURE LOOKUP**

Replaces the value with one of the 20 values in pa-nacionalidades.txt, chosen by a keyed hash.

#### EXAMPLES

| VALUE      | MASKED           |
|------------|------------------|
| Venezolana | → Estadounidense |

## 8.13 Marital status

PA\_L2\_ESTADO\_CIVIL

**COLUMN NAME** **KNOWN VALUES** **SECURE LOOKUP**

#### WHAT IT IS

Marital status, in the census categories: in a union, married, single, separated from a marriage, separated from a union, divorced, widowed.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L2\_ESTADO\_CIVIL – Path

**85%** estado\_civil edo\_civil est\_civil ecivil estado\_conyugal situacion\_conyugal  
marital\_status civil\_status

**KNOWN VALUES** PA\_L2\_ESTADO\_CIVIL – List

| LIST                         | VALUES | STRENGTH | FIRST VALUES                                   |
|------------------------------|--------|----------|------------------------------------------------|
| pa-detectar-estado-civil.txt | 18     | 70%      | soltero, soltera, casado, casada, unido, unida |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by another of the seven categories.

PA\_ESTADO\_CIVIL **SECURE LOOKUP**

Replaces the value with one of the 7 values in pa-estado-civil.txt, chosen by a keyed hash.

#### EXAMPLES

| VALUE | MASKED   |
|-------|----------|
| Unida | → Casado |

## 8.14 Occupation

PA\_L2\_OCUPACION

COLUMN NAME

SECURE LOOKUP

### WHAT IT IS

Occupation, profession or job title. A rare title ( Gerente Regional de Operaciones Bocas del Toro ) is unique in the database.

### HOW IT IS FOUND

COLUMN NAME

PA\_L2\_OCUPACION – Path

80%

ocupaci[oó]n profes[ioó]n oficio puesto(\_?(trabajo|laboral|actual))? (except when followed by votacion, venta, trabajo\_?id) cargo (except when followed by list E7) job\_?title occupation actividad\_?(laboral|economica\_?persona) categoria\_?laboral

### HOW IT IS MASKED

Replaced by one of 40 common occupations.

PA\_OCUPACION

SECURE LOOKUP

Replaces the value with one of the 40 values in pa-ocupaciones.txt , chosen by a keyed hash.

### EXAMPLES

VALUE

MASKED

Gerente Regional de Operaciones Bocas del Toro

→

Bodeguero

## 8.15 Employer

PA\_L2\_EMPLEADOR

COLUMN NAME

SECURE LOOKUP

### WHAT IT IS

The name of the employer (patrono) or workplace.

### HOW IT IS FOUND

COLUMN NAME

PA\_L2\_EMPLEADOR – Path

80%

empleador patrono nombre\_?patrono razon\_?social\_?(patrono|empleador) empresa\_?(empleadora|trabajo|laboral) lugar\_?(de\_?)?trabajo centro\_?(de\_?)?trabajo employer institucion\_?(donde\_?)?labora entidad\_?(donde\_?)?labora

### HOW IT IS MASKED

Replaced by one of 25 fictitious employers.

PA\_EMPLEADOR

SECURE LOOKUP

Replaces the value with one of the 25 values in pa-empleadores.txt , chosen by a keyed hash.

### EXAMPLES

VALUE

MASKED

Autoridad del Canal de Panamá

→

Ferretería EL Constructor, S.A.

## 8.16 Schooling

PA\_L2\_ESCOLARIDAD

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

### WHAT IT IS

Education level in the terms of the Panamanian system: primaria, premedia, media, vocational, higher education.

### HOW IT IS FOUND

COLUMN NAME

PA\_L2\_ESCOLARIDAD - Path

80%

escolaridad nivel?(educativo|educacional|escolar|de?estudios|academico|de?instruccion)  
grado?(de?)?(estudios|instruccion) grado?academico ultimo?grado(\_?aprobado)?  
education?level anos?(de?)?escolaridad

KNOWN VALUES

PA\_L2\_ESCOLARIDAD - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-escolaridad.txt

28

60%

Ningún grado, Ningun grado, Primaria incompleta, Primaria completa, Premedia incompleta, Premedia completa

A value that matches none counts 40% against the domain.

### HOW IT IS MASKED

Replaced by another of the 13 levels.

PA\_ESCOLARIDAD SECURE LOOKUP

Replaces the value with one of the 13 values in pa-escolaridad.txt , chosen by a keyed hash.

### EXAMPLES

VALUE

MASKED

Premedia completa

→

Premedia incompleta

## 8.17 School

PA\_L2\_ESCUELA

COLUMN NAME

SECURE LOOKUP

### WHAT IT IS

The name of the school, colegio, basic education center or university.

### HOW IT IS FOUND

COLUMN NAME

PA\_L2\_ESCUELA - Path

75%

escuela colegio plantel nombre?(escuela|plantel|colegio|centro?educativo|institucion?  
educativa) centro?educativo institucion?educativa universidad instituto school(\_?  
name)? cebg centro?basico

### HOW IT IS MASKED

Replaced by one of 15 fictitious schools.

PA\_ESCUELA SECURE LOOKUP

Replaces the value with one of the 15 values in pa-escuelas.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Instituto Nacional de Panamá

→ Universidad Tecnológica del Pacífico

8.18 Children and dependents

PA\_L2\_DEPENDIENTES

COLUMN NAME

DATA TYPE

REGEX DECOMPOSE

WHAT IT IS

Number of children, dependents or household members.

HOW IT IS FOUND

COLUMN NAME

PA\_L2\_DEPENDIENTES - Path

80%

dependientes n(o|um|ro)?\_?dependientes num\_?hijos n(o|um|ro)?\_?hijos cant\_?hijos  
hijos hijos\_?nacidos\_?vivos integrantes\_?(del\_)?(hogar|familia) n(o|um|ro)\_?integrantes  
tamano\_?(del\_)?hogar personas\_?(en\_)?(el\_)?hogar

DATA TYPE

PA\_L2\_DEPENDIENTES - Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

Capped at 5: large households are rare, and a rare count identifies.

PA\_CANTIDAD\_ACOTADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

9

→ 5

# Sensitive data

The categories of article 4, section 11 — ethnic origin, health, genetic and biometric data, beliefs, political opinions, union membership, sexual orientation — and, because the list is not exhaustive, gender identity and sexual life. Categorical columns keep their shape: a yes/no flag stays a flag, a numeric code gets other digits and any other value is replaced by one from its category's list. Where the values share no vocabulary, they are suppressed to Sin información .

## 9.1 Ethnic origin

PA\_L3\_ETNIA

COLUMN NAME    KNOWN VALUES    REGEX DECOMPOSE

WHAT IT IS

Belonging to an indigenous people — Ngäbe, Guna, Emberá, Buglé, Wounaan, Naso Tjërdi, Bokota, Bri Bri — or Afro-descendant self-identification — Afro-Panamanian, moreno, black, Afro-colonial, Afro-Antillean —, in the terms of the 2023 census, often as a yes/no flag.

HOW IT IS FOUND

COLUMN NAME    PA\_L3\_ETNIA – Path

90%

etnia   [eé]tnico   origen\_[eé]tnico   grupo\_[eé]tnico   pueblo\_?(indigena|originario)   grupo\_?indigena   indigena   ind[ií]lgena   afro(panameno|descendiente|panameño)?   grupo\_?afro(descendiente)?   autoidentificacion(\_?(etnica|indigena|afro))?   se\_?considera\_?(indigena|afro|afrodescendiente)   raza   race   ethnicity   ethnic\_?group   comunidad\_?indigena   pertenencia\_?(etnica|indigena)

KNOWN VALUES    PA\_L3\_ETNIA – List

| LIST                    | VALUES | STRENGTH | FIRST VALUES                                  |
|-------------------------|--------|----------|-----------------------------------------------|
| pa-detectar-pueblos.txt | 52     | 90%      | Guna, Ngäbe, Ngabe, Buglé, Bugle, Naso Tjërdi |
| pa-detectar-bandera.txt | 8      | 30%      | s, n, sí, sí, no, y                           |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag in its vocabulary, a code gets other digits and any other value is replaced by one from the census list.

PA\_PUEBLO    REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA\_PUEBLO\_LISTA .

PA\_BANDERA    REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA\_BANDERA\_TRUE\_FALSE    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA\_BANDERA\_S\_N    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA\_BANDERA\_Y\_N    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA\_BANDERA\_0\_1    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### PA\_PUEBLO\_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in `pa-pueblos.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE  |   | MASKED |
|--------|---|--------|
| Teribe | → | Moreno |
| S      | → | S      |

## 9.2 Indigenous language

PA\_L3 LENGUA INDIGENA

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

#### WHAT IT IS

The indigenous language the person speaks: Ngäbere, Buglere, Guna, Emberá, Wounmeu, Naso Tjërdi, Bribri.

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_L3 LENGUA INDIGENA - Path

90%

`lengu(?:indigena|materna|originaria)? habla?(lengu(?:indigena|originaria) idioma?(indigena|materno|originario) lengua_?que?habla dialecto native_?language`

##### KNOWN VALUES

PA\_L3 LENGUA INDIGENA - List

##### LIST

##### VALUES

##### STRENGTH

##### FIRST VALUES

pa-detectar-lenguas.txt

32

80%

Ngäbere, Ngabere, Buglere, Guna, Emberá, Embera

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Like ethnic origin: flags and codes keep their shape, names are replaced by another language from the list.

#### PA LENGUA INDIGENA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

#### PA LENGUA INDIGENA LISTA

##### PA BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PA BANDERA SI NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

##### PA BANDERA TRUE FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

##### PA BANDERA S N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

##### PA BANDERA Y N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

##### PA BANDERA 0 1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

##### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

##### PA LENGUA INDIGENA LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `pa-lenguas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Ngäbere → No habla lengua indígena

9.3 Diagnosis

PA\_L3\_SALUD\_DIAGNOSTICO

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Diagnoses, as ICD-10 codes ( F32.9 , B20 ) or text, and medical history, allergies and chronic conditions.

HOW IT IS FOUND

COLUMN NAME

PA\_L3\_SALUD\_DIAGNOSTICO - Path

85%

diagn[oó]stico[a-z]\* diag dx cie\_?10 cie\_?11 cod(igo)?\_?cie cie icd\_?10 icd  
patolog[ií]a[a-z]\* padecimiento[a-z]\* enfermedad(es)?[a-z]\* morbilidad condicion\_  
(medica|de\_salud|cronica) problema\_(de\_)?salud impresion\_diagnostica causa\_  
(incapacidad|muerte|defuncion|hospitalizacion|consulta) motivo\_(de\_)?  
(consulta|hospitalizacion|incapacidad) comorbilidad(es)? antecedentes\_  
(medicos|patologicos|personales\_patologicos|heredofamiliares|clinicos) alergias?  
enfermedad\_preexistente preexistencias? enfermedad\_cronica riesgo\_profesional

VALUE FORMAT

PA\_L3\_SALUD\_DIAGNOSTICO - Regex

PATTERN

STRENGTH

NOTES

[A-TV-Za-tv-z]\d{2}(\.\d{1,2}|\d)?

50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

An ICD-10 code is replaced by another common code of the same format, with or without a decimal; text, by a common diagnosis. The real diagnosis does not survive.

PA\_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

PA\_DIAGNOSTICO\_TEXT0 .

PA\_CIE10\_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in pa-cie10-decimal.txt , chosen by a keyed hash.

PA\_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in pa-cie10.txt , chosen by a keyed hash.

PA\_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

#### PA\_DIAGNOSTICO\_TEXTO **SECURE LOOKUP**

Replaces the value with one of the 30 values in `pa-diagnosticos.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE           |   | MASKED     |
|-----------------|---|------------|
| F32.9           | → | J02.9      |
| Depresión mayor | → | Tendinitis |

## 9.4 Medication

PA\_L3\_SALUD\_MEDICAMENTO

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

#### WHAT IT IS

Medications and active ingredients. Some reveal the diagnosis: an antiretroviral, an antipsychotic.

#### HOW IT IS FOUND

#### COLUMN NAME

PA\_L3\_SALUD\_MEDICAMENTO - Path

85%

medicamentos? f[aá]rmas? sustancia\_?activa principio\_?activo prescripci[oó]n  
receta(\_?medica)? tratamiento(\_?farmacologico)? medication drug(\_?name)? posologia  
lista\_?oficial\_?medicamentos

#### KNOWN VALUES

PA\_L3\_SALUD\_MEDICAMENTO - List

#### LIST

VALUES STRENGTH FIRST VALUES

|                                  |    |     |                                                                            |
|----------------------------------|----|-----|----------------------------------------------------------------------------|
| pa-detectar-<br>medicamentos.txt | 58 | 70% | Acetaminofén, Acetaminofen, Ibuprofeno, Naproxeno, Diclofenaco, Ketorolaco |
|----------------------------------|----|-----|----------------------------------------------------------------------------|

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Replaced by one of 26 common medications.

#### PA\_MEDICAMENTO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_MEDICAMENTO\_LISTA .

#### PA\_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

#### PA\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

#### PA\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

#### PA\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

#### PA\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

#### PA\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

#### PA\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PA\_MEDICAMENTO\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 26 values in `pa-medicamentos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Sertralina 50 mg

→ Amoxicilina

9.5 Clinical identifier

PA\_L3\_SALUD\_IDENTIFICADOR

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Medical record, file, admission, prescription or sick-leave number.

HOW IT IS FOUND

COLUMN NAME

PA\_L3\_SALUD\_IDENTIFICADOR - Path

85%

expediente\_?clinico n(o|um|ro|umero)\_?expediente(\_?clinico)? historia\_?clinica  
n(o|um|ro)\_?historia nhc n(o|um|ro)\_?episodio n(o|um|ro)\_?(de\_)?admision folio\_  
(atencion|urgencias|receta|incapacidad) n(o|um|ro)\_?incapacidad certificado\_?(de\_)?  
incapacidad n(o|um|ro)\_?(de\_)?cama n(o|um|ro)\_?interconsulta n(o|um|ro)\_?(de\_)?receta  
n(o|um|ro)\_?biopsia medical\_?record\_?number mrn

55%

expediente episodio cama

HOW IT IS MASKED

Character by character, keeping length and format.

PA\_CM\_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

HC-2024-00123

→ TW-3389-01317

9.6 Clinical text

PA\_L3\_SALUD\_TEXTO\_CLINICO

COLUMN NAME

DATA TYPE

FREE TEXT REDACTION

WHAT IT IS

Progress notes, medical history, discharge summaries and clinical summaries.

HOW IT IS FOUND

COLUMN NAME

PA\_L3\_SALUD\_TEXTO\_CLINICO - Path

85%

nota\_?(medica|de\_?evolucion|evolucion|de\_?ingreso|ingreso|de\_?egreso|egreso|de\_  
urgencias|urgencias|postoperatoria|preoperatoria|de\_?enfermeria|clinica)s? notas?  
clinicas? enfermedad\_?actual anamnesis examen\_?fisico evoluci[oó]n(\_?(medica|clinica))?  
epicrisis indicaciones(\_?medicas)? resumen\_?clinico resumen\_?(de\_)?egreso historia\_  
(de\_?la\_)?enfermedad plan\_?(de\_)?tratamiento pronostico hallazgos interpretacion  
clinical\_?notes? progress\_?notes?

DATA TYPE

PA\_L3\_SALUD\_TEXTO\_CLINICO - Type

Accepts text (length 30 or more); any other type rules the domain out.

#### HOW IT IS MASKED

Free Text Redaction, as for free text: cédulas, contact details and names are redacted. The clinical content stays; if it must go too, assign the column to a suppression domain.

##### PA\_TEXTO\_LIBRE FREE TEXT REDACTION

Replaces the text matching 7 patterns with [DATO], and the 1,094 terms of pa-texto-nombres.txt with [NOMBRE].

#### EXAMPLES

##### VALUE

Paciente Batista, cédula 8-442-445, acude por cefalea.

##### MASKED

Paciente [NOMBRE], cédula [DATO], acude por cefalea.

## 9.7 Test result

PA\_L3\_SALUD\_RESULTADO\_EXAMEN

##### COLUMN NAME

##### CHARACTER MAPPING

#### WHAT IT IS

Laboratory and imaging results: HIV, viral load, glucose, pregnancy or drug tests.

#### HOW IT IS FOUND

##### COLUMN NAME PA\_L3\_SALUD\_RESULTADO\_EXAMEN - Path

80%

resultado\_?

(estudio|examen|laboratorio|lab|prueba|pcr|antigeno|biopsia|imagen|vih|covid|papanicolaou|mamografia) examen(es)?\_(de\_)?laboratorio vih hiv carga\_?viral cd4 glucosa glicemia hemoglobina(\_?glicosilada)? hba1c colesterol trigliceridos presion\_?arterial imc prueba\_?(covid|embarazo|vih|antidoping|toxicologica) antidoping lab\_?result

50%

resultado examen

#### HOW IT IS MASKED

Character by character, keeping length and format.

##### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

##### VALUE

Reactivo

##### MASKED

→ Quilqaju

## 9.8 Blood group

PA\_L3\_SALUD\_GRUPO\_SANGUINEO

##### COLUMN NAME

##### KNOWN VALUES

##### SECURE LOOKUP

#### WHAT IT IS

ABO group and Rh factor.

#### HOW IT IS FOUND

##### COLUMN NAME PA\_L3\_SALUD\_GRUPO\_SANGUINEO - Path

90%

grupo\_?sangu[ií]neo grupo\_?sang tipo\_?(de\_)?sangre tipo\_?sanguineo factor\_?rh rh blood\_?type blood\_?group

**KNOWN VALUES** PA\_L3\_SALUD\_GRUPO\_SANGUINEO – List

| LIST                            | VALUES | STRENGTH | FIRST VALUES             |
|---------------------------------|--------|----------|--------------------------|
| pa-detectar-grupo-sanguineo.txt | 26     | 90%      | A+, A-, B+, B-, AB+, AB- |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by another of the eight groups.

**PA\_GRUPO\_SANGUINEO** **SECURE LOOKUP**

Replaces the value with one of the 8 values in `pa-grupo-sanguineo.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE | MASKED |
|-------|--------|
| 0+    | → A-   |

## 9.9 Disability

PA\_L3\_SALUD\_DISCAPACIDAD

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

#### WHAT IT IS

Type or degree of disability, in the types SENADIS certifies: physical, visual, hearing, intellectual, psychosocial, visceral.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_L3\_SALUD\_DISCAPACIDAD – Path

**90%** `discapacidad[a-z]* tipo_(de_)?discapacidad grado_(de_)?discapacidad certificado_(de_)?discapacidad carne_senadis senadis persona_con_discapacidad pcd invalidez porcentaje_(de_)?invalidez pension_(de_)?invalidez limitacion(_?(fisica|actividad))? disability dependencia_funcional nee necesidades_educativas_especiales`

**KNOWN VALUES** PA\_L3\_SALUD\_DISCAPACIDAD – List

| LIST                         | VALUES | STRENGTH | FIRST VALUES                                               |
|------------------------------|--------|----------|------------------------------------------------------------|
| pa-detectar-discapacidad.txt | 26     | 60%      | Física, Física, Visual, Auditiva, Intelectual, Psicosocial |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Flags and codes keep their shape; the type is replaced by another from the list.

**PA\_DISCAPACIDAD** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

**PA\_DISCAPACIDAD\_LISTA**.

**PA\_BANDERA** **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

**PA\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

**PA\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

**PA\_BANDERA\_S\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

**PA\_BANDERA\_Y\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_0\_1** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PA\_DISCAPACIDAD\_LISTA** **SECURE LOOKUP**

Replaces the value with one of the 8 values in `pa-discapacidad.txt` , chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Discapacidad psicosocial severa

→ **Auditiva**

## 9.10 Sexual and reproductive health

PA\_L3\_SALUD\_SEXUAL\_REPRODUCTIVA

### COLUMN NAME

### REGEX DECOMPOSE

#### WHAT IT IS

Pregnancy, weeks of gestation, abortions, contraception, fertility, sexually transmitted infections.

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_L3\_SALUD\_SEXUAL\_REPRODUCTIVA - Path

90%

embarazo embarazada gestaci[oó]n semanas\_?(de\_)?gestacion edad\_?gestacional fum  
fecha\_?ultima\_?menstruacion aborto anticonceptivos? metodo\_?(de\_)?planificacion(\_?  
familiar)? planificacion\_?familiar fertilidad infertilidad reproduccion\_?asistida  
gestas partos cesareas? puerperio its ets infeccion\_?(de\_)?transmision\_?sexual  
prep pregnan[a-z]\* contracepti[a-z]\*

#### HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información` .

**PA\_CATEGORIA\_SUPRIMIDA** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR` .

**PA\_BANDERA** **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

**PA\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

**PA\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

**PA\_BANDERA\_S\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_Y\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_0\_1** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PA\_SUPRIMIR** **SECURE LOOKUP**

Replaces the value with one of the 1 values in `pa-sin-informacion.txt` , chosen by a keyed hash.

## EXAMPLES

### VALUE

### MASKED

Embarazo de 12 semanas

→

Sin información

S

→

S

## 9.11 Mental health and addiction

PA\_L3\_SALUD\_MENTAL\_ADICCIONES

### COLUMN NAME

### REGEX DECOMPOSE

### WHAT IT IS

Mental disorders, suicide attempts, substance use and addiction treatment.

### HOW IT IS FOUND

#### COLUMN NAME

PA\_L3\_SALUD\_MENTAL\_ADICCIONES - Path

90%

salud\_?mental psiquiatri[a-z]\* psicologi[a-z]\* trastorno[a-z]\* depresi[oó]n ansiedad  
suicid[a-z]\* autolesi[oó]n adicci[oó]n(es)? consumo\_?(de\_?)?  
(drogas|alcohol|sustancias|tabaco) alcoholismo farmacodependencia drogadiccion  
toxicomania dependencia\_?(alcohol|drogas|sustancias) tratamiento\_?  
(adicciones|drogas|alcohol) rehabilitacion\_?(adicciones|drogas) mental\_?health substance\_?  
(use|abuse)

### HOW IT IS MASKED

Suppressed to Sin información .

#### PA\_CATEGORIA\_SUPRIMIDA

#### REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA\_SUPRIMIR .

#### PA\_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

#### PA\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

#### PA\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

#### PA\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

#### PA\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

#### PA\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

#### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### PA\_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in pa-sin-informacion.txt , chosen by a keyed hash.

## EXAMPLES

### VALUE

### MASKED

Trastorno bipolar

→

Sin información

## 9.12 Biometric data

PA\_L3\_BIOMETRICO

**COLUMN NAME** **DATA TYPE** **REGEX DECOMPOSE**

### WHAT IT IS

Fingerprint, face, iris or voice templates stored as text (hash, base64). The law names them sensitive and Decree 285 defines them.

### HOW IT IS FOUND

**COLUMN NAME** PA\_L3\_BIOMETRICO - Path

**90%** huella(\_?(dactilar|digital))? huellas dactilar biometri[a-z]\* template\_?  
(facial|biometrico|huella) rostro reconocimiento\_?facial facial\_?(template|id) iris  
retina voz\_?(huella|biometrica) firma\_?(digitalizada|biometrica|autografa\_?digital)  
fingerprint face\_?(id|template|encoding) palm\_?print minucias

**DATA TYPE** PA\_L3\_BIOMETRICO - Type

Accepts text; any other type rules the domain out.

### HOW IT IS MASKED

Every character becomes X . Binary columns are out of reach: see *Limits*.

**PA\_REDACTAR** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

### EXAMPLES

| VALUE        | MASKED         |
|--------------|----------------|
| A1B2C3D4E5F6 | → XXXXXXXXXXXX |

## 9.13 Genetic data

PA\_L3\_INFORMACION\_GENETICA

**COLUMN NAME** **REGEX DECOMPOSE**

### WHAT IT IS

Genetic profiles, sequences, mutations, karyotypes, paternity tests.

### HOW IT IS FOUND

**COLUMN NAME** PA\_L3\_INFORMACION\_GENETICA - Path

**90%** adn dna genoma genotipo genetic[a-z]\* gen[eé]tic[ao] informacion\_genetica  
secuencia\_genetica marcador\_genetico perfil\_?(genetico|biologico) mutacion(\_?  
genetica)? snp brca haplotipo cariotipo tamizaje\_?(genetico|neonatal|metabolico)  
prueba\_?(de\_)?paternidad

### HOW IT IS MASKED

Every character becomes X .

**PA\_REDACTAR** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

### EXAMPLES

| VALUE          | MASKED         |
|----------------|----------------|
| BRCA1 positivo | → XXXXXXXXXXXX |

## 9.14 Sexual orientation

PA\_L3\_ORIENTACION\_SEXUAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

### WHAT IT IS

Sexual orientation or preference, which the law names explicitly.

### HOW IT IS FOUND

COLUMN NAME

PA\_L3\_ORIENTACION\_SEXUAL - Path

95%

orientaci[oó]n?sexual preferencia?sexual sexual?(orientation|preference) orient\_?  
sexual pref?sexual

KNOWN VALUES

PA\_L3\_ORIENTACION\_SEXUAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-orientacion-  
sexual.txt

17

80%

Heterosexual, Homosexual, Lesbiana, Gay, Bisexual,  
Pansexual

A value that matches none counts 40% against the domain.

### HOW IT IS MASKED

Flags and codes keep their shape; the value is replaced by another from the list.

PA\_ORIENTACION\_SEXUAL

REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_ORIENTACION\_SEXUAL\_LISTA .

PA\_BANDERA

REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO

SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA\_BANDERA\_TRUE\_FALSE

SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA\_BANDERA\_S\_N

SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA\_BANDERA\_Y\_N

SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA\_BANDERA\_0\_1

SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA\_CM\_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA\_ORIENTACION\_SEXUAL\_LISTA

SECURE LOOKUP

Replaces the value with one of the 9 values in pa-orientacion-sexual.txt , chosen by a keyed hash.

### EXAMPLES

VALUE

MASKED

Bisexual

→

Lesbiana

## 9.15 Gender identity

PA\_L3\_IDENTIDAD\_GENERO

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

### WHAT IT IS

Gender identity, name changes for identity, pronouns.

HOW IT IS FOUND

COLUMN NAME

PA\_L3\_IDENTIDAD\_GENERO – Path

95%

identidad\_(de\_)?g[eé]nero

gender\_identity

g[eé]nero?(identidad|social|autopercibido)

transg[eé]nero

transsexual

persona\_?trans

cambio\_(de\_)?(sexo|genero|nombre\_?por\_?

genero)

pronombres?

KNOWN VALUES

PA\_L3\_IDENTIDAD\_GENERO – List

| LIST                             | VALUES | STRENGTH | FIRST VALUES                                                        |
|----------------------------------|--------|----------|---------------------------------------------------------------------|
| pa-detectar-identidad-genero.txt | 17     | 70%      | Mujer, Hombre, Mujer trans, Hombre trans, No binario, Género fluido |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the value is replaced by another from the list.

PA\_IDENTIDAD\_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_IDENTIDAD\_GENERO\_LISTA .

PA\_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA\_IDENTIDAD\_GENERO\_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in pa-identidad-genero.txt , chosen by a keyed hash.

EXAMPLES

| VALUE        | MASKED       |
|--------------|--------------|
| Hombre trans | → No binario |

9.16 Sexual life

PA\_L3\_VIDA\_SEXUAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Sexual activity, behavior and history.

HOW IT IS FOUND

| COLUMN NAME | PA_L3_VIDA_SEXUAL – Path                                                                                                                                                                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90%         | vida_?sexual   conducta_?sexual   actividad_?sexual   parejas?_?sexuales?   practicas_?sexuales   inicio_?(de_?)(la_?)?vida_?sexual   sexual_?(activity behavior history)   historia_?sexual   comportamiento_?sexual |

HOW IT IS MASKED

Suppressed to Sin información .

PA\_CATEGORIA\_SUPRIMIDA   **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA\_SUPRIMIR .

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| PA_BANDERA <b>REGEX DECOMPOSE</b>                                                                   |
| Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep. |
| PA_BANDERA_SI_NO <b>SECURE LOOKUP</b>                                                               |
| Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.       |
| PA_BANDERA_TRUE_FALSE <b>SECURE LOOKUP</b>                                                          |
| Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.  |
| PA_BANDERA_S_N <b>SECURE LOOKUP</b>                                                                 |
| Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.         |
| PA_BANDERA_Y_N <b>SECURE LOOKUP</b>                                                                 |
| Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.         |
| PA_BANDERA_0_1 <b>SECURE LOOKUP</b>                                                                 |
| Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.         |
| PA_CM_ALFANUM <b>CHARACTER MAPPING</b>                                                              |
| Maps each character to another of its group (7 groups), keeping length and format.                  |
| PA_SUPRIMIR <b>SECURE LOOKUP</b>                                                                    |
| Replaces the value with one of the 1 values in pa-sin-informacion.txt , chosen by a keyed hash.     |

EXAMPLES

| VALUE  | MASKED            |
|--------|-------------------|
| Activa | → Sin información |

9.17 Religion

PA\_L3\_RELIGION

|             |              |                 |
|-------------|--------------|-----------------|
| COLUMN NAME | KNOWN VALUES | REGEX DECOMPOSE |
|-------------|--------------|-----------------|

WHAT IT IS

Religion or denomination: Catholic, Evangelical, Adventist, Jehovah's Witness, Bahá'í, Jewish, Muslim, and the indigenous religions Mamatata and Ibeorgun.

HOW IT IS FOUND

| COLUMN NAME | PA_L3_RELIGION – Path                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 95%         | religi[oó]n[a-z]*   credo   culto   iglesia   creencia_?religiosa   confesion_?religiosa   denominacion_?religiosa   religious_?affiliation   comunidad_?religiosa   congregacion |

**KNOWN VALUES** PA\_L3\_RELIGION – List

| LIST                       | VALUES | STRENGTH | FIRST VALUES                                                                                       |
|----------------------------|--------|----------|----------------------------------------------------------------------------------------------------|
| pa-detectar-religiones.txt | 61     | 90%      | Católica, Catolica, Evangélica, Evangelica, Adventista del Séptimo Día, Adventista del Septimo Dia |

A value that matches none counts 40% against the domain.

**HOW IT IS MASKED**

Flags and codes keep their shape; the value is replaced by another from the list.

**PA\_RELIGION** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

**PA\_RELIGION\_LISTA** .**PA\_BANDERA** **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

**PA\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

**PA\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

**PA\_BANDERA\_S\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_Y\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_0\_1** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PA\_RELIGION\_LISTA** **SECURE LOOKUP**

Replaces the value with one of the 14 values in `pa-religiones.txt` , chosen by a keyed hash.

**EXAMPLES**

| VALUE     | MASKED      |
|-----------|-------------|
| Episcopal | → Musúľmana |

## 9.18 Philosophical and moral beliefs

PA\_L3\_CREENCIA\_FILOSOFICA\_MORAL

**COLUMN NAME** **REGEX DECOMPOSE****WHAT IT IS**

Personal convictions, conscientious objection, ethical positions.

**HOW IT IS FOUND****COLUMN NAME** PA\_L3\_CREENCIA\_FILOSOFICA\_MORAL – Path

90% `creencias?_(filosoficas?|morales?|personales)` `convicci[oó]n(es)?(_?(filosoficas?|morales?|eticas?|personales))?` `objecion?(de_)?conciencia` `postura?(etica|moral)` `valores?`  
`personales` `filosofia?(de_)?vida` `cosmovision`

**HOW IT IS MASKED**

Suppressed to Sin información .

**PA\_CATEGORIA\_SUPRIMIDA** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR`.

`PA_BANDERA` **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

`PA_BANDERA_SI_NO` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

`PA_BANDERA_TRUE_FALSE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

`PA_BANDERA_S_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

`PA_BANDERA_Y_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

`PA_BANDERA_0_1` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_SUPRIMIR` **SECURE LOOKUP**

Replaces the value with one of the 1 values in `pa-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE                 | MASKED            |
|-----------------------|-------------------|
| Rechaza transfusiones | → Sin información |

9.19 Political opinion

`PA_L3_OPINION_POLITICA`

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

WHAT IT IS

Ideology, electoral preference, voting intention.

HOW IT IS FOUND

**COLUMN NAME** `PA_L3_OPINION_POLITICA` - Path

**90%** `opini[oó]n(es)?_pol[ií]ticas?` `ideolog[ií]a(_?politica)?` `posicion_?politica` `postura_?`  
`politica` `tendencia_?politica` `orientacion_?politica` `preferencia_?`  
`(politica|electoral|partidista)` `intencion_?(de_)?voto` `simpatia_?(politica|partidista)`  
`political_?(view|orientation|opinion)`

**KNOWN VALUES** `PA_L3_OPINION_POLITICA` - List

| LIST                                             | VALUES | STRENGTH | FIRST VALUES                                                                  |
|--------------------------------------------------|--------|----------|-------------------------------------------------------------------------------|
| <code>pa-detectar-opiniones-politicas.txt</code> | 14     | 70%      | Izquierda, Centro izquierda, Centro, Centro derecha, Derecha, Sin preferencia |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another position on a generic scale.

`PA_OPINION_POLITICA` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_OPINION_POLITICA_LISTA`.

`PA_BANDERA` **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

PA\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

PA\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

PA\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA\_OPINION\_POLITICA\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 7 values in `pa-opiniones-politicas.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE          | MASKED          |
|----------------|-----------------|
| Centro derecha | → <b>Centro</b> |

9.20 Political affiliation

PA\_L3\_AFILIACION\_POLITICA

**COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

The party a person is registered with or supports. The Electoral Tribunal counted more than 1.5 million party members in 2026.

HOW IT IS FOUND

| COLUMN NAME              | PA_L3_AFILIACION_POLITICA - Path                                                                                                                                                                                                                                           |          |                                                                                                                                                       |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 95%                      | partido(?:pol[í]tico)? militancia militante adherente inscrito?partido<br>afiliaci[oó]n?(politica partidista partido) afiliado?(a_)?partido padron?(de_)?<br>(militantes adherentes) political?party party?affiliation alianza?electoral<br>simpatizante libre?postulacion |          |                                                                                                                                                       |
| KNOWN VALUES             | PA_L3_AFILIACION_POLITICA - List                                                                                                                                                                                                                                           |          |                                                                                                                                                       |
| LIST                     | VALUES                                                                                                                                                                                                                                                                     | STRENGTH | FIRST VALUES                                                                                                                                          |
| pa-detectar-partidos.txt | 35                                                                                                                                                                                                                                                                         | 80%      | Partido Revolucionario Democrático, Partido Revolucionario Democratico, Realizando Metas, Cambio Democrático, Cambio Democratico, Partido Panameñista |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by one of the legally constituted parties — PRD, Realizando Metas, Cambio Democrático, Panameñista, MOLIRENA, MOCA, FAD, Alianza, Partido Popular — or by independent candidacy (*libre postulación*) or no party.

PA\_PARTIDO\_POLITICO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_PARTIDO\_POLITICO\_LISTA .

PA\_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

**PA\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

**PA\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

**PA\_BANDERA\_S\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_Y\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_0\_1** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

**PA\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PA\_PARTIDO\_POLITICO\_LISTA** **SECURE LOOKUP**

Replaces the value with one of the 11 values in `pa-partidos.txt` , chosen by a keyed hash.

**EXAMPLES**

**VALUE**

**MASKED**

Partido Liberal

→ Libre postulación

## 9.21 Union membership

PA\_L3\_AFILIACION\_SINDICAL

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

**WHAT IT IS**

Membership of a union, guild or workers' association, union dues, collective agreement.

**HOW IT IS FOUND**

**COLUMN NAME**

PA\_L3\_AFILIACION\_SINDICAL - Path

95%

`sindicato[a-z]*` `sindical` `sindicalizado` `afiliaci[oó]n?sindical` `cuota?sindical`  
`delegado?sindical` `seccional?sindical` `gremio` `gremial` `asociacion?(de?)?`  
`(docentes|educadores|profesores|empleados)` `union?member(ship)?` `trade?union` `convencion?`  
`colectiva`

**KNOWN VALUES**

PA\_L3\_AFILIACION\_SINDICAL - List

**LIST**

**VALUES**

**STRENGTH**

**FIRST VALUES**

`pa-detectar-sindicatos.txt`

15

60%

SUNTRACS, CONATO, CONUSI, CONATLAB, ASOPROF, A EVE

A value that matches none counts 30% against the domain.

**HOW IT IS MASKED**

Flags keep their shape; any other value becomes `Sin información` .

**PA\_CATEGORIA\_SUPRIMIDA** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR` .

**PA\_BANDERA** **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

**PA\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

**PA\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

**PA\_BANDERA\_S\_N SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_Y\_N SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

**PA\_BANDERA\_0\_1 SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

**PA\_CM\_ALFANUM CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PA\_SUPRIMIR SECURE LOOKUP**

Replaces the value with one of the 1 values in `pa-sin-informacion.txt` , chosen by a keyed hash.

**EXAMPLES**

**VALUE**

**MASKED**

SUNTRACS

→

**Sin información**

## Economic and credit data

The law does not call them sensitive, but it requires prior consent for economic, financial, banking and commercial data (art. 8, 3), protects data subjects from automated decisions about their credit (art. 19), and credit references have a law of their own, Law 24 of 2002. They go beyond the bank account: an APC credit reference, a pension amount, a MIDES program or a poverty status say as much about a person as their salary. Categories are replaced within their own notation; amounts are mapped into a plausible range unrelated to the real value.

### 10.1 Credit reference

PA\_FIN\_REFERENCIA\_CREDITO

#### COLUMN NAME

#### REGEX DECOMPOSE

#### WHAT IT IS

APC scores and references, days past due, delinquent portfolio, collection lawsuits, written-off accounts.

#### HOW IT IS FOUND

##### COLUMN NAME

PA\_FIN\_REFERENCIA\_CREDITO – Path

90%

```
apc referencia_(de_)?credito referencias_crediticias historial_(de_)?credito
historial_crediticio score(_?(apc|crediticio|credito|riesgo))? puntaje_
(crediticio|apc|riesgo|credito) calificacion_(crediticia|apc|riesgo) morosidad moroso
morosa dias_(de_)?(atraso|mora) cartera_(vencida|morosa) cobro_judicial
castigad[oa] estatus_(credito|crediticio|cartera) estado_(de_)?(cuenta_)?
(credito|crediticio|cartera) credit_score delinquen[a-z]*
```

#### HOW IT IS MASKED

A score gets other digits, a flag stays a flag and any status is replaced by another credit status.

##### PA\_APC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

##### PA\_REFERENCIA\_CREDITO .

##### PA\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

##### PA\_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PA\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

##### PA\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

##### PA\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

##### PA\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

##### PA\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

##### PA\_REFERENCIA\_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `pa-referencia-credito.txt`, chosen by a keyed hash.

## EXAMPLES

### VALUE

### MASKED

|                      |   |                      |
|----------------------|---|----------------------|
| Morosidad de 90 días | → | Morosidad de 60 días |
| 745                  | → | 471                  |

## 10.2 Debt amount

PA\_FIN\_MONTO\_DEUDA

### COLUMN NAME

### DATA TYPE

### NUMERIC MAPPING

### WHAT IT IS

Balances of loans, cards and mortgages, past-due amounts, credit limits, installments.

### HOW IT IS FOUND

#### COLUMN NAME

PA\_FIN\_MONTO\_DEUDA - Path

85%

monto\_?(deuda|adeudado|vencido|moroso|credito|prestamo|financiado|hipoteca) deuda\_(?  
(total|vigente|vencida))? adeudo saldo\_  
(deudor|insoluto|vencido|moroso|credito|prestamo|tarjeta|hipoteca) limite\_?(de\_)?credito  
linea\_?(de\_)?credito pago\_?minimo letra\_?(mensual|prestamo|hipoteca) debt(\_?amount)?  
outstanding\_?balance

#### DATA TYPE

PA\_FIN\_MONTO\_DEUDA - Type

Accepts number; any other type rules the domain out.

### HOW IT IS MASKED

Mapped to an amount between B/.100 and B/.500,000.

#### PA\_MONTO\_DEUDA

### NUMERIC MAPPING

Maps the number to another between 100 and 500,000.

## EXAMPLES

### VALUE

### MASKED

|      |   |      |
|------|---|------|
| 8500 | → | 5134 |
|------|---|------|

## 10.3 Salary and income

PA\_FIN\_INGRESO\_MENSUAL

### COLUMN NAME

### DATA TYPE

### NUMERIC MAPPING

### WHAT IT IS

Salary, thirteenth-month pay, household income.

### HOW IT IS FOUND

#### COLUMN NAME

PA\_FIN\_INGRESO\_MENSUAL - Path

85%

salario[a-z]\* sueldo[a-z]\* remuneraci[oó]n[a-z]\* ingreso\_  
(mensual|neto|bruto|familiar|del\_?hogar|hogar|per\_  
capita|total|promedio|declarado|laboral|anual) ingresos(\_?[a-z]+)? decimo\_?tercer\_?mes  
xiii\_?mes total\_?devengado neto\_?a\_?pagar planilla\_?(neta|bruta) gasto\_  
(mensual|familiar) capacidad\_?(de\_)?pago salary wage income gross\_?pay net\_?pay

**DATA TYPE** PA\_FIN\_INGRESO\_MENSUAL – Type

Accepts number; any other type rules the domain out.

#### HOW IT IS MASKED

Mapped to an amount between B/.350 and B/.25,000.

PA\_INGRESO\_MENSUAL **NUMERIC MAPPING**

Maps the number to another between 350 and 25,000.

#### EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 1850  | → | 1958   |

## 10.4 Assets

PA\_FIN\_PATRIMONIO

**COLUMN NAME** **DATA TYPE** **NUMERIC MAPPING**

#### WHAT IT IS

Cadastral or market value of real estate and vehicles, savings, term deposits, net worth.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_FIN\_PATRIMONIO – Path

85% patrimonio(\_?(neto|total))? valor\_  
(catastral|comercial|avaluo|inmueble|propiedad|finca|vivienda|vehiculo|de\_la\_garantia)  
avaluo bienes?(inmuebles|muebles) valor\_bienes activos?(personales|totales) saldo\_  
(inversion|ahorro|ahorros|cuenta|plazo\_fijo) monto?(inversion|ahorro|plazo\_fijo) net\_  
worth assets

**DATA TYPE** PA\_FIN\_PATRIMONIO – Type

Accepts number; any other type rules the domain out.

#### HOW IT IS MASKED

Mapped to an amount between B/.10,000 and B/.3,000,000.

PA\_PATRIMONIO **NUMERIC MAPPING**

Maps the number to another between 10,000 and 3,000,000.

#### EXAMPLES

| VALUE  |   | MASKED |
|--------|---|--------|
| 185000 | → | 477515 |

## 10.5 Pension and individual account

PA\_FIN\_PENSION

**COLUMN NAME** **DATA TYPE** **NUMERIC MAPPING**

#### WHAT IT IS

CSS retirement or pension amount, balance of the individual account, of SIACAP or of the severance fund.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_FIN\_PENSION – Path

**90%** monto\_?(pension|jubilacion) pension\_?(mensual|monto|vejez|invalides|sobreviviente)  
jubilacion\_?(mensual|monto) saldo\_?(cuenta\_?individual|siacap|retiro|fondo\_  
complementario) cuenta\_?individual siacap fondo\_?(de\_?)?cesantia prima\_?de\_?antiguedad  
retirement\_?balance

**DATA TYPE** PA\_FIN\_PENSION – Type

Accepts number; any other type rules the domain out.

#### HOW IT IS MASKED

Mapped to an amount between B/.120 and B/.6,000.

PA\_MONTO\_PENSION **NUMERIC MAPPING**

Maps the number to another between 120 and 6,000.

#### EXAMPLES

| VALUE | MASKED |
|-------|--------|
| 680   | → 227  |

## 10.6 Social security coverage

PA\_FIN\_SEGURO\_SOCIAL

**COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

#### WHAT IT IS

A person's standing with the Social Security Fund, in the census categories: direct insured, beneficiary, retired or pensioner, no social security. It reveals formal or informal work.

#### HOW IT IS FOUND

**COLUMN NAME** PA\_FIN\_SEGURO\_SOCIAL – Path

**90%** tipo\_?(de\_?)?(asegurado|seguro\_?social|cobertura) condicion\_?(de\_?)?asegurado cobertura\_  
(css|salud|seguro\_?social) afiliacion\_?(a\_?la\_?)?(css|caja\_?de\_?seguro\_?social|seguridad\_  
social) asegurado\_?css es\_?asegurado seguro\_?(medico|de\_?salud|privado) health\_  
insurance

**KNOWN VALUES** PA\_FIN\_SEGURO\_SOCIAL – List

| LIST                          | VALUES | STRENGTH | FIRST VALUES                                                                                                                                                                    |
|-------------------------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pa-detectar-seguro-social.txt | 20     | 80%      | Asegurado directo, Beneficiario, Jubilado o pensionado por vejez, Pensionado por enfermedad o accidente, Jubilado o pensionado de otro país, Jubilado o pensionado de otro país |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Flags and codes keep their shape; the category is replaced by another.

PA\_SEGURO\_SOCIAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_SEGURO\_SOCIAL\_LISTA .

PA\_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

`PA_BANDERA_TRUE_FALSE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

`PA_BANDERA_S_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

`PA_BANDERA_Y_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

`PA_BANDERA_0_1` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_SEGURO_SOCIAL_LISTA` **SECURE LOOKUP**

Replaces the value with one of the 6 values in `pa-seguro-social.txt` , chosen by a keyed hash.

#### EXAMPLES

VALUE

MASKED

Beneficiario

→ Jubilado o pensionado de otro país

## 10.7 Social program

PA\_FIN\_PROGRAMA\_SOCIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

#### WHAT IT IS

Beneficiaries of transfer programs: MIDES's 120 a los 65, Red de Oportunidades, Ángel Guardián and Bono Alimenticio Nutricional, IFARHU's PASE-U and scholarships. Being a beneficiary reveals poverty, old age or disability.

#### HOW IT IS FOUND

COLUMN NAME

PA\_FIN\_PROGRAMA\_SOCIAL - Path

90%

```
programa_?(social|de_?transferencia|de_?apoyo|gobierno|mides) programas_?sociales
beneficiario_?(de_?)?(programa|bono|beca|subsidio|mides) bono_?
(solidario|alimenticio|nutricional|digital) vale_?digital subsidio_?
(estatal|social|senadis) transferencia_?monetaria red_?(de_?)?oportunidades angel_?
guardian 120_?a_?los_?65 pase_?u beca_?universal ifarhu senapan padron_?(de_?)?
beneficiarios social_?program
```

KNOWN VALUES

PA\_FIN\_PROGRAMA\_SOCIAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-  
programas.txt

23

80%

120 a los 65, Red de Oportunidades, Ángel Guardián, Angel Guardian, Bono Alimenticio Nutricional, PASE-U

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Flags and codes keep their shape; the program is replaced by another.

`PA_PROGRAMA_SOCIAL` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_PROGRAMA_SOCIAL_LISTA` .

`PA_BANDERA` **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

`PA_BANDERA_SI_NO` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

`PA_BANDERA_TRUE_FALSE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

`PA_BANDERA_S_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

`PA_BANDERA_Y_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

`PA_BANDERA_0_1` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_PROGRAMA_SOCIAL_LISTA` **SECURE LOOKUP**

Replaces the value with one of the 9 values in `pa-programas-sociales.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Vale Digital

→ Auxilio económico IFARHU

10.8 Socioeconomic level

`PA_FIN_NIVEL_SOCIOECONOMICO`

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Poverty status, income quintile or decile, multidimensional poverty index.

HOW IT IS FOUND

COLUMN NAME

`PA_FIN_NIVEL_SOCIOECONOMICO` - Path

90%

`nivel_?socioeconomico nivel_?socio_?economico estrato(_?socioeconomico)? clase_?social  
decil(_?(de_?)?ingresos)? quintil(_?(de_?)?ingresos)? pobreza(_?  
(extrema|general|multidimensional))? indice_?(de_?)?pobreza ipm vulnerabilidad(_?  
(social|economica))? carencias? socioeconomic_?level ficha_?socioeconomica`

KNOWN VALUES

`PA_FIN_NIVEL_SOCIOECONOMICO` - List

LIST

VALUES

STRENGTH

FIRST VALUES

`pa-detectar-  
pobreza.txt`

10

40%

`pobreza extrema, pobreza general, pobre, no pobre, indigente,  
vulnerable`

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A quintile or decile becomes another number of the same range, with its prefix; a status, another from the list.

`PA_NIVEL_SOCIOECONOMICO` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_POBREZA`.

`PA_QUINTIL` **SECURE LOOKUP**

Replaces the value with one of the 5 values in `pa-quintil.txt`, chosen by a keyed hash.

`PA_DECIL_ALTO` **SECURE LOOKUP**

Replaces the value with one of the 5 values in `pa-decil-alto.txt`, chosen by a keyed hash.

#### PA\_DECIL **SECURE LOOKUP**

Replaces the value with one of the 10 values in `pa-decil.txt` , chosen by a keyed hash.

#### PA\_POBREZA **SECURE LOOKUP**

Replaces the value with one of the 4 values in `pa-pobreza.txt` , chosen by a keyed hash.

#### EXAMPLES

| VALUE           |   | MASKED                 |
|-----------------|---|------------------------|
| Pobreza extrema | → | <b>Pobreza general</b> |
| Quintil 2       | → | <b>Quintil 4</b>       |

## 10.9 Housing tenure

PA\_FIN\_VIVIENDA

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

#### WHAT IT IS

Whether the home is owned, mortgaged, rented, lent or an informal settlement, and its conditions.

#### HOW IT IS FOUND

#### COLUMN NAME

PA\_FIN\_VIVIENDA - Path

95%

```
tenencia_(de_)?(la_)?vivienda tipo_(de_)?vivienda situacion_(de_?la_)?vivienda
vivienda_(propia|alquilada|prestada) condicion_(de_)?(la_)?vivienda material_(de_)?
(la_)?vivienda piso_de_?tierra hacinamiento precarista invasion housing_?
(type|tenure|status)
```

#### KNOWN VALUES

PA\_FIN\_VIVIENDA - List

#### LIST

#### VALUES

#### STRENGTH

#### FIRST VALUES

|                          |    |     |                                                                                  |
|--------------------------|----|-----|----------------------------------------------------------------------------------|
| pa-detectar-vivienda.txt | 17 | 60% | Propia, Propia hipotecada, Alquilada, Cedida o prestada, Condena, Otra situación |
|--------------------------|----|-----|----------------------------------------------------------------------------------|

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Flags and codes keep their shape; the tenure is replaced by another from the census list.

#### PA\_VIVIENDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

#### PA\_VIVIENDA\_LISTA .

#### PA\_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

#### PA\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

#### PA\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

#### PA\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

#### PA\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

#### PA\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

#### PA\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PA\_VIVIENDA\_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `pa-vivienda.txt` , chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Invasión

→

**Propia hipotecada**

# Criminal records

Criminal convictions have their own rules in the law (art. 30) and its regulation (Decree 285, art. 7), and are among the data most likely to give rise to discrimination. They get the same care as sensitive data.

## 11.1 Criminal record

PA\_PENAL\_ANTECEDENTES

COLUMN NAME    REGEX DECOMPOSE

WHAT IT IS

Police record (*récord policivo*), criminal record, sentences, offences, detentions and precautionary measures.

HOW IT IS FOUND

COLUMN NAME    PA\_PENAL\_ANTECEDENTES – Path

95%

antecedentes(\_?(penales|judiciales|policivos|policiales|criminales))? *(except when followed by list E8)*  
record\_(policivo|policial|penal|criminal)    certificado\_(de\_)?antecedentes    sentencia(\_?  
condenatoria)?    sentenciad[oa]    condena(do|da)?    delitos?    tipo\_(de\_)?delito    reincidente  
reincidencia    imputad[oa]    privado\_de\_libertad    detencion\_provisional    medida\_cautelar  
centro\_penitenciario    pena    reclusion    libertad\_condicional    criminal\_record  
conviction    offen[cs]e    detenid[oa]    violencia\_domestica

HOW IT IS MASKED

A flag stays a flag; any other value becomes a generic status: with a record, without a record, or no information.

PA\_ANTECEDENTES    REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_ANTECEDENTES\_LISTA .

PA\_BANDERA    REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA\_BANDERA\_SI\_NO    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA\_BANDERA\_TRUE\_FALSE    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA\_BANDERA\_S\_N    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA\_BANDERA\_Y\_N    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA\_BANDERA\_0\_1    SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA\_CM\_ALFANUM    CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA\_ANTECEDENTES\_LISTA    SECURE LOOKUP

Replaces the value with one of the 3 values in pa-antecedentes.txt , chosen by a keyed hash.

EXAMPLES

| VALUE               |   | MASKED                   |
|---------------------|---|--------------------------|
| Condenado por hurto | → | Sin antecedentes penales |

11.2 Case number

PA\_PENAL\_CAUSA

| COLUMN NAME | VALUE FORMAT | CHARACTER MAPPING |
|-------------|--------------|-------------------|
|-------------|--------------|-------------------|

WHAT IT IS

Case file (*carpetilla*), investigation folder, case, court file or complaint numbers.

HOW IT IS FOUND

| COLUMN NAME   | PA_PENAL_CAUSA – Path                                                                                                                                                                                                                                                 |       |  |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--|
| 90%           | carpetilla n(o um ro)?carpetilla carpeta?(de_)?investigacion n(o um ro)?carpeta causa?penal n(o um ro umero)?(de_)?causa expediente?(judicial penal) n(o um ro)? expediente?(judicial penal) n(o um ro)?(de_)?denuncia folio?denuncia n(o um ro)? proceso case?number |       |  |
| VALUE FORMAT  | PA_PENAL_CAUSA – Regex                                                                                                                                                                                                                                                |       |  |
| PATTERN       | STRENGTH                                                                                                                                                                                                                                                              | NOTES |  |
| \d{12,15}     |                                                                                                                                                                                                                                                                       | 40%   |  |
| \d{1,6}–\d{4} |                                                                                                                                                                                                                                                                       | 50%   |  |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

| PA_CM_ALFANUM | CHARACTER MAPPING |
|---------------|-------------------|
|---------------|-------------------|

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE        | MASKED         |
|--------------|----------------|
| 202400012345 | → 917747298420 |

## The RUC check digit, computed by the set

A natural person's RUC check digit is two digits that the DGI computes over a 21-position table: the cédula fields right-aligned in fixed widths — asiento 5, tomo 3, a two-digit code for the letters ( AV 15, PI 79, E 50, N 40, PE 75), province 2 — and a constant 5. Weights grow from 2 at the right, modulo 11, and a remainder of 0 or 1 gives 0; the second digit repeats the computation with the first one appended. A field longer than its width pushes the fields to its left, and a one-digit province with AV or PI is padded with two zeros.

No framework of the plugin does this: Check Digit can add weighted digits, but not pad fields or weigh letters. The set does it with tables:

- The cédula is masked with the same algorithm as the cédula domain. Letters and separators are swapped for one-character markers, and the old DV is replaced by an initial state.
- The state — weighted sum, digit sum and position — travels through the value from right to left, one character per step. Each step is a Data Cleansing that turns *character + state* into *new state + character*.
- At the left end, a table turns the state into the two digits of the DV, and a second pass carries them back to where the old DV was.
- The markers become letters and separators again.

The tables are Data Cleansing files generated with the set, and the steps are Regex Decompose algorithms joined in String Algorithm Chains of at most eight — the engine's limit. Every masked RUC has a valid DV and the mapping stays one-to-one. The price is a long algorithm tree under the domain and a step table of more than 30,000 lines.

## Where people live: districts and corregimientos

With the data of the 2023 XII Population Census (INEC), the set knows the population, district, province and location of each of the 699 corregimientos.

- A **corregimiento** under 10,000 inhabitants becomes the nearest corregimiento of its province or comarca with at least 10,000. Darién, Los Santos and the Emberá-Wounaan and Ngäbe-Buglé comarcas have none: theirs become the nearest one in the country.
- A **district** under 20,000 inhabitants becomes the nearest district of its province or comarca with at least 20,000, or of the country in the Emberá-Wounaan comarca.
- Distance is measured between centroids, a district's weighted by the population of its corregimientos.
- Sixty corregimiento names repeat — there are five San José —: a name is generalized only when every place bearing it is small. INEC codes have no such ambiguity.

The result is always a real place, near the original, and every group of people who end up sharing a name holds at least 10,000 inhabitants. Names that are also given names or surnames are left out of the detection list, so that a surname column is not taken for corregimientos.

# Limits

- **Binary columns** — photos of the cédula, fingerprint images, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **A DV in its own column** is not recomputed: it is masked, and no longer matches the masked RUC. Where the application validates a RUC and DV stored separately, validation fails.
- **The cédula letters stay.** `PI` still marks a registration as an indigenous Panamanian, and `E`, `N` and `PE` the status of foreigner, naturalized or born abroad. Changing them would break the format or make different cédulas collide. If that is sensitive in your case, handle it together with the ethnic origin or nationality column.
- **Cédulas with a zero-padded province** ( `08-442-445` ) or without hyphens are masked character by character, without recomputing the DV. An `NT` RUC is also masked without recomputing.
- **A legal entity's RUC stays.** A company with a single owner is still a legal entity under the law.
- **Data Cleansing leaves unknown values as they are:** a corregimiento spelled in a way the list does not have passes unchanged.
- **Free text** is redacted token by token: a phone written with spaces is not found, and a given name that is also a common word ( `Luz` , `Rosa` ) is redacted wherever it appears.
- **Minors** — a birth date moves at most 60 days, so few people cross their 18th birthday, but some do. Where that matters, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, *barriada*, *corregimiento* and district in separate columns are masked independently and no longer form a coherent address.
- **Plates and VINs** keep their format, but not the VIN check digit.

---

15

## Verification

The set is generated and checked by two scripts kept next to it in the repository:

- `build.mjs` generates the set, its lookup tables and the DV tables from the definitions and the hand-kept lists.
- `verify.mjs` first confirms that its DV computation reproduces the examples of the published algorithm. It reviews every classifier configuration and profiles 138 columns with the whole set at its threshold — well named, badly named and 24 that are not personal data. It then masks 255 sample values and 210 placeholders such as `N/D` through the running app: RUCs with a DV in the four separators and the five letter types, checking each DV; 150 cédulas and 150 RUCs for collisions; and 20 people whose cédula and RUC must still agree.

In the last build, every column was assigned as expected and every value was masked as expected.

---

16

## Sources

- Law 81 of 26 March 2019, on Personal Data Protection — Gaceta Oficial 28743-A, 29 March 2019 (National Assembly, Legispan).
- Executive Decree 285 of 28 May 2021, regulating Law 81 of 2019.

- National Authority for Transparency and Access to Information (ANTAI), Personal Data Protection Directorate.
- Algorithm for the RUC check digit — Dirección General de Ingresos and the National Authority for Government Innovation.
- Format of the cédula and its letters — Electoral Tribunal, National Directorate of Identity Cards. Party membership, July and August 2026 — Electoral Tribunal.
- Population, code and location of the 699 corregimientos; indigenous and Afro-descendant groups; marital status; social security; schooling — XII National Population and VIII Housing Census 2023 (INEC), in the geographic service by corregimiento published with Esri Panamá.
- Most frequent surnames and given names — Civil Registry of the Electoral Tribunal.
- Certified disability types — National Secretariat for Disability (SENADIS). Transfer programs, 2026 — Ministry of Social Development (MIDES) and IFARHU (PASE-U).
- Credit references — Law 24 of 2002 and APC. Religion — 2015 national survey. Plates — Land Transit and Transport Authority (2013 format). Phone numbering — ASEP, National Numbering Plan. Postal codes — Correos y Telégrafos.

## APPENDIX A

# Algorithms

Every algorithm of the set in alphabetical order, with its configuration. The one a domain names is the top of its tree; the others are building blocks.

### PA\_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_L2\_ANIO\_NACIMIENTO

#### PATTERNS, TRIED IN ORDER

|             |                      |
|-------------|----------------------|
| (\d{3})(\d) | keep · PA_CM_ALFANUM |
| otherwise   | keep                 |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

### PA\_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_ANTECEDENTES\_LISTA .

Used by: PA\_PENAL\_ANTECEDENTES

#### PATTERNS, TRIED IN ORDER

|                                   |                       |
|-----------------------------------|-----------------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA            |
| (\d{1,6})                         | PA_CM_ALFANUM         |
| otherwise                         | PA_ANTECEDENTES_LISTA |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

### PA\_ANTECEDENTES\_LISTA SECURE LOOKUP

Replaces the value with one of the 3 values in pa-antecedentes.txt , chosen by a keyed hash.

Used by: PA\_ANTECEDENTES

#### CONFIGURATION

|                            |                     |
|----------------------------|---------------------|
| lookupFile                 | pa-antecedentes.txt |
| hashMethod                 | SHA256              |
| maskedValueCase            | PRESERVE_INPUT      |
| inputCaseSensitive         | no                  |
| trimWhitespaceFromInput    | yes                 |
| trimWhitespaceInLookupFile | yes                 |

### PA\_APC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_REFERENCIA\_CREDITO .

Used by: PA\_FIN\_REFERENCIA\_CREDITO

PATTERNS, TRIED IN ORDER

|                                   |                       |
|-----------------------------------|-----------------------|
| (\d{2,3})                         | PA_CM_ALFANUM         |
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA            |
| otherwise                         | PA_REFERENCIA_CREDITO |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_APELLIDO NAME

Replaces the name with one of the 200 names in `pa-apellidos.txt`.

Used by: `PA_APELLIDOS`, `PA_NOMBRE_COMPLETO`

CONFIGURATION

|                    |                  |
|--------------------|------------------|
| lookupFile         | pa-apellidos.txt |
| maskedValueCase    | PRESERVE_INPUT   |
| filterAccent       | yes              |
| inputCaseSensitive | no               |

PA\_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM`.

Used by: `PA_L1_APELLIDO`, `PA_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

|                                                           |                                                |
|-----------------------------------------------------------|------------------------------------------------|
| (\S+)                                                     | PA_APELLIDO                                    |
| (\S+)\s+(\S+)                                             | PA_APELLIDO · PA_APELLIDO                      |
| (\S+)(\s+)([Dd][Ee] [Vv][Dd][Aa]\.?\s+[Dd][Ee])(\s+)(\S+) | PA_APELLIDO · keep · keep · keep · PA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+)                                     | PA_APELLIDO · PA_APELLIDO · PA_APELLIDO        |
| otherwise                                                 | PA_CM_ALFANUM                                  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `keep`.

Used by: `PA_CATEGORIA_SUPRIMIDA`, `PA_PUEBLO`, `PA LENGUA_INDIGENA`, `PA_DIAGNOSTICO`, `PA_MEDICAMENTO`, `PA_DISCAPACIDAD`, `PA_ORIENTACION_SEXUAL`, `PA_IDENTIDAD_GENERO`, `PA_RELIGION`, `PA_OPINION_POLITICA`, `PA_PARTIDO_POLITICO`, `PA_APC`, `PA_SEGURO_SOCIAL`, `PA_PROGRAMA_SOCIAL`, `PA_VIVIENDA`, `PA_ANTECEDENTES`

PATTERNS, TRIED IN ORDER

|                  |                       |
|------------------|-----------------------|
| (?i)(s[ií] no)   | PA_BANDERA_SI_NO      |
| (?i)(true false) | PA_BANDERA_TRUE_FALSE |
| (?i)([sn])       | PA_BANDERA_S_N        |
| (?i)(y)          | PA_BANDERA_Y_N        |
| ([01])           | PA_BANDERA_0_1        |
| otherwise        | keep                  |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

#### CONFIGURATION

|                            |                                 |
|----------------------------|---------------------------------|
| lookupFile                 | <code>pa-bandera-0-1.txt</code> |
| hashMethod                 | SHA256                          |
| maskedValueCase            | PRESERVE_INPUT                  |
| inputCaseSensitive         | no                              |
| trimWhitespaceFromInput    | yes                             |
| trimWhitespaceInLookupFile | yes                             |

#### PA\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

#### CONFIGURATION

|                            |                                 |
|----------------------------|---------------------------------|
| lookupFile                 | <code>pa-bandera-s-n.txt</code> |
| hashMethod                 | SHA256                          |
| maskedValueCase            | PRESERVE_INPUT                  |
| inputCaseSensitive         | no                              |
| trimWhitespaceFromInput    | yes                             |
| trimWhitespaceInLookupFile | yes                             |

#### PA\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

#### CONFIGURATION

|                            |                                   |
|----------------------------|-----------------------------------|
| lookupFile                 | <code>pa-bandera-si-no.txt</code> |
| hashMethod                 | SHA256                            |
| maskedValueCase            | PRESERVE_INPUT                    |
| inputCaseSensitive         | no                                |
| trimWhitespaceFromInput    | yes                               |
| trimWhitespaceInLookupFile | yes                               |

#### PA\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

#### CONFIGURATION

|                 |                                        |
|-----------------|----------------------------------------|
| lookupFile      | <code>pa-bandera-true-false.txt</code> |
| hashMethod      | SHA256                                 |
| maskedValueCase | PRESERVE_INPUT                         |

|                            |     |
|----------------------------|-----|
| inputCaseSensitive         | no  |
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

#### PA\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

##### CONFIGURATION

|                            |                                 |
|----------------------------|---------------------------------|
| lookupFile                 | <code>pa-bandera-y-n.txt</code> |
| hashMethod                 | <code>SHA256</code>             |
| maskedValueCase            | <code>PRESERVE_INPUT</code>     |
| inputCaseSensitive         | no                              |
| trimWhitespaceFromInput    | yes                             |
| trimWhitespaceInLookupFile | yes                             |

#### PA\_BARRIADA SECURE LOOKUP

Replaces the value with one of the 75 values in `pa-barriadas.txt` , chosen by a keyed hash.

Used by: `PA_L2_BARRIADA`

##### CONFIGURATION

|                            |                                   |
|----------------------------|-----------------------------------|
| lookupFile                 | <code>pa-barriadas.txt</code>     |
| hashMethod                 | <code>SHA256</code>               |
| maskedValueCase            | <code>PRESERVE_LOOKUP_FILE</code> |
| inputCaseSensitive         | no                                |
| trimWhitespaceFromInput    | yes                               |
| trimWhitespaceInLookupFile | yes                               |

#### PA\_CANTIDAD\_ACOTADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PA_L2_DEPENDIENTES`

##### PATTERNS, TRIED IN ORDER

|                      |                             |
|----------------------|-----------------------------|
| <code>([0-5])</code> | keep                        |
| <code>(\d+)</code>   | replace with <code>5</code> |
| otherwise            | keep                        |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_CATEGORIA\_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR` .

Used by: `PA_L3_SALUD_SEXUAL_REPRODUCTIVA` , `PA_L3_SALUD_MENTAL_ADICCIONES` , `PA_L3_VIDA_SEXUAL` , `PA_L3_CREENCIA_FILOSOFICA_MORAL` , `PA_L3_AFILIACION_SINDICAL`

##### PATTERNS, TRIED IN ORDER

|                                                |                            |
|------------------------------------------------|----------------------------|
| <code>(?i)(s[ií] no true false [sny01])</code> | <code>PA_BANDERA</code>    |
| <code>(\d{1,6})</code>                         | <code>PA_CM_ALFANUM</code> |

|                      |             |
|----------------------|-------------|
| otherwise            | PA_SUPRIMIR |
| <b>CONFIGURATION</b> |             |
| trimInput            | yes         |
| requireMask          | no          |

**PA\_CEDULA** REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

Used by: PA\_L1\_CEDULA , PA\_RUC

**PATTERNS, TRIED IN ORDER**

|                                                                                    |                                                                          |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| ( [1-9]   1[0-3] ) ( [\\s\\-] ) ( \\d{1,4} ) ( [\\s\\-] ) ( \\d{1,6} )             | PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM        |
| ( [1-9]   1[0-3] ) ( AV   PI ) ( [\\s\\-] ) ( \\d{1,4} ) ( [\\s\\-] ) ( \\d{1,6} ) | PA_PROVINCIA_CEDULA · keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM |
| ( E ) ( [\\s\\-] ) ( [1-9]   1[0-3] ) ( [\\s\\-] ) ( \\d{1,6} )                    | keep · keep · PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM                 |
| ( N   PE ) ( [\\s\\-] ) ( \\d{1,4} ) ( [\\s\\-] ) ( \\d{1,6} )                     | keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM                       |
| ( [1-9]   1[0-3] ) ( [\\s\\-] NT [\\s\\-] ) ( \\d{1,4} ) ( [\\s\\-] ) ( \\d{1,6} ) | PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM        |
| otherwise                                                                          | PA_CM_ALFANUM                                                            |

|                      |     |
|----------------------|-----|
| <b>CONFIGURATION</b> |     |
| trimInput            | yes |
| requireMask          | no  |

**PA\_CIE10** SECURE LOOKUP

Replaces the value with one of the 37 values in pa-cie10.txt , chosen by a keyed hash.

Used by: PA\_DIAGNOSTICO

|                            |                      |
|----------------------------|----------------------|
| <b>CONFIGURATION</b>       |                      |
| lookupFile                 | pa-cie10.txt         |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

**PA\_CIE10\_DECIMAL** SECURE LOOKUP

Replaces the value with one of the 33 values in pa-cie10-decimal.txt , chosen by a keyed hash.

Used by: PA\_DIAGNOSTICO

|                            |                      |
|----------------------------|----------------------|
| <b>CONFIGURATION</b>       |                      |
| lookupFile                 | pa-cie10-decimal.txt |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

**PA\_CM\_ALFANUM** CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: PA\_L1\_PASAPORTE , PA\_L1\_DOCUMENTO\_MIGRATORIO , PA\_L1\_SEGURO\_SOCIAL , PA\_L1\_IDONEIDAD , PA\_L1\_DIRECCION\_COMPLEMENTO , PA\_L1\_CUENTA\_BANCARIA , PA\_L1\_VIN , PA\_L1\_FINCA , PA\_L1\_NUMERO\_CONTRATO , PA\_L1\_USUARIO\_RRSS , PA\_L3\_SALUD\_IDENTIFICADOR , PA\_L3\_SALUD\_RESULTADO\_EXAMEN , PA\_PENAL\_CAUSA , PA\_CATEGORIA\_SUPRIMIDA , PA\_CEDULA , PA\_RUC\_DV\_PREPARAR , PA\_RUC , PA\_DV , PA\_NOMBRES , PA\_APELLIDOS , PA\_NOMBRE\_COMPLETO , PA\_EMAIL , PA\_TELEFONO , PA\_PLACA , PA\_COORDENADA , PA\_ANIO , PA\_EDAD , PA\_CODIGO\_POSTAL , PA\_PUEBLO , PA LENGUA\_INDIGENA , PA\_MEDICAMENTO , PA\_DISCAPACIDAD , PA\_ORIENTACION\_SEXUAL , PA\_IDENTIDAD\_GENERO , PA\_RELIGION , PA\_OPINION\_POLITICA , PA\_PARTIDO\_POLITICO , PA\_APC , PA\_SEGURO\_SOCIAL , PA\_PROGRAMA\_SOCIAL , PA\_VIVIENDA , PA\_ANTECEDENTES

**CONFIGURATION**

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| characterGroups    | 0123456789 AEIOU BCDFGHJKLMNPQRSTUVWXYZ aeiou<br>bcdfghjklmnpqrstvwxyz ÁÉÍÓÚÚ áéíóúú |
| caseSensitive      | yes                                                                                  |
| minMaskedPositions | 0                                                                                    |

**PA\_CM\_HEX** CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: PA\_L1\_DISPOSITIVO , PA\_IP

**CONFIGURATION**

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| characterGroups    | 0123456789 abcdef ABCDEF ghijklmnopqrstuvwxyz<br>GHIJKLMNOPQRSTUVWXYZ |
| caseSensitive      | yes                                                                   |
| minMaskedPositions | 0                                                                     |

**PA\_CODIGO\_CORREGIMIENTO** DATA CLEANSING

Replaces each value found in pa-codigo-corregimiento-generalizado.txt (1,331 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_CODIGO\_LUGAR

**CONFIGURATION**

|                |                                          |
|----------------|------------------------------------------|
| lookupFile     | pa-codigo-corregimiento-generalizado.txt |
| delimiter      | ,                                        |
| caseSensitive  | yes                                      |
| trimWhitespace | yes                                      |

**PA\_CODIGO\_DISTRITO** DATA CLEANSING

Replaces each value found in pa-codigo-distrito-generalizado.txt (148 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_CODIGO\_LUGAR

**CONFIGURATION**

|                |                                     |
|----------------|-------------------------------------|
| lookupFile     | pa-codigo-distrito-generalizado.txt |
| delimiter      | ,                                   |
| caseSensitive  | yes                                 |
| trimWhitespace | yes                                 |

**PA\_CODIGO\_LUGAR** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA\_L2\_CODIGO\_LUGAR

PATTERNS, TRIED IN ORDER

|           |                         |
|-----------|-------------------------|
| (\d{5,6}) | PA_CODIGO_CORREGIMIENTO |
| (\d{3,4}) | PA_CODIGO_DISTRITO      |
| otherwise | keep                    |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_CODIGO\_POSTAL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

Used by: PA\_L2\_CODIGO\_POSTAL

PATTERNS, TRIED IN ORDER

|                  |                           |
|------------------|---------------------------|
| (\d{2})(\d{2,3}) | keep · each character → 0 |
| otherwise        | PA_CM_ALFANUM             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA\_L1\_GEOLOCALIZACION

PATTERNS, TRIED IN ORDER

|                                                 |                                             |
|-------------------------------------------------|---------------------------------------------|
| (-?\d{1,3}\.\d)(\d+)\s*,\s*(-?\d{1,3}\.\d)(\d+) | keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM |
| (-?\d{1,3}\.\d)(\d+)                            | keep · PA_CM_ALFANUM                        |
| otherwise                                       | keep                                        |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_CORREGIMIENTO DATA CLEANSING

Replaces each value found in pa-corregimiento-generalizado.txt (1,760 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_L2\_CORREGIMIENTO

CONFIGURATION

|                |                                   |
|----------------|-----------------------------------|
| lookupFile     | pa-corregimiento-generalizado.txt |
| delimiter      |                                   |
| caseSensitive  | yes                               |
| trimWhitespace | yes                               |

PA\_DECIL SECURE LOOKUP

Replaces the value with one of the 10 values in pa-decil.txt , chosen by a keyed hash.

Used by: PA\_NIVEL\_SOCIOECONOMICO

**CONFIGURATION**

|                            |                |
|----------------------------|----------------|
| lookupFile                 | pa-decil.txt   |
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PA\_DECIL\_ALTO** SECURE LOOKUP

Replaces the value with one of the 5 values in pa-decil-alto.txt , chosen by a keyed hash.

Used by: PA\_NIVEL\_SOCIOECONOMICO

**CONFIGURATION**

|                            |                   |
|----------------------------|-------------------|
| lookupFile                 | pa-decil-alto.txt |
| hashMethod                 | SHA256            |
| maskedValueCase            | PRESERVE_INPUT    |
| inputCaseSensitive         | no                |
| trimWhitespaceFromInput    | yes               |
| trimWhitespaceInLookupFile | yes               |

**PA\_DIAGNOSTICO** REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

PA\_DIAGNOSTICO\_TEXTO .

Used by: PA\_L3\_SALUD\_DIAGNOSTICO

**PATTERNS, TRIED IN ORDER**

|                                   |                      |
|-----------------------------------|----------------------|
| ([A-TV-Za-tv-z]\d{2}\.\d{1,2})    | PA_CIE10_DECIMAL     |
| ([A-TV-Za-tv-z]\d{2,3})           | PA_CIE10             |
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA           |
| otherwise                         | PA_DIAGNOSTICO_TEXTO |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_DIAGNOSTICO\_TEXTO** SECURE LOOKUP

Replaces the value with one of the 30 values in pa-diagnostics.txt , chosen by a keyed hash.

Used by: PA\_DIAGNOSTICO

**CONFIGURATION**

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pa-diagnostics.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

**PA\_DIRECCION** SECURE LOOKUP

Replaces the value with one of the 3,000 values in `pa-direcciones.txt` , chosen by a keyed hash.

Used by: `PA_L1_DIRECCION`

CONFIGURATION

|                            |                      |
|----------------------------|----------------------|
| lookupFile                 | pa-direcciones.txt   |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

**PA\_DISCAPACIDAD** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_DISCAPACIDAD_LISTA` .

Used by: `PA_L3_SALUD_DISCAPACIDAD`

PATTERNS, TRIED IN ORDER

|                                                |                                    |
|------------------------------------------------|------------------------------------|
| <code>(?i)(s[ií] no true false [sny01])</code> | <code>PA_BANDERA</code>            |
| <code>(\d{1,6})</code>                         | <code>PA_CM_ALFANUM</code>         |
| otherwise                                      | <code>PA_DISCAPACIDAD_LISTA</code> |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_DISCAPACIDAD\_LISTA** SECURE LOOKUP

Replaces the value with one of the 8 values in `pa-discapacidad.txt` , chosen by a keyed hash.

Used by: `PA_DISCAPACIDAD`

CONFIGURATION

|                            |                     |
|----------------------------|---------------------|
| lookupFile                 | pa-discapacidad.txt |
| hashMethod                 | SHA256              |
| maskedValueCase            | PRESERVE_INPUT      |
| inputCaseSensitive         | no                  |
| trimWhitespaceFromInput    | yes                 |
| trimWhitespaceInLookupFile | yes                 |

**PA\_DISTRITO** DATA CLEANSING

Replaces each value found in `pa-distrito-generalizado.txt` (234 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PA_L2_DISTRITO`

CONFIGURATION

|                |                              |
|----------------|------------------------------|
| lookupFile     | pa-distrito-generalizado.txt |
| delimiter      |                              |
| caseSensitive  | yes                          |
| trimWhitespace | yes                          |

**PA\_DV** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

Used by: PA\_L1\_RUC\_DV

PATTERNS, TRIED IN ORDER

|           |               |
|-----------|---------------|
| (\d{1,2}) | PA_CM_ALFANUM |
| otherwise | PA_CM_ALFANUM |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_EDAD** REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA\_L2\_EDAD

PATTERNS, TRIED IN ORDER

|                    |                             |
|--------------------|-----------------------------|
| (\d{3,})           | replace with 90             |
| ([1-9])(\d)(\.\d+) | keep · PA_CM_ALFANUM · keep |
| ([1-9])(\d)        | keep · PA_CM_ALFANUM        |
| (\d)               | PA_CM_ALFANUM               |
| otherwise          | keep                        |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_EMAIL** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

Used by: PA\_L1\_EMAIL

PATTERNS, TRIED IN ORDER

|                                       |                                                   |
|---------------------------------------|---------------------------------------------------|
| ([^\s@]+)@([^\s@]+\.\.[A-Za-z]{2,24}) | PA_CM_ALFANUM · PA_CM_ALFANUM · replace with test |
| otherwise                             | PA_CM_ALFANUM                                     |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_EMPLEADOR** SECURE LOOKUP

Replaces the value with one of the 25 values in pa-empleadores.txt , chosen by a keyed hash.

Used by: PA\_L2\_EMPLEADOR

CONFIGURATION

|                         |                    |
|-------------------------|--------------------|
| lookupFile              | pa-empleadores.txt |
| hashMethod              | SHA256             |
| maskedValueCase         | PRESERVE_INPUT     |
| inputCaseSensitive      | no                 |
| trimWhitespaceFromInput | yes                |

|                            |     |
|----------------------------|-----|
| trimWhitespaceInLookupFile | yes |
|----------------------------|-----|

#### PA\_ESCOLARIDAD SECURE LOOKUP

Replaces the value with one of the 13 values in `pa-escolaridad.txt` , chosen by a keyed hash.

Used by: `PA_L2_ESCOLARIDAD`

##### CONFIGURATION

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pa-escolaridad.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

#### PA\_ESCUELA SECURE LOOKUP

Replaces the value with one of the 15 values in `pa-escuelas.txt` , chosen by a keyed hash.

Used by: `PA_L2_ESCUELA`

##### CONFIGURATION

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pa-escuelas.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

#### PA\_ESTADO\_CIVIL SECURE LOOKUP

Replaces the value with one of the 7 values in `pa-estado-civil.txt` , chosen by a keyed hash.

Used by: `PA_L2_ESTADO_CIVIL`

##### CONFIGURATION

|                            |                     |
|----------------------------|---------------------|
| lookupFile                 | pa-estado-civil.txt |
| hashMethod                 | SHA256              |
| maskedValueCase            | PRESERVE_INPUT      |
| inputCaseSensitive         | no                  |
| trimWhitespaceFromInput    | yes                 |
| trimWhitespaceInLookupFile | yes                 |

#### PA\_FECHA\_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: `PA_L2_FECHA_EVENTO`

##### CONFIGURATION

|          |      |
|----------|------|
| minRange | -30  |
| maxRange | 30   |
| unit     | DAYS |
| roll     | no   |

**PA\_FECHA\_NACIMIENTO** DATE SHIFT

Moves the date by -60 to 60 days.

Used by: PA\_L2\_FECHA\_NACIMIENTO

CONFIGURATION

|          |      |
|----------|------|
| minRange | -60  |
| maxRange | 60   |
| unit     | DAYS |
| roll     | no   |

**PA\_GRUPO\_SANGUINEO** SECURE LOOKUP

Replaces the value with one of the 8 values in pa-grupo-sanguineo.txt , chosen by a keyed hash.

Used by: PA\_L3\_SALUD\_GRUPO\_SANGUINEO

CONFIGURATION

|                            |                        |
|----------------------------|------------------------|
| lookupFile                 | pa-grupo-sanguineo.txt |
| hashMethod                 | SHA256                 |
| maskedValueCase            | PRESERVE_LOOKUP_FILE   |
| inputCaseSensitive         | no                     |
| trimWhitespaceFromInput    | yes                    |
| trimWhitespaceInLookupFile | yes                    |

**PA\_IDENTIDAD\_GENERO** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_IDENTIDAD\_GENERO\_LISTA .

Used by: PA\_L3\_IDENTIDAD\_GENERO

PATTERNS, TRIED IN ORDER

|                                   |                           |
|-----------------------------------|---------------------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA                |
| (\d{1,6})                         | PA_CM_ALFANUM             |
| otherwise                         | PA_IDENTIDAD_GENERO_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_IDENTIDAD\_GENERO\_LISTA** SECURE LOOKUP

Replaces the value with one of the 8 values in pa-identidad-genero.txt , chosen by a keyed hash.

Used by: PA\_IDENTIDAD\_GENERO

CONFIGURATION

|                            |                         |
|----------------------------|-------------------------|
| lookupFile                 | pa-identidad-genero.txt |
| hashMethod                 | SHA256                  |
| maskedValueCase            | PRESERVE_INPUT          |
| inputCaseSensitive         | no                      |
| trimWhitespaceFromInput    | yes                     |
| trimWhitespaceInLookupFile | yes                     |

**PA\_INGRESO\_MENSUAL** NUMERIC MAPPING

Maps the number to another between 350 and 25,000.

Used by: PA\_FIN\_INGRESO\_MENSUAL

CONFIGURATION

|          |       |
|----------|-------|
| minValue | 350   |
| maxValue | 25000 |

**PA\_IP** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA\_CM\_HEX .

Used by: PA\_L1\_IP

PATTERNS, TRIED IN ORDER

|                                               |                                               |
|-----------------------------------------------|-----------------------------------------------|
| (\d{1,3})\.\(\d{1,3})\.\(\d{1,3})\.\(\d{1,3}) | PA_OCTETO · PA_OCTETO · PA_OCTETO · PA_OCTETO |
| otherwise                                     | PA_CM_HEX                                     |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA LENGUA INDIGENA** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA LENGUA INDIGENA LISTA .

Used by: PA\_L3 LENGUA INDIGENA

PATTERNS, TRIED IN ORDER

|                                   |                          |
|-----------------------------------|--------------------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA               |
| (\d{1,6})                         | PA_CM_ALFANUM            |
| otherwise                         | PA LENGUA INDIGENA LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA LENGUA INDIGENA LISTA** SECURE LOOKUP

Replaces the value with one of the 9 values in pa-lenguas.txt , chosen by a keyed hash.

Used by: PA LENGUA INDIGENA

CONFIGURATION

|                            |                |
|----------------------------|----------------|
| lookupFile                 | pa-lenguas.txt |
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PA\_LUGAR\_POBLADO** SECURE LOOKUP

Replaces the value with one of the 52 values in pa-lugares-poblados.txt , chosen by a keyed hash.

Used by: PA\_L2\_LUGAR\_POBLADO

#### CONFIGURATION

|                            |                         |
|----------------------------|-------------------------|
| lookupFile                 | pa-lugares-poblados.txt |
| hashMethod                 | SHA256                  |
| maskedValueCase            | PRESERVE_LOOKUP_FILE    |
| inputCaseSensitive         | no                      |
| trimWhitespaceFromInput    | yes                     |
| trimWhitespaceInLookupFile | yes                     |

#### PA\_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_MEDICAMENTO\_LISTA .

Used by: PA\_L3\_SALUD\_MEDICAMENTO

#### PATTERNS, TRIED IN ORDER

|                                   |                      |
|-----------------------------------|----------------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA           |
| (\d{1,6})                         | PA_CM_ALFANUM        |
| otherwise                         | PA_MEDICAMENTO_LISTA |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_MEDICAMENTO\_LISTA SECURE LOOKUP

Replaces the value with one of the 26 values in pa-medicamentos.txt , chosen by a keyed hash.

Used by: PA\_MEDICAMENTO

#### CONFIGURATION

|                            |                     |
|----------------------------|---------------------|
| lookupFile                 | pa-medicamentos.txt |
| hashMethod                 | SHA256              |
| maskedValueCase            | PRESERVE_INPUT      |
| inputCaseSensitive         | no                  |
| trimWhitespaceFromInput    | yes                 |
| trimWhitespaceInLookupFile | yes                 |

#### PA\_MONTO\_DEUDA NUMERIC MAPPING

Maps the number to another between 100 and 500,000.

Used by: PA\_FIN\_MONTO\_DEUDA

#### CONFIGURATION

|          |        |
|----------|--------|
| minValue | 100    |
| maxValue | 500000 |

#### PA\_MONTO\_PENSION NUMERIC MAPPING

Maps the number to another between 120 and 6,000.

Used by: PA\_FIN\_PENSION

#### CONFIGURATION

|          |      |
|----------|------|
| minValue | 120  |
| maxValue | 6000 |

**PA\_NACIONALIDAD** SECURE LOOKUP

Replaces the value with one of the 20 values in `pa-nacionalidades.txt` , chosen by a keyed hash.

Used by: `PA_L2_NACIONALIDAD`

**CONFIGURATION**

|                            |                       |
|----------------------------|-----------------------|
| lookupFile                 | pa-nacionalidades.txt |
| hashMethod                 | SHA256                |
| maskedValueCase            | PRESERVE_INPUT        |
| inputCaseSensitive         | no                    |
| trimWhitespaceFromInput    | yes                   |
| trimWhitespaceInLookupFile | yes                   |

**PA\_NIVEL\_SOCIOECONOMICO** REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_POBREZA` .

Used by: `PA_FIN_NIVEL_SOCIOECONOMICO`

**PATTERNS, TRIED IN ORDER**

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| <code>( [1-5] )</code>                            | <code>PA_QUINTIL</code>        |
| <code>( [6-9]   10 )</code>                       | <code>PA_DECIL_ALTO</code>     |
| <code>( ?i ) ( decil \s* ) ( \d { 1, 2 } )</code> | <code>keep · PA_DECIL</code>   |
| <code>( ?i ) ( quintil \s* ) ( [ 1-5 ] )</code>   | <code>keep · PA_QUINTIL</code> |
| otherwise                                         | <code>PA_POBREZA</code>        |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_NOMBRE\_COMPLETO** REGEX DECOMPOSE

Tries 8 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

Used by: `PA_L1_NOMBRE_COMPLETO`

**PATTERNS, TRIED IN ORDER**

|                                                                                                                                              |                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>( [ ^ , ] + ) , \s* ( . + )</code>                                                                                                     | <code>PA_APELLIDOS · PA_NOMBRES</code>                                                                      |
| <code>( \S+ )</code>                                                                                                                         | <code>PA_NOMBRE_PILA</code>                                                                                 |
| <code>( \S+ ) \s+ ( \S+ )</code>                                                                                                             | <code>PA_NOMBRE_PILA · PA_APELLIDO</code>                                                                   |
| <code>( \S+ ) ( \S+ ) ( \S+ ) ( \S+ ) ( [ Dd ] [ Ee ]   [ Vv ] [ Dd ] [ Aa ] \. ? \s+ [ Dd ] [ Ee ] ) ( \S+ ) ( \S+ )</code>                 | <code>PA_NOMBRE_PILA · keep · PA_APELLIDO · keep · keep · keep · PA_APELLIDO</code>                         |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code>                                                                                                 | <code>PA_NOMBRE_PILA · PA_APELLIDO · PA_APELLIDO</code>                                                     |
| <code>( \S+ ) ( \S+ ) ( \S+ ) ( \S+ ) ( \S+ ) ( \S+ ) ( [ Dd ] [ Ee ]   [ Vv ] [ Dd ] [ Aa ] \. ? \s+ [ Dd ] [ Ee ] ) ( \S+ ) ( \S+ )</code> | <code>PA_NOMBRE_PILA · keep · PA_NOMBRE_PILA · keep · PA_APELLIDO · keep · keep · keep · PA_APELLIDO</code> |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code>                                                                                     | <code>PA_NOMBRE_PILA · PA_NOMBRE_PILA · PA_APELLIDO · PA_APELLIDO</code>                                    |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code>                                                                         | <code>PA_NOMBRE_PILA · PA_NOMBRE_PILA · PA_NOMBRE_PILA · PA_APELLIDO · PA_APELLIDO</code>                   |
| otherwise                                                                                                                                    | <code>PA_CM_ALFANUM</code>                                                                                  |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_NOMBRE\_PILA** NAME

Replaces the name with one of the 253 names in `pa-nombres.txt` .

Used by: `PA_NOMBRES` , `PA_NOMBRE_COMPLETO`

**CONFIGURATION**

|                    |                             |
|--------------------|-----------------------------|
| lookupFile         | <code>pa-nombres.txt</code> |
| maskedValueCase    | <code>PRESERVE_INPUT</code> |
| filterAccent       | <code>yes</code>            |
| inputCaseSensitive | <code>no</code>             |

**PA\_NOMBRES** REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

Used by: `PA_L1_NOMBRE` , `PA_NOMBRE_COMPLETO`

**PATTERNS, TRIED IN ORDER**

|                                    |                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------|
| <code>(\S+)</code>                 | <code>PA_NOMBRE_PILA</code>                                                             |
| <code>(\S+)\s+(\S+)</code>         | <code>PA_NOMBRE_PILA</code> · <code>PA_NOMBRE_PILA</code>                               |
| <code>(\S+)\s+(\S+)\s+(\S+)</code> | <code>PA_NOMBRE_PILA</code> · <code>PA_NOMBRE_PILA</code> · <code>PA_NOMBRE_PILA</code> |
| otherwise                          | <code>PA_CM_ALFANUM</code>                                                              |

**CONFIGURATION**

|             |                  |
|-------------|------------------|
| trimInput   | <code>yes</code> |
| requireMask | <code>no</code>  |

**PA\_OCTETO** NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: `PA_IP`

**CONFIGURATION**

|          |                  |
|----------|------------------|
| minValue | <code>1</code>   |
| maxValue | <code>254</code> |

**PA\_OCUPACION** SECURE LOOKUP

Replaces the value with one of the 40 values in `pa-ocupaciones.txt` , chosen by a keyed hash.

Used by: `PA_L2_OCUPACION`

**CONFIGURATION**

|                            |                                 |
|----------------------------|---------------------------------|
| lookupFile                 | <code>pa-ocupaciones.txt</code> |
| hashMethod                 | <code>SHA256</code>             |
| maskedValueCase            | <code>PRESERVE_INPUT</code>     |
| inputCaseSensitive         | <code>no</code>                 |
| trimWhitespaceFromInput    | <code>yes</code>                |
| trimWhitespaceInLookupFile | <code>yes</code>                |

**PA\_OPINION\_POLITICA** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_OPINION_POLITICA_LISTA` .

Used by: `PA_L3_OPINION_POLITICA`

PATTERNS, TRIED IN ORDER

|                                   |                           |
|-----------------------------------|---------------------------|
| (?i)(s[ii] no true false [sny01]) | PA_BANDERA                |
| (\d{1,6})                         | PA_CM_ALFANUM             |
| otherwise                         | PA_OPINION_POLITICA_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_OPINION\_POLITICA\_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in pa-opiniones-politicas.txt , chosen by a keyed hash.

Used by: PA\_OPINION\_POLITICA

CONFIGURATION

|                            |                            |
|----------------------------|----------------------------|
| lookupFile                 | pa-opiniones-politicas.txt |
| hashMethod                 | SHA256                     |
| maskedValueCase            | PRESERVE_INPUT             |
| inputCaseSensitive         | no                         |
| trimWhitespaceFromInput    | yes                        |
| trimWhitespaceInLookupFile | yes                        |

PA\_ORIENTACION\_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_ORIENTACION\_SEXUAL\_LISTA .

Used by: PA\_L3\_ORIENTACION\_SEXUAL

PATTERNS, TRIED IN ORDER

|                                   |                             |
|-----------------------------------|-----------------------------|
| (?i)(s[ii] no true false [sny01]) | PA_BANDERA                  |
| (\d{1,6})                         | PA_CM_ALFANUM               |
| otherwise                         | PA_ORIENTACION_SEXUAL_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_ORIENTACION\_SEXUAL\_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in pa-orientacion-sexual.txt , chosen by a keyed hash.

Used by: PA\_ORIENTACION\_SEXUAL

CONFIGURATION

|                            |                           |
|----------------------------|---------------------------|
| lookupFile                 | pa-orientacion-sexual.txt |
| hashMethod                 | SHA256                    |
| maskedValueCase            | PRESERVE_INPUT            |
| inputCaseSensitive         | no                        |
| trimWhitespaceFromInput    | yes                       |
| trimWhitespaceInLookupFile | yes                       |

PA\_PARTIDO\_POLITICO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_PARTIDO\_POLITICO\_LISTA .

Used by: PA\_L3\_AFILIACION\_POLITICA

PATTERNS, TRIED IN ORDER

|                                   |                           |
|-----------------------------------|---------------------------|
| (?i)(s[ii] no true false [sny01]) | PA_BANDERA                |
| (\d{1,6})                         | PA_CM_ALFANUM             |
| otherwise                         | PA_PARTIDO_POLITICO_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_PARTIDO\_POLITICO\_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in pa-partidos.txt , chosen by a keyed hash.

Used by: PA\_PARTIDO\_POLITICO

CONFIGURATION

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pa-partidos.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

PA\_PATRIMONIO NUMERIC MAPPING

Maps the number to another between 10,000 and 3,000,000.

Used by: PA\_FIN\_PATRIMONIO

CONFIGURATION

|          |         |
|----------|---------|
| minValue | 10000   |
| maxValue | 3000000 |

PA\_PLACA REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

Used by: PA\_L1\_PLACA

PATTERNS, TRIED IN ORDER

|                                             |                                      |
|---------------------------------------------|--------------------------------------|
| (ADM MB MI PR BC CD CC M T B G D E)([\s-]?) | keep · keep · PA_CM_ALFANUM          |
| (\d{3,6})                                   |                                      |
| ([A-Z]{2})([\s-]?)(\d{4})                   | PA_CM_ALFANUM · keep · PA_CM_ALFANUM |
| (\d{6})                                     | PA_CM_ALFANUM                        |
| otherwise                                   | PA_CM_ALFANUM                        |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

### PA\_POBREZA SECURE LOOKUP

Replaces the value with one of the 4 values in `pa-pobreza.txt` , chosen by a keyed hash.

Used by: `PA_NIVEL_SOCIOECONOMICO`

#### CONFIGURATION

|                            |                      |
|----------------------------|----------------------|
| lookupFile                 | pa-pobreza.txt       |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

### PA\_PROGRAMA\_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_PROGRAMA_SOCIAL_LISTA` .

Used by: `PA_FIN_PROGRAMA_SOCIAL`

#### PATTERNS, TRIED IN ORDER

|                                                |                                       |
|------------------------------------------------|---------------------------------------|
| <code>(?i)(s[ií] no true false [sny01])</code> | <code>PA_BANDERA</code>               |
| <code>(\d{1,6})</code>                         | <code>PA_CM_ALFANUM</code>            |
| otherwise                                      | <code>PA_PROGRAMA_SOCIAL_LISTA</code> |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

### PA\_PROGRAMA\_SOCIAL\_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `pa-programas-sociales.txt` , chosen by a keyed hash.

Used by: `PA_PROGRAMA_SOCIAL`

#### CONFIGURATION

|                            |                           |
|----------------------------|---------------------------|
| lookupFile                 | pa-programas-sociales.txt |
| hashMethod                 | SHA256                    |
| maskedValueCase            | PRESERVE_INPUT            |
| inputCaseSensitive         | no                        |
| trimWhitespaceFromInput    | yes                       |
| trimWhitespaceInLookupFile | yes                       |

### PA\_PROVINCIA\_CEDULA NUMERIC MAPPING

Maps the number to another between 1 and 13.

Used by: `PA_CEDULA` , `PA_RUC_DV_PREPARAR` , `PA_RUC`

#### CONFIGURATION

|          |    |
|----------|----|
| minValue | 1  |
| maxValue | 13 |

### PA\_PUEBLO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_PUEBLO_LISTA` .

Used by: `PA_L3_ETNIA`

#### PATTERNS, TRIED IN ORDER

|                                   |                 |
|-----------------------------------|-----------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA      |
| (\d{1,6})                         | PA_CM_ALFANUM   |
| otherwise                         | PA_PUEBLO_LISTA |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_PUEBLO\_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in `pa-pueblos.txt`, chosen by a keyed hash.

Used by: `PA_PUEBLO`

#### CONFIGURATION

|                            |                |
|----------------------------|----------------|
| lookupFile                 | pa-pueblos.txt |
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

#### PA\_QUINTIL SECURE LOOKUP

Replaces the value with one of the 5 values in `pa-quintil.txt`, chosen by a keyed hash.

Used by: `PA_NIVEL_SOCIOECONOMICO`

#### CONFIGURATION

|                            |                |
|----------------------------|----------------|
| lookupFile                 | pa-quintil.txt |
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

#### PA\_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PA_L1_CREDENCIAL`, `PA_L3_BIOMETRICO`, `PA_L3_INFORMACION_GENETICA`

#### PATTERNS, TRIED IN ORDER

|           |                    |
|-----------|--------------------|
| (.+)      | each character → X |
| otherwise | keep               |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_REFERENCIA\_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `pa-referencia-credito.txt`, chosen by a keyed hash.

Used by: `PA_APC`

CONFIGURATION

|                            |                           |
|----------------------------|---------------------------|
| lookupFile                 | pa-referencia-credito.txt |
| hashMethod                 | SHA256                    |
| maskedValueCase            | PRESERVE_INPUT            |
| inputCaseSensitive         | no                        |
| trimWhitespaceFromInput    | yes                       |
| trimWhitespaceInLookupFile | yes                       |

PA\_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA\_RELIGION\_LISTA .

Used by: PA\_L3\_RELIGION

PATTERNS, TRIED IN ORDER

|                                   |                   |
|-----------------------------------|-------------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA        |
| (\d{1,6})                         | PA_CM_ALFANUM     |
| otherwise                         | PA_RELIGION_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RELIGION\_LISTA SECURE LOOKUP

Replaces the value with one of the 14 values in pa-religiones.txt , chosen by a keyed hash.

Used by: PA\_RELIGION

CONFIGURATION

|                            |                   |
|----------------------------|-------------------|
| lookupFile                 | pa-religiones.txt |
| hashMethod                 | SHA256            |
| maskedValueCase            | PRESERVE_INPUT    |
| inputCaseSensitive         | no                |
| trimWhitespaceFromInput    | yes               |
| trimWhitespaceInLookupFile | yes               |

PA\_RUC REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PA\_CM\_ALFANUM .

Used by: PA\_L1\_RUC

PATTERNS, TRIED IN ORDER

|                                                                                                      |                                                                                   |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| ((?:([1-9] 1[0-3])(?:AV PI)?-\d{1,4}-\d{1,6} (?:E N PE)-\d{1,4}-\d{1,6}))(?: +DV +  +DV  -  +)\d{2}) | PA_RUC_DV                                                                         |
| ((?:([1-9] 1[0-3])(?:AV PI)?-\d{1,4}-\d{1,6} (?:E N PE)-\d{1,4}-\d{1,6}))                            | PA_CEDULA                                                                         |
| ([1-9] 1[0-3])(-NT-)(\d{1,4})(-)(\d{1,6})(.*)                                                        | PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM · PA_CM_ALFANUM |
| (\d{3,10}-\d{1,4}-\d{1,7}.*)                                                                         | keep                                                                              |
| otherwise                                                                                            | PA_CM_ALFANUM                                                                     |

CONFIGURATION

|           |     |
|-----------|-----|
| trimInput | yes |
|-----------|-----|

|             |    |
|-------------|----|
| requireMask | no |
|-------------|----|

**PA\_RUC\_DV** STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_1 → PA\_RUC\_DV\_2 → PA\_RUC\_DV\_3 → PA\_RUC\_DV\_4 → PA\_RUC\_DV\_5 , in this order.

Used by: PA\_RUC

**CONFIGURATION**

|                        |             |
|------------------------|-------------|
| algorithmReferences[1] | PA_RUC_DV_1 |
| algorithmReferences[2] | PA_RUC_DV_2 |
| algorithmReferences[3] | PA_RUC_DV_3 |
| algorithmReferences[4] | PA_RUC_DV_4 |
| algorithmReferences[5] | PA_RUC_DV_5 |

**PA\_RUC\_DV\_1** STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_PREPARAR → PA\_RUC\_DV\_IDA\_01 → PA\_RUC\_DV\_IDA\_02 → PA\_RUC\_DV\_IDA\_03 → PA\_RUC\_DV\_IDA\_04 → PA\_RUC\_DV\_IDA\_05 → PA\_RUC\_DV\_IDA\_06 → PA\_RUC\_DV\_IDA\_07 , in this order.

Used by: PA\_RUC\_DV

**CONFIGURATION**

|                        |                    |
|------------------------|--------------------|
| algorithmReferences[1] | PA_RUC_DV_PREPARAR |
| algorithmReferences[2] | PA_RUC_DV_IDA_01   |
| algorithmReferences[3] | PA_RUC_DV_IDA_02   |
| algorithmReferences[4] | PA_RUC_DV_IDA_03   |
| algorithmReferences[5] | PA_RUC_DV_IDA_04   |
| algorithmReferences[6] | PA_RUC_DV_IDA_05   |
| algorithmReferences[7] | PA_RUC_DV_IDA_06   |
| algorithmReferences[8] | PA_RUC_DV_IDA_07   |

**PA\_RUC\_DV\_2** STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_IDA\_08 → PA\_RUC\_DV\_IDA\_09 → PA\_RUC\_DV\_IDA\_10 → PA\_RUC\_DV\_IDA\_11 → PA\_RUC\_DV\_IDA\_12 → PA\_RUC\_DV\_IDA\_13 → PA\_RUC\_DV\_IDA\_14 → PA\_RUC\_DV\_IDA\_15 , in this order.

Used by: PA\_RUC\_DV

**CONFIGURATION**

|                        |                  |
|------------------------|------------------|
| algorithmReferences[1] | PA_RUC_DV_IDA_08 |
| algorithmReferences[2] | PA_RUC_DV_IDA_09 |
| algorithmReferences[3] | PA_RUC_DV_IDA_10 |
| algorithmReferences[4] | PA_RUC_DV_IDA_11 |
| algorithmReferences[5] | PA_RUC_DV_IDA_12 |
| algorithmReferences[6] | PA_RUC_DV_IDA_13 |
| algorithmReferences[7] | PA_RUC_DV_IDA_14 |
| algorithmReferences[8] | PA_RUC_DV_IDA_15 |

**PA\_RUC\_DV\_3** STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_CALCULAR → PA\_RUC\_DV\_VUELTA\_01 → PA\_RUC\_DV\_VUELTA\_02 → PA\_RUC\_DV\_VUELTA\_03 → PA\_RUC\_DV\_VUELTA\_04 → PA\_RUC\_DV\_VUELTA\_05 → PA\_RUC\_DV\_VUELTA\_06 → PA\_RUC\_DV\_VUELTA\_07 , in this order.

Used by: PA\_RUC\_DV

CONFIGURATION

|                        |                     |
|------------------------|---------------------|
| algorithmReferences[1] | PA_RUC_DV_CALCULAR  |
| algorithmReferences[2] | PA_RUC_DV_VUELTA_01 |
| algorithmReferences[3] | PA_RUC_DV_VUELTA_02 |
| algorithmReferences[4] | PA_RUC_DV_VUELTA_03 |
| algorithmReferences[5] | PA_RUC_DV_VUELTA_04 |
| algorithmReferences[6] | PA_RUC_DV_VUELTA_05 |
| algorithmReferences[7] | PA_RUC_DV_VUELTA_06 |
| algorithmReferences[8] | PA_RUC_DV_VUELTA_07 |

PA\_RUC\_DV\_4 STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_VUELTA\_08 → PA\_RUC\_DV\_VUELTA\_09 → PA\_RUC\_DV\_VUELTA\_10 → PA\_RUC\_DV\_VUELTA\_11 → PA\_RUC\_DV\_VUELTA\_12 → PA\_RUC\_DV\_VUELTA\_13 → PA\_RUC\_DV\_VUELTA\_14 → PA\_RUC\_DV\_VUELTA\_15 , in this order.

Used by: PA\_RUC\_DV

CONFIGURATION

|                        |                     |
|------------------------|---------------------|
| algorithmReferences[1] | PA_RUC_DV_VUELTA_08 |
| algorithmReferences[2] | PA_RUC_DV_VUELTA_09 |
| algorithmReferences[3] | PA_RUC_DV_VUELTA_10 |
| algorithmReferences[4] | PA_RUC_DV_VUELTA_11 |
| algorithmReferences[5] | PA_RUC_DV_VUELTA_12 |
| algorithmReferences[6] | PA_RUC_DV_VUELTA_13 |
| algorithmReferences[7] | PA_RUC_DV_VUELTA_14 |
| algorithmReferences[8] | PA_RUC_DV_VUELTA_15 |

PA\_RUC\_DV\_5 STRING ALGORITHM CHAIN

Applies PA\_RUC\_DV\_VUELTA\_16 → PA\_RUC\_DV\_RESTAURAR , in this order.

Used by: PA\_RUC\_DV

CONFIGURATION

|                        |                     |
|------------------------|---------------------|
| algorithmReferences[1] | PA_RUC_DV_VUELTA_16 |
| algorithmReferences[2] | PA_RUC_DV_RESTAURAR |

PA\_RUC\_DV\_CALCULAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

PATTERNS, TRIED IN ORDER

|           |                            |
|-----------|----------------------------|
| (.{3}).*) | PA_RUC_DV_RESULTADO · keep |
| otherwise | keep                       |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_01 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

PATTERNS, TRIED IN ORDER

|                |                              |
|----------------|------------------------------|
| (.*)({4})({1}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise      | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_02 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

PATTERNS, TRIED IN ORDER

|                |                              |
|----------------|------------------------------|
| (.*)({4})({2}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise      | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_03 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

PATTERNS, TRIED IN ORDER

|                |                              |
|----------------|------------------------------|
| (.*)({4})({3}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise      | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_04 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

PATTERNS, TRIED IN ORDER

|                |                              |
|----------------|------------------------------|
| (.*)({4})({4}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise      | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_05 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

PATTERNS, TRIED IN ORDER

|                |                              |
|----------------|------------------------------|
| (.*)({4})({5}) | keep · PA_RUC_DV_PAS0 · keep |
|----------------|------------------------------|

|                      |      |
|----------------------|------|
| otherwise            | keep |
| <b>CONFIGURATION</b> |      |
| trimInput            | yes  |
| requireMask          | no   |

**PA\_RUC\_DV\_IDA\_06** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

|                                 |                              |
|---------------------------------|------------------------------|
| <b>PATTERNS, TRIED IN ORDER</b> |                              |
| (.*)({4})({6})                  | keep · PA_RUC_DV_PAS0 · keep |
| otherwise                       | keep                         |
| <b>CONFIGURATION</b>            |                              |
| trimInput                       | yes                          |
| requireMask                     | no                           |

**PA\_RUC\_DV\_IDA\_07** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_1

|                                 |                              |
|---------------------------------|------------------------------|
| <b>PATTERNS, TRIED IN ORDER</b> |                              |
| (.*)({4})({7})                  | keep · PA_RUC_DV_PAS0 · keep |
| otherwise                       | keep                         |
| <b>CONFIGURATION</b>            |                              |
| trimInput                       | yes                          |
| requireMask                     | no                           |

**PA\_RUC\_DV\_IDA\_08** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

|                                 |                              |
|---------------------------------|------------------------------|
| <b>PATTERNS, TRIED IN ORDER</b> |                              |
| (.*)({4})({8})                  | keep · PA_RUC_DV_PAS0 · keep |
| otherwise                       | keep                         |
| <b>CONFIGURATION</b>            |                              |
| trimInput                       | yes                          |
| requireMask                     | no                           |

**PA\_RUC\_DV\_IDA\_09** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

|                                 |                              |
|---------------------------------|------------------------------|
| <b>PATTERNS, TRIED IN ORDER</b> |                              |
| (.*)({4})({9})                  | keep · PA_RUC_DV_PAS0 · keep |
| otherwise                       | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_10 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

PATTERNS, TRIED IN ORDER

|                 |                              |
|-----------------|------------------------------|
| (.*)({4})({10}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise       | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_11 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

PATTERNS, TRIED IN ORDER

|                 |                              |
|-----------------|------------------------------|
| (.*)({4})({11}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise       | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_12 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

PATTERNS, TRIED IN ORDER

|                 |                              |
|-----------------|------------------------------|
| (.*)({4})({12}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise       | keep                         |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_IDA\_13 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

PATTERNS, TRIED IN ORDER

|                 |                              |
|-----------------|------------------------------|
| (.*)({4})({13}) | keep · PA_RUC_DV_PAS0 · keep |
| otherwise       | keep                         |

CONFIGURATION

|           |     |
|-----------|-----|
| trimInput | yes |
|-----------|-----|

|             |    |
|-------------|----|
| requireMask | no |
|-------------|----|

#### PA\_RUC\_DV\_IDA\_14 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

##### PATTERNS, TRIED IN ORDER

|                   |                              |
|-------------------|------------------------------|
| (.*)((.{4})的前14}) | keep · PA_RUC_DV_PASO · keep |
| otherwise         | keep                         |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_RUC\_DV\_IDA\_15 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_2

##### PATTERNS, TRIED IN ORDER

|                   |                              |
|-------------------|------------------------------|
| (.*)((.{4})的前15}) | keep · PA_RUC_DV_PASO · keep |
| otherwise         | keep                         |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PA\_RUC\_DV\_LETRAS DATA CLEANSING

Replaces each value found in pa-ruc-dv-letras.txt (5 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_RUC\_DV\_PREPARAR

##### CONFIGURATION

|                |                      |
|----------------|----------------------|
| lookupFile     | pa-ruc-dv-letras.txt |
| delimiter      | ,                    |
| caseSensitive  | yes                  |
| trimWhitespace | yes                  |

#### PA\_RUC\_DV\_LETRAS\_DE\_VUELTA DATA CLEANSING

Replaces each value found in pa-ruc-dv-letras-de-vuelta.txt (5 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_RUC\_DV\_RESTAURAR

##### CONFIGURATION

|                |                                |
|----------------|--------------------------------|
| lookupFile     | pa-ruc-dv-letras-de-vuelta.txt |
| delimiter      | ,                              |
| caseSensitive  | yes                            |
| trimWhitespace | yes                            |

## PA\_RUC\_DV\_PASO DATA CLEANSING

Replaces each value found in `pa-ruc-dv-paso.txt` (32,317 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PA_RUC_DV_IDA_01`, `PA_RUC_DV_IDA_02`, `PA_RUC_DV_IDA_03`, `PA_RUC_DV_IDA_04`, `PA_RUC_DV_IDA_05`, `PA_RUC_DV_IDA_06`, `PA_RUC_DV_IDA_07`, `PA_RUC_DV_IDA_08`, `PA_RUC_DV_IDA_09`, `PA_RUC_DV_IDA_10`, `PA_RUC_DV_IDA_11`, `PA_RUC_DV_IDA_12`, `PA_RUC_DV_IDA_13`, `PA_RUC_DV_IDA_14`, `PA_RUC_DV_IDA_15`

### CONFIGURATION

|                |                    |
|----------------|--------------------|
| lookupFile     | pa-ruc-dv-paso.txt |
| delimiter      | ,                  |
| caseSensitive  | yes                |
| trimWhitespace | yes                |

## PA\_RUC\_DV\_PREPARAR REGEX DECOMPOSE

Tries 16 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PA_RUC_DV_1`

### PATTERNS, TRIED IN ORDER

|                                                                           |                                                                                                                                                                                                            |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})( +DV +)(\d{2})</code>        | <code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove                                  |
| <code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})( +DV +)(\d{2})</code> | <code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove  |
| <code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})( +DV +)(\d{2})</code>              | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove                               |
| <code>(N PE)(-)(\d{1,4})(-)(\d{1,6})( +DV +)(\d{2})</code>                | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove                                     |
| <code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})( +DV)(\d{2})</code>          | <code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove                                  |
| <code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})( +DV)(\d{2})</code>   | <code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove  |
| <code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})( +DV)(\d{2})</code>                | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove                               |
| <code>(N PE)(-)(\d{1,4})(-)(\d{1,6})( +DV)(\d{2})</code>                  | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove                                     |
| <code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})(-)(\d{2})</code>             | <code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove                                 |
| <code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})(-)(\d{2})</code>      | <code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove |
| <code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})(-)(\d{2})</code>                   | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove                              |
| <code>(N PE)(-)(\d{1,4})(-)(\d{1,6})(-)(\d{2})</code>                     | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove                                    |
| <code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})( +)(\d{2})</code>            | <code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove                                  |
| <code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})( +)(\d{2})</code>     | <code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove  |
| <code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})( +)(\d{2})</code>                  | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove                               |
| <code>(N PE)(-)(\d{1,4})(-)(\d{1,6})( +)(\d{2})</code>                    | <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove                                     |
| otherwise                                                                 | keep                                                                                                                                                                                                       |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_RESTAURAR REGEX DECOMPOSE

Tries 12 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_5

PATTERNS, TRIED IN ORDER

|                                                   |                                                                                                  |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------|
| (\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(!)(\d\d)        | keep · keep · keep · replace with - · keep · replace with DV · keep                              |
| (\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(!)(\d\d)  | keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep |
| ([enq])(-)(\d{1,4})(_)(\d{1,6})(!)(\d\d)          | PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep        |
| (\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(#)(\d\d)        | keep · keep · keep · replace with - · keep · replace with DV · keep                              |
| (\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(#)(\d\d)  | keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep |
| ([enq])(-)(\d{1,4})(_)(\d{1,6})(#)(\d\d)          | PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep        |
| (\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(\$)(\d\d)       | keep · keep · keep · replace with - · keep · replace with - · keep                               |
| (\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(\$)(\d\d) | keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with - · keep  |
| ([enq])(-)(\d{1,4})(_)(\d{1,6})(\$)(\d\d)         | PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with - · keep         |
| (\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(%)(\d\d)        | keep · keep · keep · replace with - · keep · replace with · keep                                 |
| (\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(%)(\d\d)  | keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with · keep    |
| ([enq])(-)(\d{1,4})(_)(\d{1,6})(%)(\d\d)          | PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with · keep           |
| otherwise                                         | keep                                                                                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_RESULTADO DATA CLEANSING

Replaces each value found in pa-ruc-dv-resultado.txt (1,815 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_RUC\_DV\_CALCULAR

CONFIGURATION

|                |                         |
|----------------|-------------------------|
| lookupFile     | pa-ruc-dv-resultado.txt |
| delimiter      | ,                       |
| caseSensitive  | yes                     |
| trimWhitespace | yes                     |

PA\_RUC\_DV TRASLADO DATA CLEANSING

Replaces each value found in pa-ruc-dv-traslado.txt (2,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA\_RUC\_DV\_VUELTA\_01 , PA\_RUC\_DV\_VUELTA\_02 , PA\_RUC\_DV\_VUELTA\_03 , PA\_RUC\_DV\_VUELTA\_04 , PA\_RUC\_DV\_VUELTA\_05 , PA\_RUC\_DV\_VUELTA\_06 , PA\_RUC\_DV\_VUELTA\_07 , PA\_RUC\_DV\_VUELTA\_08 , PA\_RUC\_DV\_VUELTA\_09 , PA\_RUC\_DV\_VUELTA\_10 ,

PA\_RUC\_DV\_VUELTA\_11 , PA\_RUC\_DV\_VUELTA\_12 , PA\_RUC\_DV\_VUELTA\_13 , PA\_RUC\_DV\_VUELTA\_14 , PA\_RUC\_DV\_VUELTA\_15 , PA\_RUC\_DV\_VUELTA\_16

CONFIGURATION

|                |                        |
|----------------|------------------------|
| lookupFile     | pa-ruc-dv-traslado.txt |
| delimiter      | ,                      |
| caseSensitive  | yes                    |
| trimWhitespace | yes                    |

PA\_RUC\_DV\_VUELTA\_01 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

PATTERNS, TRIED IN ORDER

|                  |                                  |
|------------------|----------------------------------|
| (.{0})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise        | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_02 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

PATTERNS, TRIED IN ORDER

|                  |                                  |
|------------------|----------------------------------|
| (.{1})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise        | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_03 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

PATTERNS, TRIED IN ORDER

|                  |                                  |
|------------------|----------------------------------|
| (.{2})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise        | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_04 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

PATTERNS, TRIED IN ORDER

|                  |                                  |
|------------------|----------------------------------|
| (.{3})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
|------------------|----------------------------------|

|                      |      |
|----------------------|------|
| otherwise            | keep |
| <b>CONFIGURATION</b> |      |
| trimInput            | yes  |
| requireMask          | no   |

**PA\_RUC\_DV\_VUELTA\_05** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

**PATTERNS, TRIED IN ORDER**

|                               |                                  |
|-------------------------------|----------------------------------|
| <code>(.{4})(.{3})(.*)</code> | keep · PA_RUC_DV TRASLADO · keep |
| otherwise                     | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_RUC\_DV\_VUELTA\_06** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

**PATTERNS, TRIED IN ORDER**

|                               |                                  |
|-------------------------------|----------------------------------|
| <code>(.{5})(.{3})(.*)</code> | keep · PA_RUC_DV TRASLADO · keep |
| otherwise                     | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_RUC\_DV\_VUELTA\_07** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_3

**PATTERNS, TRIED IN ORDER**

|                               |                                  |
|-------------------------------|----------------------------------|
| <code>(.{6})(.{3})(.*)</code> | keep · PA_RUC_DV TRASLADO · keep |
| otherwise                     | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_RUC\_DV\_VUELTA\_08** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

**PATTERNS, TRIED IN ORDER**

|                               |                                  |
|-------------------------------|----------------------------------|
| <code>(.{7})(.{3})(.*)</code> | keep · PA_RUC_DV TRASLADO · keep |
| otherwise                     | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_09 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

PATTERNS, TRIED IN ORDER

|                  |                                  |
|------------------|----------------------------------|
| (.{8})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise        | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_10 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

PATTERNS, TRIED IN ORDER

|                  |                                  |
|------------------|----------------------------------|
| (.{9})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise        | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_11 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

PATTERNS, TRIED IN ORDER

|                   |                                  |
|-------------------|----------------------------------|
| (.{10})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise         | keep                             |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PA\_RUC\_DV\_VUELTA\_12 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

PATTERNS, TRIED IN ORDER

|                   |                                  |
|-------------------|----------------------------------|
| (.{11})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise         | keep                             |

CONFIGURATION

|           |     |
|-----------|-----|
| trimInput | yes |
|-----------|-----|

|             |    |
|-------------|----|
| requireMask | no |
|-------------|----|

**PA\_RUC\_DV\_VUELTA\_13** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

**PATTERNS, TRIED IN ORDER**

|                   |                                  |
|-------------------|----------------------------------|
| (.{12})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise         | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_RUC\_DV\_VUELTA\_14** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

**PATTERNS, TRIED IN ORDER**

|                   |                                  |
|-------------------|----------------------------------|
| (.{13})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise         | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_RUC\_DV\_VUELTA\_15** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_4

**PATTERNS, TRIED IN ORDER**

|                   |                                  |
|-------------------|----------------------------------|
| (.{14})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise         | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_RUC\_DV\_VUELTA\_16** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA\_RUC\_DV\_5

**PATTERNS, TRIED IN ORDER**

|                   |                                  |
|-------------------|----------------------------------|
| (.{15})(.{3})(.*) | keep · PA_RUC_DV TRASLADO · keep |
| otherwise         | keep                             |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_SEGURO\_SOCIAL** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA\_SEGURO\_SOCIAL\_LISTA .

Used by: PA\_FIN\_SEGURO\_SOCIAL

PATTERNS, TRIED IN ORDER

|                                   |                        |
|-----------------------------------|------------------------|
| (?i)(s[ii] no true false [sny01]) | PA_BANDERA             |
| (\d{1,2})                         | PA_CM_ALFANUM          |
| otherwise                         | PA_SEGURO_SOCIAL_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_SEGURO\_SOCIAL\_LISTA** SECURE LOOKUP

Replaces the value with one of the 6 values in pa-seguro-social.txt , chosen by a keyed hash.

Used by: PA\_SEGURO\_SOCIAL

CONFIGURATION

|                            |                      |
|----------------------------|----------------------|
| lookupFile                 | pa-seguro-social.txt |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

**PA\_SEXO** REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA\_L2\_SEX0

PATTERNS, TRIED IN ORDER

|                          |                      |
|--------------------------|----------------------|
| (?i)(femenino masculino) | PA_SEX0_PALABRA      |
| (?i)(mujer hombre)       | PA_SEX0_MUJER_HOMBRE |
| (?i)(female male)        | PA_SEX0_INGLES       |
| (?i)([fm])               | PA_SEX0_F_M          |
| (?i)([hm])               | PA_SEX0_H_M          |
| otherwise                | keep                 |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_SEX0\_F\_M** SECURE LOOKUP

Replaces the value with one of the 2 values in pa-sexo-f-m.txt , chosen by a keyed hash.

Used by: PA\_SEX0

CONFIGURATION

|            |                 |
|------------|-----------------|
| lookupFile | pa-sexo-f-m.txt |
| hashMethod | SHA256          |

|                            |                |
|----------------------------|----------------|
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PA\_SEXO\_H\_M** SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-h-m.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

**CONFIGURATION**

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pa-sexo-h-m.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

**PA\_SEXO\_INGLES** SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-ingles.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

**CONFIGURATION**

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pa-sexo-ingles.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

**PA\_SEXO\_MUJER\_HOMBRE** SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-mujer-hombre.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

**CONFIGURATION**

|                            |                          |
|----------------------------|--------------------------|
| lookupFile                 | pa-sexo-mujer-hombre.txt |
| hashMethod                 | SHA256                   |
| maskedValueCase            | PRESERVE_INPUT           |
| inputCaseSensitive         | no                       |
| trimWhitespaceFromInput    | yes                      |
| trimWhitespaceInLookupFile | yes                      |

**PA\_SEXO\_PALABRA** SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-palabra.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

**CONFIGURATION**

|            |                     |
|------------|---------------------|
| lookupFile | pa-sexo-palabra.txt |
| hashMethod | SHA256              |

|                            |                |
|----------------------------|----------------|
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PA\_SUPRIMIR** SECURE LOOKUP

Replaces the value with one of the 1 values in `pa-sin-informacion.txt` , chosen by a keyed hash.

Used by: `PA_CATEGORIA_SUPRIMIDA`

**CONFIGURATION**

|                            |                        |
|----------------------------|------------------------|
| lookupFile                 | pa-sin-informacion.txt |
| hashMethod                 | SHA256                 |
| maskedValueCase            | PRESERVE_LOOKUP_FILE   |
| inputCaseSensitive         | no                     |
| trimWhitespaceFromInput    | yes                    |
| trimWhitespaceInLookupFile | yes                    |

**PA\_TARJETA** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PA_L1_TARJETA`

**PATTERNS, TRIED IN ORDER**

|                                    |                              |
|------------------------------------|------------------------------|
| <code>(\d[\d\s\~]{11,22}\d)</code> | <code>PA_TARJETA_LUHN</code> |
| otherwise                          | keep                         |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PA\_TARJETA\_LUHN** PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: `PA_TARJETA`

**CONFIGURATION**

|                    |   |
|--------------------|---|
| minMaskedPositions | 6 |
| preserve           | 0 |

**PA\_TELEFONO** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

Used by: `PA_L1_TELEFONO`

**PATTERNS, TRIED IN ORDER**

|                                                                       |                                                                 |
|-----------------------------------------------------------------------|-----------------------------------------------------------------|
| <code>(\+?(?07\)?[\s\~]?)?(6)(\d{3})([\s\~]?)<br/>(\d{4})</code>      | <code>keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM</code> |
| <code>(\+?(?07\)?[\s\~]?)?([2-9])(\d{2})<br/>([\s\~]?) (\d{4})</code> | <code>keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM</code> |
| otherwise                                                             | <code>PA_CM_ALFANUM</code>                                      |

**CONFIGURATION**

|           |     |
|-----------|-----|
| trimInput | yes |
|-----------|-----|

|             |    |
|-------------|----|
| requireMask | no |
|-------------|----|

### PA\_TEXTO\_LIBRE FREE TEXT REDACTION

Replaces the text matching 7 patterns with [DATO] , and the 1,094 terms of pa-texto-nombres.txt with [NOMBRE] .

Used by: PA\_L1\_TEXTO\_LIBRE , PA\_L3\_SALUD\_TEXTO\_CLINICO

#### PATTERNS, TRIED IN ORDER

|                                                                          |
|--------------------------------------------------------------------------|
| (?<![\w\~])(( [1-9]   1[0-3] )(AV PI)? E N PE)-\d{1,4}-\d{1,6}(?![\w\~]) |
| [\w.+~]+@[~\w\~]+(\. [\w\~]+)+                                           |
| (?<!\d)(\+?507)?6\d{3}-?\d{4}(?! \d)                                     |
| (?<![\d\~]) [2-9]\d{2}-\d{4}(?! [\d\~])                                  |
| (?<!\d)\d{13,19}(?! \d)                                                  |
| (?<![A-Za-z]) [A-Z]{2}\d{4}(?! [A-Za-z\d])                               |
| (?<!\d)(\d{1,3}\.){3}\d{1,3}(?! \d)                                      |

#### CONFIGURATION

|                       |                      |
|-----------------------|----------------------|
| regExRedactValue      | [DATO]               |
| lookupFile            | pa-texto-nombres.txt |
| lookupFileRedactValue | [NOMBRE]             |
| isDenyList            | yes                  |

### PA\_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA\_VIVIENDA\_LISTA .

Used by: PA\_FIN\_VIVIENDA

#### PATTERNS, TRIED IN ORDER

|                                   |                   |
|-----------------------------------|-------------------|
| (?i)(s[ií] no true false [sny01]) | PA_BANDERA        |
| (\d{1,6})                         | PA_CM_ALFANUM     |
| otherwise                         | PA_VIVIENDA_LISTA |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

### PA\_VIVIENDA\_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in pa-vivienda.txt , chosen by a keyed hash.

Used by: PA\_VIVIENDA

#### CONFIGURATION

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pa-vivienda.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

APPENDIX B

Files

Lookup and detection files travel with the set: loading it copies them into the files folder, and sending it to Delphix uploads them.

| FILE                                             | LINES | USED BY                                                     | FIRST LINES                                                                                                |
|--------------------------------------------------|-------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| pa-<br>antecedentes.txt                          | 3     | PA_ANTECEDENTES_LIST<br>A                                   | Sin antecedentes penales · Con antecedentes penales · Sin información                                      |
| pa-apellidos.txt                                 | 200   | PA_APELLIDO                                                 | Rodríguez · González · Sánchez · Martínez                                                                  |
| pa-bandera-0-<br>1.txt                           | 2     | PA_BANDERA_0_1                                              | 0 · 1                                                                                                      |
| pa-bandera-s-<br>n.txt                           | 2     | PA_BANDERA_S_N                                              | S · N                                                                                                      |
| pa-bandera-si-<br>no.txt                         | 2     | PA_BANDERA_SI_NO                                            | Sí · No                                                                                                    |
| pa-bandera-true-<br>false.txt                    | 2     | PA_BANDERA_TRUE_FALS<br>E                                   | true · false                                                                                               |
| pa-bandera-y-<br>n.txt                           | 2     | PA_BANDERA_Y_N                                              | Y · N                                                                                                      |
| pa-barriadas.txt                                 | 75    | PA_BARRIADA                                                 | Villa Lucre · Condado del Rey · Brisas del Golf · Altos de Panamá                                          |
| pa-cie10-<br>decimal.txt                         | 33    | PA_CIE10_DECIMAL                                            | E11.9 · E78.5 · J45.9 · J06.9                                                                              |
| pa-cie10.txt                                     | 37    | PA_CIE10                                                    | I10 · E11 · E78 · J45                                                                                      |
| pa-codigo-<br>corregimiento-<br>generalizado.txt | 1,331 | PA_CODIGO_CORREGIMIEN<br>TO                                 | 010101,010206 · 10101,10206 · 010102,010206 · 10102,10206                                                  |
| pa-codigo-<br>distrito-<br>generalizado.txt      | 148   | PA_CODIGO_DISTRITO                                          | 0101,0104 · 101,104 · 0102,0102 · 102,102                                                                  |
| pa-<br>corregimiento-<br>generalizado.txt        | 1,760 | PA_CORREGIMIENTO                                            | Bocas del Toro El Empalme · BOCAS DEL TORO EL EMPALME · Bastimentos El<br>Empalme · BASTIMENTOS EL EMPALME |
| pa-decil-<br>alto.txt                            | 5     | PA_DECIL_ALTO                                               | 6 · 7 · 8 · 9                                                                                              |
| pa-decil.txt                                     | 10    | PA_DECIL                                                    | 1 · 2 · 3 · 4                                                                                              |
| pa-detectar-<br>apellidos.txt                    | 258   | PA_L1_APELLIDO -<br>List<br>PA_L1_NOMBRE_COMPLETO<br>- List | Rodríguez · Rodriguez · González · Gonzalez                                                                |
| pa-detectar-<br>bandera.txt                      | 8     | PA_L3_ETNIA - List                                          | s · n · sí · si                                                                                            |
| pa-detectar-<br>codigo-lugar.txt                 | 1,405 | PA_L2_CODIGO_LUGAR -<br>List                                | 010101 · 10101 · 010102 · 10102                                                                            |

| FILE                                | LINES | USED BY                                             | FIRST LINES                                                                                                     |
|-------------------------------------|-------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| pa-detectar-corregimientos.txt      | 814   | PA_L2_CORREGIMIENTO – List                          | Bocas del Toro · Bastimentos · Punta Laurel · Tierra Oscura                                                     |
| pa-detectar-discapacidad.txt        | 26    | PA_L3_SALUD_DISCAPACIDAD – List                     | Física · Fisica · Visual · Auditiva                                                                             |
| pa-detectar-distritos.txt           | 111   | PA_L2_DISTRITO – List                               | Bocas del Toro · Changuinola · Chiriquí Grande · Chiriqui Grande                                                |
| pa-detectar-escolaridad.txt         | 28    | PA_L2_ESCOLARIDAD – List                            | Ningún grado · Ningun grado · Primaria incompleta · Primaria completa                                           |
| pa-detectar-estado-civil.txt        | 18    | PA_L2_ESTADO_CIVIL – List                           | soltero · soltera · casado · casada                                                                             |
| pa-detectar-grupo-sanguineo.txt     | 26    | PA_L3_SALUD_GRUPO_SANGUINEO – List                  | A+ · A- · B+ · B-                                                                                               |
| pa-detectar-identidad-genero.txt    | 17    | PA_L3_IDENTIDAD_GENERO – List                       | Mujer · Hombre · Mujer trans · Hombre trans                                                                     |
| pa-detectar-lenguas.txt             | 32    | PA_L3 LENGUA_INDIGENA – List                        | Ngäbere · Ngabere · Buglere · Guna                                                                              |
| pa-detectar-medicamentos.txt        | 58    | PA_L3_SALUD_MEDICAMENTO – List                      | Acetaminofén · Acetaminofen · Ibuprofeno · Naproxeno                                                            |
| pa-detectar-nacionalidades.txt      | 63    | PA_L2_NACIONALIDAD – List                           | Panameña · Panamena · Colombiana · Venezolana                                                                   |
| pa-detectar-nombres.txt             | 291   | PA_L1_NOMBRE – List<br>PA_L1_NOMBRE_COMPLETO – List | María · Maria · José · Jose                                                                                     |
| pa-detectar-opiniones-politicas.txt | 14    | PA_L3_OPINION_POLITICA – List                       | Izquierda · Centro izquierda · Centro · Centro derecha                                                          |
| pa-detectar-orientacion-sexual.txt  | 17    | PA_L3_ORIENTACION_SEXUAL – List                     | Heterosexual · Homosexual · Lesbiana · Gay                                                                      |
| pa-detectar-partidos.txt            | 35    | PA_L3_AFILIACION POLITICA – List                    | Partido Revolucionario Democrático · Partido Revolucionario Democratico · Realizando Metas · Cambio Democrático |
| pa-detectar-pobreza.txt             | 10    | PA_FIN_NIVEL_SOCIOECONOMICO – List                  | pobreza extrema · pobreza general · pobre · no pobre                                                            |
| pa-detectar-programas.txt           | 23    | PA_FIN_PROGRAMA_SOCIAL – List                       | 120 a los 65 · Red de Oportunidades · Ángel Guardián · Angel Guardian                                           |
| pa-detectar-pueblos.txt             | 52    | PA_L3_ETNIA – List                                  | Guna · Ngäbe · Ngabe · Buglé                                                                                    |
| pa-detectar-religiones.txt          | 61    | PA_L3_RELIGION – List                               | Católica · Catolica · Evangélica · Evangelica                                                                   |
| pa-detectar-seguro-social.txt       | 20    | PA_FIN_SEGURO_SOCIAL – List                         | Asegurado directo · Beneficiario · Jubilado o pensionado por vejez · Pensionado por enfermedad o accidente      |

| FILE                         | LINES | USED BY                          | FIRST LINES                                                                                                                                                         |
|------------------------------|-------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pa-detectar-sexo.txt         | 11    | PA_L2_SEX0 – List                | h · m · f · hombre                                                                                                                                                  |
| pa-detectar-sindicatos.txt   | 15    | PA_L3_AFILIACION_SINDICAL – List | SUNTRACS · CONATO · CONUSI · CONATLAB                                                                                                                               |
| pa-detectar-vivienda.txt     | 17    | PA_FIN_VIVIENDA – List           | Propia · Propia hipotecada · Alquilada · Cedida o prestada                                                                                                          |
| pa-diagnosticos.txt          | 30    | PA_DIAGNOSTICO_TEXTO             | Hipertensión arterial · Diabetes mellitus tipo 2 · Asma bronquial · Lumbalgia                                                                                       |
| pa-direcciones.txt           | 3,000 | PA_DIRECCION                     | Barriada Villa del Carmen, Calle 5 D, Casa 389 · Don Bosco, Calle 14 B, Casa 147 · Calle 63, Edificio Torre del Mar, Piso 10, Apto. 16E · Villa Grecia, Casa No. 46 |
| pa-discapacidad.txt          | 8     | PA_DISCAPACIDAD_LISTA            | Física · Visual · Auditiva · Intelectual                                                                                                                            |
| pa-distrito-generalizado.txt | 234   | PA_DISTRITO                      | Bocas del Toro Almirante · BOCAS DEL TORO ALMIRANTE · Changuinola Changuinola · CHANGUINOLA CHANGUINOLA                                                             |
| pa-empleadores.txt           | 25    | PA_EMPLEADOR                     | Comercializadora del Istmo, S.A. · Servicios Integrales del Pacífico, S.A. · Constructora Cerro Ancón, S.A. · Transportes Canal Seguro, S.A.                        |
| pa-escolaridad.txt           | 13    | PA_ESCOLARIDAD                   | Ningún grado · Primaria incompleta · Primaria completa · Premedia incompleta                                                                                        |
| pa-escuelas.txt              | 15    | PA_ESCUELA                       | Escuela República de Colombia · Escuela Juan Demóstenes Arosemena · Centro Básico General Las Mañanitas · Centro de Educación Básica General Guararé                |
| pa-estado-civil.txt          | 7     | PA_ESTADO_CIVIL                  | Unido · Casado · Soltero · Separado de matrimonio                                                                                                                   |
| pa-grupo-sanguineo.txt       | 8     | PA_GRUPO_SANGUINEO               | A+ · A- · B+ · B-                                                                                                                                                   |
| pa-identidad-genero.txt      | 8     | PA_IDENTIDAD_GENERO_LISTA        | Mujer · Hombre · Mujer trans · Hombre trans                                                                                                                         |
| pa-lenguas.txt               | 9     | PA LENGUA_INDIGENA_LISTA         | Ngäbere · Buglere · Guna · Emberá                                                                                                                                   |
| pa-lugares-poblados.txt      | 52    | PA_LUGAR_POBLADO                 | Penonomé · Cativá · Sabanitas · San Juan                                                                                                                            |
| pa-medicamentos.txt          | 26    | PA_MEDICAMENTO_LISTA             | Acetaminofén · Ibuprofeno · Naproxeno · Diclofenaco                                                                                                                 |
| pa-nacionalidades.txt        | 20    | PA_NACIONALIDAD                  | Panameña · Colombiana · Venezolana · Nicaragüense                                                                                                                   |
| pa-nombres.txt               | 253   | PA_NOMBRE_PILA                   | María · José · Luis · Carlos                                                                                                                                        |
| pa-ocupaciones.txt           | 40    | PA_OCUPACION                     | Oficinista · Comerciante · Vendedor · Obrero de la construcción                                                                                                     |
| pa-opiniones-politicas.txt   | 7     | PA_OPINION_POLITICA_LISTA        | Izquierda · Centro izquierda · Centro · Centro derecha                                                                                                              |
| pa-orientacion-sexual.txt    | 9     | PA_ORIENTACION_SEXUAL_LISTA      | Heterosexual · Homosexual · Lesbiana · Gay                                                                                                                          |
| pa-partidos.txt              | 11    | PA_PARTIDO_POLITICO_LISTA        | Partido Revolucionario Democrático · Realizando Metas · Cambio Democrático · Partido Panameñista                                                                    |

| FILE                           | LINES  | USED BY                    | FIRST LINES                                                                                                |
|--------------------------------|--------|----------------------------|------------------------------------------------------------------------------------------------------------|
| pa-pobreza.txt                 | 4      | PA_POBREZA                 | Pobreza extrema · Pobreza general · No pobre · Vulnerable                                                  |
| pa-programas-sociales.txt      | 9      | PA_PROGRAMA_SOCIAL_LISTA   | 120 a los 65 · Red de Oportunidades · Ángel Guardián · Bono Alimenticio Nutricional                        |
| pa-pueblos.txt                 | 16     | PA_PUEBLO_LISTA            | Guna · Ngäbe · Buglé · Naso Tjërdi                                                                         |
| pa-quintil.txt                 | 5      | PA_QUINTIL                 | 1 · 2 · 3 · 4                                                                                              |
| pa-referencia-credito.txt      | 10     | PA_REFERENCIA_CREDITO      | Al día · Morosidad de 30 días · Morosidad de 60 días · Morosidad de 90 días                                |
| pa-religiones.txt              | 14     | PA_RELIGION_LISTA          | Católica · Evangélica · Adventista del Séptimo Día · Testigo de Jehová                                     |
| pa-ruc-dv-letras-de-vuelta.txt | 5      | PA_RUC_DV_LETRAS_DE_VUELTA | v,AV · i,PI · e,E · n,N                                                                                    |
| pa-ruc-dv-letras.txt           | 5      | PA_RUC_DV_LETRAS           | AV,v · PI,i · E,e · N,n                                                                                    |
| pa-ruc-dv-paso.txt             | 32,317 | PA_RUC_DV_PASO             | 000a,00b0 · 100a,21b1 · 200a,42b2 · 300a,63b3                                                              |
| pa-ruc-dv-resultado.txt        | 1,815  | PA_RUC_DV_RESULTADO        | A8J,81 · 40J,33 · 41J,32 · 4Au,99                                                                          |
| pa-ruc-dv-traslado.txt         | 2,100  | PA_RUC_DV TRASLADO         | 000,000 · 001,100 · 002,200 · 003,300                                                                      |
| pa-seguro-social.txt           | 6      | PA_SEGURO_SOCIAL_LISTA     | Asegurado directo · Beneficiario · Jubilado o pensionado por vejez · Pensionado por enfermedad o accidente |
| pa-sexo-f-m.txt                | 2      | PA_SEX0_F_M                | F · M                                                                                                      |
| pa-sexo-h-m.txt                | 2      | PA_SEX0_H_M                | H · M                                                                                                      |
| pa-sexo-ingles.txt             | 2      | PA_SEX0_INGLES             | Female · Male                                                                                              |
| pa-sexo-mujer-hombre.txt       | 2      | PA_SEX0_MUJER_HOMBRE       | Mujer · Hombre                                                                                             |
| pa-sexo-palabra.txt            | 2      | PA_SEX0_PALABRA            | Femenino · Masculino                                                                                       |
| pa-sin-informacion.txt         | 1      | PA_SUPRIMIR                | Sin información                                                                                            |
| pa-texto-nombres.txt           | 1,094  | PA_TEXTO_LIBRE             | María · MARÍA · Maria · MARIA                                                                              |
| pa-vivienda.txt                | 6      | PA_VIVIENDA_LISTA          | Propia · Propia hipotecada · Alquilada · Cedida o prestada                                                 |

## Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

**E1** producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar\_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro\_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto cliente cte cli paciente pac trabajador trab empleado colaborador funcionario servidor\_? publico alumno estudiante afiliado asegurado cotizante jubilado pensionado beneficiario benef titular madre padre acudiente tutor conyuge pareja medico doctor profesional responsable contacto emergencia fiador codeudor deudor remitente destinatario receptor firmante testigo representante apoderado propietario arrendatario inquilino contratante victima denunciante imputado socio donante heredero persona pers conductor chofer vendedor ejecutivo agente solicitante ciudadano elector votante contribuyente jefe\_?(de\_?)?hogar hijo hija familiar dependiente completos? y\_?apellidos? apellidos?

**E2** producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar\_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro\_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto cliente cte cli paciente pac trabajador trab empleado colaborador funcionario servidor\_? publico alumno estudiante afiliado asegurado cotizante jubilado pensionado beneficiario benef titular madre padre acudiente tutor conyuge pareja medico doctor profesional responsable contacto emergencia fiador codeudor deudor remitente destinatario receptor firmante testigo representante apoderado propietario arrendatario inquilino contratante victima denunciante imputado socio donante heredero persona pers conductor chofer vendedor ejecutivo agente solicitante ciudadano elector votante contribuyente jefe\_?(de\_?)?hogar hijo hija familiar dependiente completo compl full y\_? apellidos? apellidos?

- 
- E3** producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar\_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro\_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto completo
- 
- E4** producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar\_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro\_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto
- 
- E5** ip mac web url correo mail electronica email
- 
- E6** producto prod articulo sucursal interna tipo registro primaria foranea social banco pais moneda catastral ubicacion zona area postal
- 
- E7** fijo variable monto adicional servicio tarifa valor cuenta extra automatico recurrente
- 
- E8** medicos patologicos personales\_?patologicos heredofamiliares gineco obstetricos familiares clinicos academicos laborales crediticios comerciales generales
-

## APPENDIX D

# Profile set and classifiers

The profile set **PA – Ley 81 de 2019 – Datos personales** runs these 135 classifiers, with an assignment threshold of 60%.

Identificadores directos, cuasi-identificadores, datos sensibles (art. 4, 11), datos económicos y de crédito, y antecedentes penales, conforme a la Ley 81 de 2019 y el Decreto Ejecutivo 285 de 2021.

| CLASSIFIER                         | KIND         | DOMAIN                      |
|------------------------------------|--------------|-----------------------------|
| PA_L1_CEDULA – Path                | COLUMN NAME  | PA_L1_CEDULA                |
| PA_L1_CEDULA – Type                | DATA TYPE    | PA_L1_CEDULA                |
| PA_L1_CEDULA – Regex               | VALUE FORMAT | PA_L1_CEDULA                |
| PA_L1_RUC – Path                   | COLUMN NAME  | PA_L1_RUC                   |
| PA_L1_RUC – Type                   | DATA TYPE    | PA_L1_RUC                   |
| PA_L1_RUC – Regex                  | VALUE FORMAT | PA_L1_RUC                   |
| PA_L1_RUC_DV – Path                | COLUMN NAME  | PA_L1_RUC_DV                |
| PA_L1_RUC_DV – Type                | DATA TYPE    | PA_L1_RUC_DV                |
| PA_L1_PASAPORTE – Path             | COLUMN NAME  | PA_L1_PASAPORTE             |
| PA_L1_PASAPORTE – Regex            | VALUE FORMAT | PA_L1_PASAPORTE             |
| PA_L1_DOCUMENTO_MIGRATORIO – Path  | COLUMN NAME  | PA_L1_DOCUMENTO_MIGRATORIO  |
| PA_L1_SEGURO_SOCIAL – Path         | COLUMN NAME  | PA_L1_SEGURO_SOCIAL         |
| PA_L1_IDONEIDAD – Path             | COLUMN NAME  | PA_L1_IDONEIDAD             |
| PA_L1_NOMBRE – Path                | COLUMN NAME  | PA_L1_NOMBRE                |
| PA_L1_NOMBRE – List                | KNOWN VALUES | PA_L1_NOMBRE                |
| PA_L1_APELLIDO – Path              | COLUMN NAME  | PA_L1_APELLIDO              |
| PA_L1_APELLIDO – List              | KNOWN VALUES | PA_L1_APELLIDO              |
| PA_L1_NOMBRE_COMPLETO – Path       | COLUMN NAME  | PA_L1_NOMBRE_COMPLETO       |
| PA_L1_NOMBRE_COMPLETO – List       | KNOWN VALUES | PA_L1_NOMBRE_COMPLETO       |
| PA_L1_EMAIL – Path                 | COLUMN NAME  | PA_L1_EMAIL                 |
| PA_L1_EMAIL – Type                 | DATA TYPE    | PA_L1_EMAIL                 |
| PA_L1_EMAIL – Regex                | VALUE FORMAT | PA_L1_EMAIL                 |
| PA_L1_TELEFONO – Path              | COLUMN NAME  | PA_L1_TELEFONO              |
| PA_L1_TELEFONO – Regex             | VALUE FORMAT | PA_L1_TELEFONO              |
| PA_L1_DIRECCION – Path             | COLUMN NAME  | PA_L1_DIRECCION             |
| PA_L1_DIRECCION – Type             | DATA TYPE    | PA_L1_DIRECCION             |
| PA_L1_DIRECCION – Regex            | VALUE FORMAT | PA_L1_DIRECCION             |
| PA_L1_DIRECCION_COMPLEMENTO – Path | COLUMN NAME  | PA_L1_DIRECCION_COMPLEMENTO |
| PA_L1_CUENTA_BANCARIA – Path       | COLUMN NAME  | PA_L1_CUENTA_BANCARIA       |
| PA_L1_TARJETA – Path               | COLUMN NAME  | PA_L1_TARJETA               |

| CLASSIFIER                    | KIND         | DOMAIN                 |
|-------------------------------|--------------|------------------------|
| PA_L1_TARJETA – Regex         | VALUE FORMAT | PA_L1_TARJETA          |
| PA_L1_IP – Path               | COLUMN NAME  | PA_L1_IP               |
| PA_L1_IP – Regex              | VALUE FORMAT | PA_L1_IP               |
| PA_L1_DISPOSITIVO – Path      | COLUMN NAME  | PA_L1_DISPOSITIVO      |
| PA_L1_DISPOSITIVO – Regex     | VALUE FORMAT | PA_L1_DISPOSITIVO      |
| PA_L1_PLACA – Path            | COLUMN NAME  | PA_L1_PLACA            |
| PA_L1_PLACA – Regex           | VALUE FORMAT | PA_L1_PLACA            |
| PA_L1_VIN – Path              | COLUMN NAME  | PA_L1_VIN              |
| PA_L1_VIN – Regex             | VALUE FORMAT | PA_L1_VIN              |
| PA_L1_FINCA – Path            | COLUMN NAME  | PA_L1_FINCA            |
| PA_L1_NUMERO_CONTRATO – Path  | COLUMN NAME  | PA_L1_NUMERO_CONTRATO  |
| PA_L1_USUARIO_RRSS – Path     | COLUMN NAME  | PA_L1_USUARIO_RRSS     |
| PA_L1_CREDENCIAL – Path       | COLUMN NAME  | PA_L1_CREDENCIAL       |
| PA_L1_GEOLOCALIZACION – Path  | COLUMN NAME  | PA_L1_GEOLOCALIZACION  |
| PA_L1_GEOLOCALIZACION – Regex | VALUE FORMAT | PA_L1_GEOLOCALIZACION  |
| PA_L1_TEXTO_LIBRE – Path      | COLUMN NAME  | PA_L1_TEXTO_LIBRE      |
| PA_L1_TEXTO_LIBRE – Type      | DATA TYPE    | PA_L1_TEXTO_LIBRE      |
| PA_L1_TEXTO_LIBRE – Regex     | VALUE FORMAT | PA_L1_TEXTO_LIBRE      |
| PA_L2_FECHA_NACIMIENTO – Path | COLUMN NAME  | PA_L2_FECHA_NACIMIENTO |
| PA_L2_FECHA_NACIMIENTO – Type | DATA TYPE    | PA_L2_FECHA_NACIMIENTO |
| PA_L2_ANIO_NACIMIENTO – Path  | COLUMN NAME  | PA_L2_ANIO_NACIMIENTO  |
| PA_L2_ANIO_NACIMIENTO – Type  | DATA TYPE    | PA_L2_ANIO_NACIMIENTO  |
| PA_L2_EDAD – Path             | COLUMN NAME  | PA_L2_EDAD             |
| PA_L2_EDAD – Type             | DATA TYPE    | PA_L2_EDAD             |
| PA_L2_FECHA_EVENTO – Path     | COLUMN NAME  | PA_L2_FECHA_EVENTO     |
| PA_L2_FECHA_EVENTO – Type     | DATA TYPE    | PA_L2_FECHA_EVENTO     |
| PA_L2_SEX0 – Path             | COLUMN NAME  | PA_L2_SEX0             |
| PA_L2_SEX0 – List             | KNOWN VALUES | PA_L2_SEX0             |
| PA_L2_DISTRITO – Path         | COLUMN NAME  | PA_L2_DISTRITO         |
| PA_L2_DISTRITO – List         | KNOWN VALUES | PA_L2_DISTRITO         |
| PA_L2_CORREGIMIENTO – Path    | COLUMN NAME  | PA_L2_CORREGIMIENTO    |
| PA_L2_CORREGIMIENTO – List    | KNOWN VALUES | PA_L2_CORREGIMIENTO    |
| PA_L2_CODIGO_LUGAR – Path     | COLUMN NAME  | PA_L2_CODIGO_LUGAR     |
| PA_L2_CODIGO_LUGAR – Type     | DATA TYPE    | PA_L2_CODIGO_LUGAR     |
| PA_L2_CODIGO_LUGAR – List     | KNOWN VALUES | PA_L2_CODIGO_LUGAR     |
| PA_L2_LUGAR_POBLADO – Path    | COLUMN NAME  | PA_L2_LUGAR_POBLADO    |
| PA_L2_BARRIADA – Path         | COLUMN NAME  | PA_L2_BARRIADA         |

| CLASSIFIER                             | KIND         | DOMAIN                          |
|----------------------------------------|--------------|---------------------------------|
| PA_L2_CODIGO_POSTAL - Path             | COLUMN NAME  | PA_L2_CODIGO_POSTAL             |
| PA_L2_CODIGO_POSTAL - Regex            | VALUE FORMAT | PA_L2_CODIGO_POSTAL             |
| PA_L2_NACIONALIDAD - Path              | COLUMN NAME  | PA_L2_NACIONALIDAD              |
| PA_L2_NACIONALIDAD - List              | KNOWN VALUES | PA_L2_NACIONALIDAD              |
| PA_L2_ESTADO_CIVIL - Path              | COLUMN NAME  | PA_L2_ESTADO_CIVIL              |
| PA_L2_ESTADO_CIVIL - List              | KNOWN VALUES | PA_L2_ESTADO_CIVIL              |
| PA_L2_OCUPACION - Path                 | COLUMN NAME  | PA_L2_OCUPACION                 |
| PA_L2_EMPLEADOR - Path                 | COLUMN NAME  | PA_L2_EMPLEADOR                 |
| PA_L2_ESCOLARIDAD - Path               | COLUMN NAME  | PA_L2_ESCOLARIDAD               |
| PA_L2_ESCOLARIDAD - List               | KNOWN VALUES | PA_L2_ESCOLARIDAD               |
| PA_L2_ESCUELA - Path                   | COLUMN NAME  | PA_L2_ESCUELA                   |
| PA_L2_DEPENDIENTES - Path              | COLUMN NAME  | PA_L2_DEPENDIENTES              |
| PA_L2_DEPENDIENTES - Type              | DATA TYPE    | PA_L2_DEPENDIENTES              |
| PA_L3_ETNIA - Path                     | COLUMN NAME  | PA_L3_ETNIA                     |
| PA_L3_ETNIA - List                     | KNOWN VALUES | PA_L3_ETNIA                     |
| PA_L3 LENGUA_INDIGENA - Path           | COLUMN NAME  | PA_L3 LENGUA_INDIGENA           |
| PA_L3 LENGUA_INDIGENA - List           | KNOWN VALUES | PA_L3 LENGUA_INDIGENA           |
| PA_L3_SALUD_DIAGNOSTICO - Path         | COLUMN NAME  | PA_L3_SALUD_DIAGNOSTICO         |
| PA_L3_SALUD_DIAGNOSTICO - Regex        | VALUE FORMAT | PA_L3_SALUD_DIAGNOSTICO         |
| PA_L3_SALUD_MEDICAMENTO - Path         | COLUMN NAME  | PA_L3_SALUD_MEDICAMENTO         |
| PA_L3_SALUD_MEDICAMENTO - List         | KNOWN VALUES | PA_L3_SALUD_MEDICAMENTO         |
| PA_L3_SALUD_IDENTIFICADOR - Path       | COLUMN NAME  | PA_L3_SALUD_IDENTIFICADOR       |
| PA_L3_SALUD_TEXTO_CLINICO - Path       | COLUMN NAME  | PA_L3_SALUD_TEXTO_CLINICO       |
| PA_L3_SALUD_TEXTO_CLINICO - Type       | DATA TYPE    | PA_L3_SALUD_TEXTO_CLINICO       |
| PA_L3_SALUD_RESULTADO_EXAMEN - Path    | COLUMN NAME  | PA_L3_SALUD_RESULTADO_EXAMEN    |
| PA_L3_SALUD_GRUPO_SANGUINEO - Path     | COLUMN NAME  | PA_L3_SALUD_GRUPO_SANGUINEO     |
| PA_L3_SALUD_GRUPO_SANGUINEO - List     | KNOWN VALUES | PA_L3_SALUD_GRUPO_SANGUINEO     |
| PA_L3_SALUD_DISCAPACIDAD - Path        | COLUMN NAME  | PA_L3_SALUD_DISCAPACIDAD        |
| PA_L3_SALUD_DISCAPACIDAD - List        | KNOWN VALUES | PA_L3_SALUD_DISCAPACIDAD        |
| PA_L3_SALUD_SEXUAL_REPRODUCTIVA - Path | COLUMN NAME  | PA_L3_SALUD_SEXUAL_REPRODUCTIVA |
| PA_L3_SALUD_MENTAL_ADICCIONES - Path   | COLUMN NAME  | PA_L3_SALUD_MENTAL_ADICCIONES   |
| PA_L3_BIOMETRICO - Path                | COLUMN NAME  | PA_L3_BIOMETRICO                |
| PA_L3_BIOMETRICO - Type                | DATA TYPE    | PA_L3_BIOMETRICO                |
| PA_L3_INFORMACION_GENETICA - Path      | COLUMN NAME  | PA_L3_INFORMACION_GENETICA      |
| PA_L3_ORIENTACION_SEXUAL - Path        | COLUMN NAME  | PA_L3_ORIENTACION_SEXUAL        |
| PA_L3_ORIENTACION_SEXUAL - List        | KNOWN VALUES | PA_L3_ORIENTACION_SEXUAL        |
| PA_L3_IDENTIDAD_GENERO - Path          | COLUMN NAME  | PA_L3_IDENTIDAD_GENERO          |

| CLASSIFIER                             | KIND         | DOMAIN                          |
|----------------------------------------|--------------|---------------------------------|
| PA_L3_IDENTIDAD_GENERO - List          | KNOWN VALUES | PA_L3_IDENTIDAD_GENERO          |
| PA_L3_VIDA_SEXUAL - Path               | COLUMN NAME  | PA_L3_VIDA_SEXUAL               |
| PA_L3_RELIGION - Path                  | COLUMN NAME  | PA_L3_RELIGION                  |
| PA_L3_RELIGION - List                  | KNOWN VALUES | PA_L3_RELIGION                  |
| PA_L3_CREENCIA_FILOSOFICA_MORAL - Path | COLUMN NAME  | PA_L3_CREENCIA_FILOSOFICA_MORAL |
| PA_L3_OPINION_POLITICA - Path          | COLUMN NAME  | PA_L3_OPINION_POLITICA          |
| PA_L3_OPINION_POLITICA - List          | KNOWN VALUES | PA_L3_OPINION_POLITICA          |
| PA_L3_AFILIACION_POLITICA - Path       | COLUMN NAME  | PA_L3_AFILIACION_POLITICA       |
| PA_L3_AFILIACION_POLITICA - List       | KNOWN VALUES | PA_L3_AFILIACION_POLITICA       |
| PA_L3_AFILIACION_SINDICAL - Path       | COLUMN NAME  | PA_L3_AFILIACION_SINDICAL       |
| PA_L3_AFILIACION_SINDICAL - List       | KNOWN VALUES | PA_L3_AFILIACION_SINDICAL       |
| PA_FIN_REFERENCIA_CREDITO - Path       | COLUMN NAME  | PA_FIN_REFERENCIA_CREDITO       |
| PA_FIN_MONTO_DEUDA - Path              | COLUMN NAME  | PA_FIN_MONTO_DEUDA              |
| PA_FIN_MONTO_DEUDA - Type              | DATA TYPE    | PA_FIN_MONTO_DEUDA              |
| PA_FIN_INGRESO_MENSUAL - Path          | COLUMN NAME  | PA_FIN_INGRESO_MENSUAL          |
| PA_FIN_INGRESO_MENSUAL - Type          | DATA TYPE    | PA_FIN_INGRESO_MENSUAL          |
| PA_FIN_PATRIMONIO - Path               | COLUMN NAME  | PA_FIN_PATRIMONIO               |
| PA_FIN_PATRIMONIO - Type               | DATA TYPE    | PA_FIN_PATRIMONIO               |
| PA_FIN_PENSION - Path                  | COLUMN NAME  | PA_FIN_PENSION                  |
| PA_FIN_PENSION - Type                  | DATA TYPE    | PA_FIN_PENSION                  |
| PA_FIN_SEGURO_SOCIAL - Path            | COLUMN NAME  | PA_FIN_SEGURO_SOCIAL            |
| PA_FIN_SEGURO_SOCIAL - List            | KNOWN VALUES | PA_FIN_SEGURO_SOCIAL            |
| PA_FIN_PROGRAMA_SOCIAL - Path          | COLUMN NAME  | PA_FIN_PROGRAMA_SOCIAL          |
| PA_FIN_PROGRAMA_SOCIAL - List          | KNOWN VALUES | PA_FIN_PROGRAMA_SOCIAL          |
| PA_FIN_NIVEL_SOCIOECONOMICO - Path     | COLUMN NAME  | PA_FIN_NIVEL_SOCIOECONOMICO     |
| PA_FIN_NIVEL_SOCIOECONOMICO - List     | KNOWN VALUES | PA_FIN_NIVEL_SOCIOECONOMICO     |
| PA_FIN_VIVIENDA - Path                 | COLUMN NAME  | PA_FIN_VIVIENDA                 |
| PA_FIN_VIVIENDA - List                 | KNOWN VALUES | PA_FIN_VIVIENDA                 |
| PA_PENAL_ANTECEDENTES - Path           | COLUMN NAME  | PA_PENAL_ANTECEDENTES           |
| PA_PENAL_CAUSA - Path                  | COLUMN NAME  | PA_PENAL_CAUSA                  |
| PA_PENAL_CAUSA - Regex                 | VALUE FORMAT | PA_PENAL_CAUSA                  |